

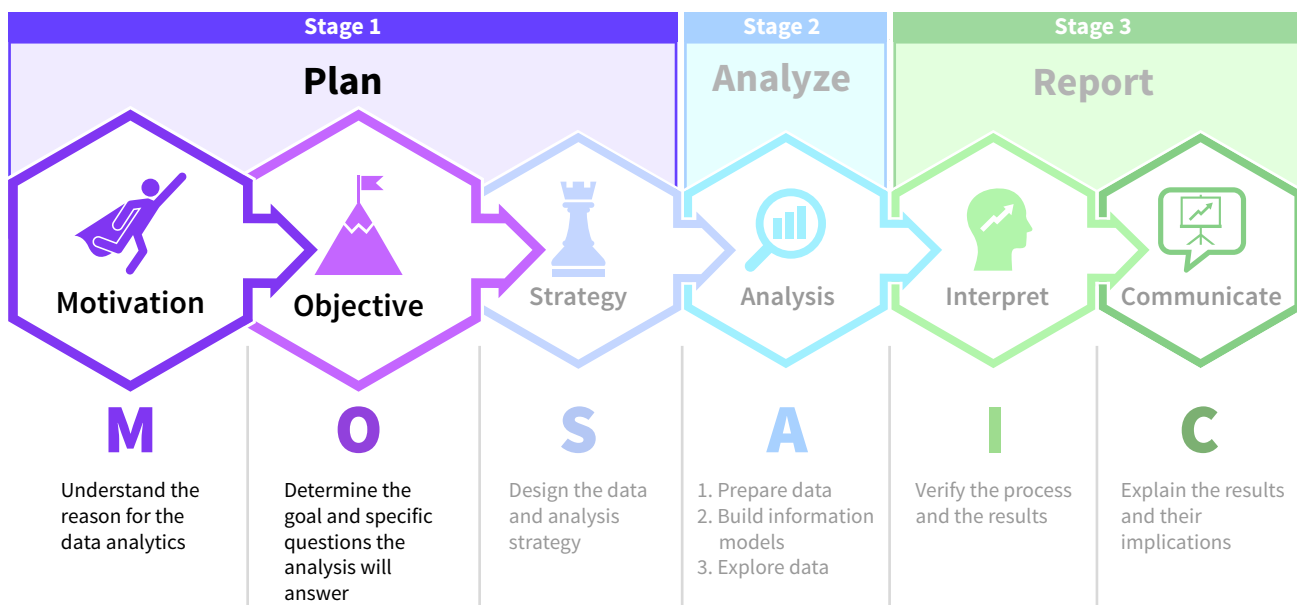
Motivations and Objectives for Data Analysis

Chapter Preview

Data analytics is a rapidly evolving and exciting field, and there is a demand within it for accounting professionals who can use new tools to analyze large amounts of data. Analyzing data and extracting information from it leads to informed decisions, and businesses today want people with these skills.

But a data analytics project does not start with a deep dive into the data. Planning the project, including understanding the motivation for performing it and developing its specific goals, is equally important. Data analytics can be costly, complex, and time consuming. A detailed plan links the project's motivation with its objective and key questions, which helps us select the best analysis method. Without a plan, it is possible to overlook important information or critical questions, which could make the analysis less effective or even useless.

The first stage of the data analytics process is planning. It includes identifying the motivation for the project, which helps determine the objective. A project's objective drives everything from the questions asked to the analysis choices we make to find the answers to them. This chapter examines the link between understanding why the project is necessary and developing a focused project objective.



Professional Insight: Why do Motivations and Objectives Matter for Data Analysis Projects?

Georgia lived with her parents while earning an undergraduate accounting degree at the local university. Immediately after graduation, she completed an auditing internship during busy season. She is entering a master's program of accounting in the fall semester.

On the first day of my internship, I learned professional accountants follow a process each time they work on a data analyses project. **First, they understand the motivation for their projects, which in turn influences their specific work objectives. When they are clear about their motivation and objectives, they can plan a successful strategy for their data analyses projects.**

I also learned that professional accountants are motivated by more than personal career goals. Auditors want to add value to their firms and clients by assuring them that their financial statements are reliable and accurate. After all, good information means better capital markets, lending decisions, government relationships, and business partnerships.

This helped me realize that I was also motivated by more than just getting that great job after college. I want to make a difference in the world, and I want to honor my parents' hard work raising me to be successful. This awareness of my motivations has helped me become more committed to my objective and my studies!

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 3.1 Summarize the relationship between motivations, objectives, and data analysis questions.	- Understanding Motivation - Clear Objectives Lead to Focused Data Analysis Questions	Link Motivation to Objectives (Example: Auditing)
LO 3.2 Demonstrate how to develop descriptive questions.	- Develop Descriptive Questions - Descriptive Analyses Examples	Describe Customers' Buying Behaviors (Example: Financial Accounting)
LO 3.3 Demonstrate how to develop diagnostic questions.	- Develop Diagnostic Questions - Diagnostic Analyses Examples	Determine the Risk of Material Misstatement of Sales (Example: Auditing)
LO 3.4 Demonstrate how to develop predictive questions.	- Develop Predictive Questions - Predictive Analyses Examples	Plan a Sales Trend Analysis (Example: Managerial Accounting)
LO 3.5 Demonstrate how to develop prescriptive questions.	- Develop Prescriptive questions - Prescriptive Analyses Examples	Prescribe Optimal Sales Mix (Example: Managerial Accounting)
LO 3.6 Describe motivations and objectives for data analytics in professional practice.	- Accounting Information Systems - Auditing - Financial Accounting - Managerial Accounting - Tax Accounting	Match Motivations to Professional Practice Areas

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

3.1 How Does Motivation Inform Objective-Based Data Analysis Questions?

LEARNING OBJECTIVE 1

Summarize the relationship between motivations, objectives, and data analysis questions.

It is easier to stay focused on a task's objective when we know why we are performing it in the first place. Understanding the motivation to perform an analysis and identifying its objective also helps create specific, objective-based questions. After all, it is impossible to get the right answers if we do not know what to ask.

Understanding Motivation

Motivation in data analytics in accounting is the reason the analysis is being performed. The “why” behind a project can vary, but data analysis projects are typically motivated by four forces:

- **Opportunity:** Evaluating new opportunities that will benefit the organization.
- **Professional issues:** Evaluating changes due to new laws, regulations, or changes in accounting practices.
- **Problem solving:** Solving a problem or issue that the organization is facing.
- **Process and performance assessment:** Understanding and improving the processes and performance of the organization.



Illustration 3.1 describes some common motivations and includes an example from a company called Super Scooters, which manufactures and sells a variety of scooters.

In each Super Scooters example, the motivation for the analysis is to either describe the situation's potential, diagnose the causal factors or determine operational change, predict sales revenues or net income, or prescribe the best course of action. Regardless of the motivation's source, the data analyses choices we make depend on what can be gained from the analysis. If its potential benefits outweigh the costs of doing it, then stakeholders will consider it a valuable analysis.

Clear Objectives Lead to Focused Data Analysis Questions

You just learned that the four common motivators for data analytics in accounting are opportunities, regulatory changes, solving problems, and process and performance evaluation. For example, a company with an underperforming business unit will evaluate its performance to understand why. A successful data analytics project depends on a more specific statement, which will then inform the questions that will be asked in the analysis.

ILLUSTRATION 3.1 Motivation for Data Analytics in Accounting

Motivation	Description	Role of Data Analysis	Super Scooters Example
Opportunities	New, potentially advantageous occasions or channels.	Reliable evidence justifies new products and services. Evidence to justify changing resource allocations and capital sourcing.	• New scooter models. • New services, such as scooter training classes. • Store expansions.
Professional Issues and Requirements	Compliance with new or upcoming regulations, laws, or changes in practices.	Understanding changes improves responses. Evaluating new accounting regulations can positively impact client financial statements.	The local government may have recently allowed scooters, unmotorized and motorized, on all paved bicycle paths and road lanes in the city.
Problem Solving	Solving problems related to risks and issues around clients, technologies, employees, operations, and supply chains.	Solving problems to minimize productivity losses.	Super Scooters may need to investigate why a particular scooter is no longer selling as well as it did, or why it is not selling as well as their other scooter products.
Process and Performance Assessment	Evaluating financial statements for material misstatements or risks of material misstatement. Inspecting operational efficiency and effectiveness at the individual, functional, unit, and business levels.	Audit data analytics can provide risk assessment and substantive audit evidence. Evaluating the success of a new strategy helps management make future decisions.	Super Scooters may need to evaluate the contribution margin and inventory turnover for each product line, or the performance of each salesperson.

Determine the Objective

The objective of a data analysis project naturally follows the motivation. Let's go back to that common motivation for data analysis, an underperforming business unit:

- The motivation is to solve a problem by evaluating the unit's performance to discover why it is underperforming.
- It is necessary to be more specific because poor performance could have several causes.
- The first step is to specify the areas of performance to investigate and articulate what those investigations should accomplish.

In other words, it is important to clarify the goal of the data analysis.

Every data analysis project begins with setting an **objective**, which is the project's goal. It is a statement that details what the project will accomplish. A new example can illustrate how to determine a project's objective. Omni Restaurants owns restaurants across five regions in the United States. The financial information for the current year by region is shown in **Illustration 3.2**.

Illustration 3.2 shows that Omni is losing money in one region:

- **Motivation:** They are concerned about the losses in Region 1 and want to understand what is driving them.
- **General objective:** Eliminate those financial losses.

Now we have moved from motivation to objective. But the objective is so general that it is difficult to decide where to begin the analysis. A specific objective would narrow the analysis' focus. In the Omni case, only one region is experiencing losses:

- **Specific objective:** Identify the factors driving the losses in Region 1.

After the objective is clear and specific, formulate the questions that will achieve that objective once they are addressed.

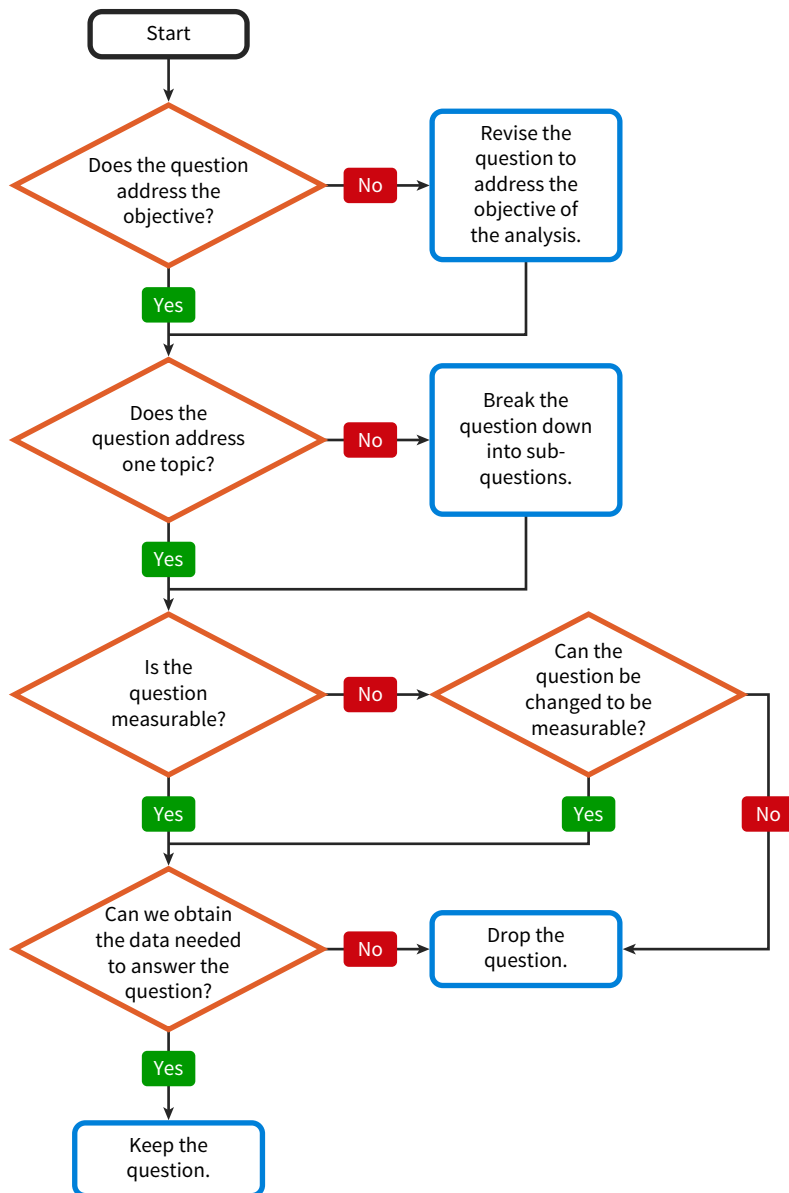
ILLUSTRATION 3.2 Omni Restaurants Financial Performance by Region

	Region 1	Region 2	Region 3	Region 4	Region 5	Total
Dine in Sales	\$226,800	\$333,396	\$367,569	\$405,245	\$446,783	\$1,779,793
Takeout Sales	\$151,200	\$222,264	\$245,046	\$270,163	\$297,855	\$1,186,528
Total Revenues	\$378,000	\$555,660	\$612,615	\$675,408	\$744,638	\$2,966,321
Cost of Goods Sold	\$ 86,940	\$122,245	\$122,523	\$128,328	\$134,035	\$ 594,071
Gross Profit	\$291,060	\$433,415	\$490,092	\$547,080	\$610,603	\$2,372,250
Lease	\$ 90,000	\$ 61,500	\$ 63,038	\$ 64,613	\$ 66,229	\$ 345,380
Marketing	\$ 30,000	\$ 25,000	\$ 25,000	\$ 25,000	\$ 25,000	\$ 130,000
Salaries	\$200,835	\$204,030	\$224,943	\$236,190	\$248,000	\$1,113,998
Other Expenses	\$ 5,250	\$ 4,000	\$ 4,500	\$ 5,000	\$ 5,500	\$ 24,250
Total S G & A Expenses	\$326,085	\$294,530	\$317,481	\$330,803	\$344,729	\$1,613,628
Depreciation	\$ 36,960	\$ 36,960	\$ 36,960	\$ 36,960	\$ 36,960	\$ 184,800
Interest	\$ 23,621	\$ 20,668	\$ 17,716	\$ 14,763	\$ 11,810	\$ 88,578
Earnings Before Tax	(\$95,606)	\$ 81,257	\$117,935	\$164,554	\$217,104	\$ 485,244
Income Tax Expenses	0	\$ 29,526	\$ 42,853	\$ 59,344	\$ 77,911	\$ 209,634
Net Income (Loss)	(\$95,606)	\$ 51,731	\$ 75,082	\$105,210	\$139,193	\$ 275,610

Articulate Questions

Asking the wrong questions leads to answers that do not address the objective. Good questions are clear, concise, and measurable. So, how are good questions developed? Evaluate each question as to whether it addresses the objective, focuses on a single issue, is measurable, and if the data necessary to answer it are available. If the question does not meet this criteria, then either revise it, break it down, or drop it. Use the flowchart in **Illustration 3.3** to develop specific data analysis questions.

ILLUSTRATION 3.3 Flowchart for Developing Data Analysis Questions

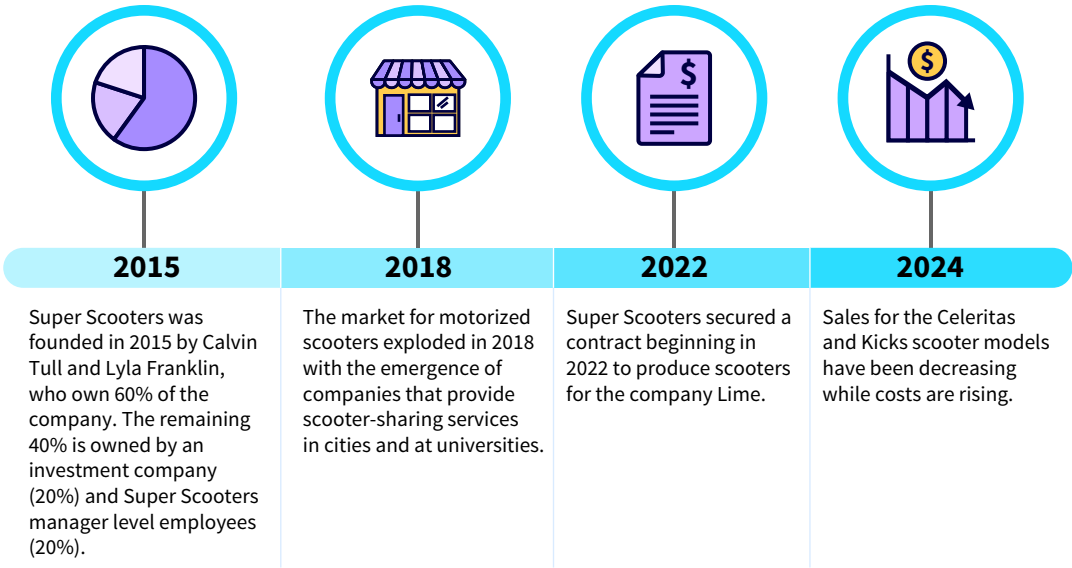


A critical thinking approach to question development can help with this process. Recall that thinking critically when performing data analytics involves considering those who are impacted by the analysis, its purpose, considering a variety of alternatives at each step, controlling for risks, acquiring the right information or knowledge, and reflecting on past and current projects.

The Role of Critical Thinking

Let’s use the Super Scooters example to show the importance of critically evaluating motivation, objectives, and questions at this stage of the data analysis process. Assume you are an accountant for the company Super Scooters. Super Scooters is a privately held company¹ that produces three models of motorized scooters (two electric and one gas-powered) and one non-motorized scooter (Illustration 3.4).

ILLUSTRATION 3.4 Super Scooters Company Timeline



Super Scooters is now in its third year of the contract with Lime, and they are unsure if they have made the right production decisions. Sales seem to be decreasing for some models, and costs are rising. Calvin and Lyla asked you to help them do three things:

- 1. Understand which models have decreasing sales and why.
- 2. Forecast future warranty expenses.
- 3. Determine the most profitable product mix for production.

Use the critical thinking framework, SPARKS, to identify the motivation, objectives, and questions relevant to the analysis Calvin and Lyla would like you to perform. Illustration 3.5 summarizes how critical thinking helps evaluate motivations, objectives, and questions.

¹A privately held company is one that does not sell their stock on public exchanges.

ILLUSTRATION 3.5 Thinking Critically About Motivations and Objectives

Critical Thinking Element		Why It Matters	Super Scooters Example
S	 Stakeholders	Considering the perspectives of stakeholders helps identify the objective and develop good questions.	The internal stakeholders are Lyla and Calvin. Their motivation is to better understand company performance. The objective is to describe current performance, diagnose issues related to poor performance, predict future sales, and make recommendations for improvements.
P	 Purpose	Critically thinking through the purpose of data analyses is the link between a defined objective and specific questions: • A purpose is the reason something is done or created. • An objective is something we plan to do, or a goal. The questions will be different depending on whether the purpose is to describe and diagnose current conditions or predict and prescribe future outcomes.	Based on the objectives, the purpose is to: • Describe what is happening now and diagnose issues related to poor performance. • Predict future sales. • Prescribe a profitable sales mix.
A	 Alternatives	Evaluating different methods of analysis increases the likelihood of selecting the best option for the project's purpose.	Develop several alternative methods for performing the analysis to describe performance over the past year. Consider how likely each option is to provide valuable information. Rank the alternatives and choose the one with the highest expected value.
R	 Risks	Being aware of potential risks makes it possible to also mitigate them when evaluating motivation, objectives, and questions.	It may be tempting to make Calvin and Lyla happy with a positive outlook in the predictive model. Be aware of that bias and develop objectives and questions that will not be influenced by it.
K	 Knowledge	Successfully completing a project requires confirming we have the necessary knowledge to do so or doing additional research to obtain that knowledge.	One purpose of the Super Scooters analysis is to predict future sales and expenses. If you do not know how to prepare a predictive analysis, you must do one of two things: • Hire someone to prepare the predictive analysis. • Learn how to do the analysis.
S	 Self-Reflection	Understanding the purpose of the analysis makes it easier to apply the experience of performing it to future data analytics contexts and tasks. Similarly, considering past experiences can help identify objectives and develop questions.	Assessing whether you should have performed the analyses may depend on whether the information lead to better decisions by management. If it did not, then consider why. Only then can you compare expected net benefits to the actual net benefits.

Apply It 3.1

Link Motivation to Objectives



Auditing Best friends Luanne and Maxine had been making baked goods for a friend who owned a local coffee shop. Their delicious baked goods soon had so many fans that Luanne and Maxine had to rent commercial kitchen space to keep up with demand. Now their business, Best Bakes Bakery, provides baked goods for a variety of coffee shops and restaurants.

They want to expand their operations into other states and are looking for potential investors. They believe a CPA's opinion on their financial statements will help convince investors to invest in their company.

The owners hired your CPA firm to prepare a review of their financial statements. Best Bakes Bakery is the first bakery business for your firm, and you are the external auditor assigned to the engagement. Your supervisor asked you to use data analytics for the risk assessment:

- You have the last three years of financial statement data and must calculate several ratios relevant to the company's operations.
- Your supervisor believes the analysis will identify areas of potentially higher risk of material misstatement.

1. Which category of motivation applies to this scenario?
2. Who are the stakeholders in your data analysis?
3. What is the objective of your analysis?

SOLUTION

1. The motivation is process and performance evaluation. The analysis of financial ratios will provide information relevant to the risk assessment of the potential for material misstatements.
2. You, your audit firm, and the client are the internal stakeholders. The client's investors and creditors are the external stakeholders.
3. The objective of the analysis is to determine the potential risk of material misstatements in the financial statements.

3.2 What are Descriptive Objectives?

LEARNING OBJECTIVE 2

Demonstrate how to develop descriptive questions.

A data analysis project with the objective to understand something that is currently happening or that has happened requires descriptive questions. These types of questions are answered using descriptive analytics. Descriptive questions are often the first asked in any analysis because understanding the data is necessary before beginning more advanced analyses.

Develop Descriptive Questions

Descriptive questions are designed to better understand data to answer business questions. To develop good descriptive questions, first identify the objective of the analysis, then break that down into questions. Remember, a good question is one that relates to the objective, is specific, measurable, and can be answered with the available data.

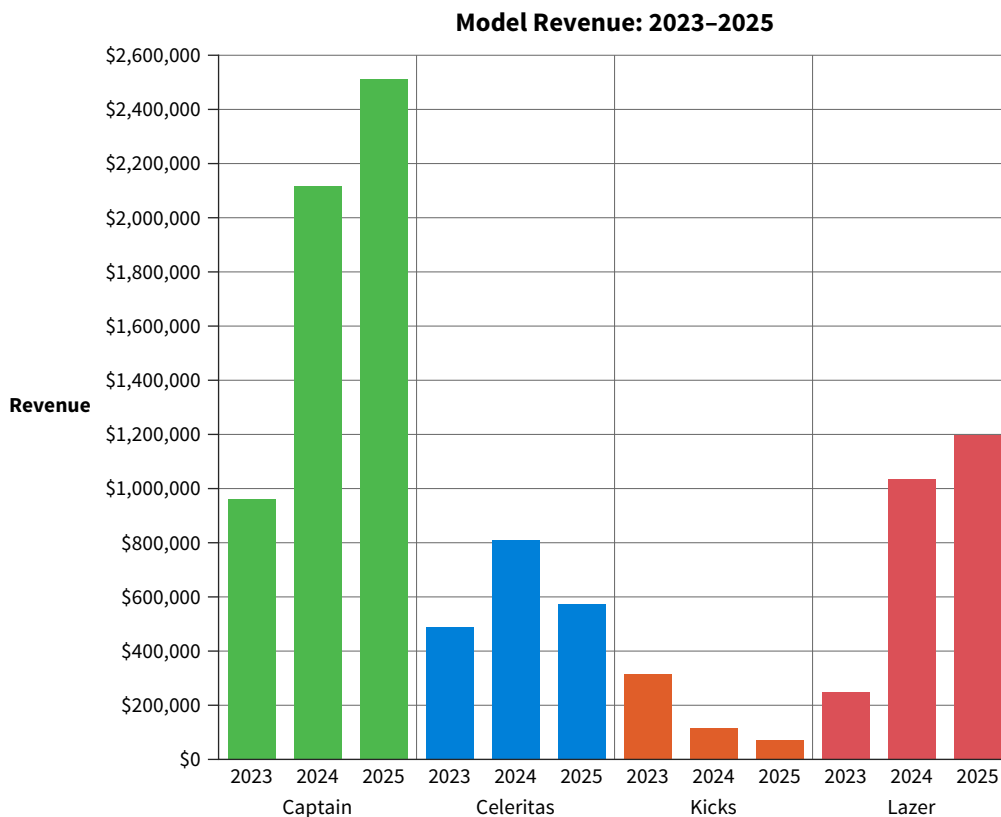
Let's continue with the Super Scooters example to illustrate. **Illustration 3.6** provides the financial information for the years 2023 through 2025.

ILLUSTRATION 3.6 Super Scooters' Income Statements: 2023–2025

	Year Ended December 31		
	2023	2024	2025
Gross Sales	\$4,924,816	\$10,506,334	\$11,378,847
Total Variable Costs	\$2,190,588	\$ 4,922,488	\$ 5,416,827
Contribution Margin	\$2,734,228	\$ 5,583,846	\$ 5,962,020
Total Fixed Costs	\$ 579,988	\$ 1,027,391	\$ 1,042,201
Net Income	\$2,154,240	\$ 4,556,455	\$ 4,919,819

While sales and net revenue have increased, two of the models have decreasing net revenue (**Illustration 3.7**).

ILLUSTRATION 3.7 Scooter Model Revenue: 2023–2025



Super Scooters wants to understand the decreases in net revenue for the Celeritas and Kicks models. What questions should they ask of the data? What possible measure(s) can be used to answer the questions?

Illustration 3.8 provides examples of both initial questions and sub-questions (in other words, more detailed questions).

ILLUSTRATION 3.8 Descriptive Questions to Understand Decreases in Revenue at Super Scooters

Objective	Initial Questions	Sub-questions	Possible Measures
Understand decreases in net revenue for the Celeritas and Kicks models.	Have sales decreased for the Celeritas and Kicks models?	1. Have sales decreased for the Celeritas model?	Gross sales dollars
		2. Have sales decreased in all locations for the Celeritas model?	Average gross sales dollars
		3. Have sales decreased for the Kicks model?	Sales volume
		4. Have sales decreased in all locations for the Kicks model?	
Have expenses increased for the Celeritas and Kicks models?		1. Have expenses increased in all locations for the Celeritas model?	Total expense dollars
		2. Have expenses increased in all locations for the Kicks model?	Average total expense dollars Percent change from prior year
Have variable expenses increased for the Celeritas and Kicks models?		1. Which variable expenses increased from prior year for the Celeritas model?	Variable expense dollars
		2. Do some locations have higher increases in variable expenses for the Celeritas model than others?	Average variable expenses
		3. Which variable expenses increased from prior year for the Kicks model?	Percent change from prior year
		4. Do some locations have higher increases in variable expenses for the Kicks model than others?	
Have fixed expenses increased for the Celeritas and Kicks models?		1. Which fixed expenses increased from prior year for the Celeritas model?	Fixed expense dollars
		2. Do some locations have higher increases in fixed expenses for the Celeritas model than others?	Average fixed expenses
		3. Which fixed expenses increased from prior year for the Kicks model?	Percent change from prior year
		4. Do some locations have higher increases in fixed expenses for the Kicks model than others?	

Notice how the questions all relate to the analysis' objective. The initial questions are fine but broad. The sub-questions in the second column narrow the focus. Moreover, each question is measurable, as indicated by the measures in the last column. Once the detailed descriptive questions are outlined, it is time to determine the necessary data and analysis method to answer them.

Descriptive Analyses Examples

Analyses commonly used to answer descriptive questions include measures of frequency, location, dispersion, and percentages:

- Frequency measures help us understand categories of data.
- Measures of location (mean, median, mode) reveal average observations in a data set.
- Measures of dispersion (minimum, maximum, range, variance, and standard deviation) show how much variance there is among the observations in the data set.
- Measures of percentage change compare results to prior periods and the percent of total.

Data Let's demonstrate an example using the questions identified in Illustration 3.8.

It is possible to discover if gross sales have decreased for the Celeritas model by using the measure gross sales. Several different analyses can determine if gross sales have decreased:

- Gross sales dollars: total measure
- Sales volume: total measure
- Average unit sales price

A PivotTable in Microsoft Excel created with the Super Scooter data prepares a descriptive analysis for this question ([Illustration 3.9](#)).

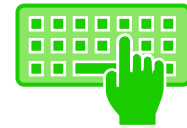
Model	Total Gross Sales	Total Sales Volume	Average of Unit Sale Price
[-] Celeritas			
+ 2023	\$ 1,256,478.00	3,595	\$ 350.25
+ 2024	\$ 2,081,584.00	5,932	\$ 350.57
+ 2025	\$ 1,488,293.00	4,258	\$ 347.92
[-] Kicks			
+ 2023	\$ 628,271.00	3,385	\$ 187.88
+ 2024	\$ 279,848.00	1,816	\$ 148.37
+ 2025	\$ 221,134.00	1,695	\$ 136.21

ILLUSTRATION 3.9 Descriptive Analysis of Celeritas and Kicks Sales

The sales data confirms the Celeritas and Kicks models each declined from 2024 to 2025. Interestingly, both sales volume and average price decreased, so the decline is a combination of both volume and price. The next step is answering the next sub-question: Did sales decrease in all locations?

[Illustration 3.10](#) shows the change in average gross sales dollars, which is a measure of location. It is the result of a descriptive analysis in Tableau that shows the change in gross sales from 2023 to 2025 by location for the Celeritas model. This is a highlight table where the darker color signals a larger decrease. ([Data](#) See How To 3.1 at the end of this chapter to learn how to create this table in Tableau.)

How To



**Change in Gross Sales from
2024 – Celeritas**

	2025
Seattle	\$ (140,206)
Dallas	\$ (106,783)
Charlotte	\$ (94,794)
Boston	\$ (88,837)
Chicago	\$ (58,619)
Miami	\$ (57,786)
Salt Lake City	\$ (46,867)
Phoenix	\$ (20,787)

Difference in Gross Sales

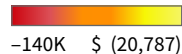


ILLUSTRATION 3.10 Descriptive Analysis of Celeritas' Change in Gross Sales by Location

The Seattle location had the largest total gross sales decrease from 2024 at \$140,206. Dallas had the second largest decrease at \$106,783. However, it is also clear that sales have dropped in all locations. The next step is to determine why sales have decreased, which will require diagnostic analytics.

Apply It 3.2

Describe Customers' Buying Behavior



Data **Financial Accounting** Best Bakes Bakery would like to better understand their top customers' buying behavior. You have been given sales transactions for the years 2022 to 2025. An excerpt from the file follows.

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930

Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Costof GoodsSold	Profit
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60
853300	Poppyseed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80

- What is the objective of the analysis?
- Develop three questions relevant to the objective, and describe the measures necessary to answer the questions.
- What analyses will you use to answer these questions?

SOLUTION

- The objective of the analysis is to identify the top customers and evaluate what products they purchase.
- | Questions | Measures |
|---|--|
| 1. Who are the top five customers? | Gross sales, sales volume, profit margin |
| 2. What are the top five products sold? | Sales volume, gross sales |
| 3. What are the spending patterns for the top five customers? | Sales volume, gross sales |
- Analyses for the three questions:
 - Descriptive analysis: Top five customers by year for each of the measures compared to average for all customers.
 - Descriptive analysis: Five highest selling products by year for each of the measures compared to average for all products.
 - Descriptive analysis: Analysis showing sales by month or quarter for each year for the top five customers. A bar chart or a line chart could be used.

3.3 What are Diagnostic Objectives?

LEARNING OBJECTIVE 3

Demonstrate how to develop diagnostic questions.

Once we know what has happened, the next step is to determine *why*. Diagnostic questions build on descriptive analyses and further explore the data to find the outcome's cause. Diagnostic analytics do this by looking for anomalies, correlations, patterns, or trends.

Develop Diagnostic Questions

Diagnostic questions identify a problem or issue to understand why an outcome occurred.

Illustration 3.11 shows diagnostic questions based on the results of the descriptive analysis of Super Scooters' sales in the previous section.

ILLUSTRATION 3.11 Diagnostic Questions About Super Scooters' Sales Decline

Objective	Initial Question	Sub-questions	Possible Measures
Determine why sales are decreasing for the Celeritas and Kicks models.	What is driving the sales volume decrease?	1. Are there anomalies in sales volume for the Celeritas model? 2. Are there anomalies in sales volume for the Kicks model? 3. Are there identifiable patterns in sales volume for the Celeritas model? 4. Are there identifiable patterns in sales volume for the Kicks model?	Sales volume
Understand the large decrease in sales at the Seattle location.	What factors are driving the decrease in sales at the Seattle locations?	1. Are there unusual patterns in the Seattle location sales of Celeritas models? 2. Are there unusual patterns in the Seattle location sales of the Kicks model?	Gross sales dollars Average gross sales dollars Sales volume

Why are sales decreasing, and specifically, why did the Seattle location have the largest decrease? The questions and sub-questions dive deeper into the sales decrease by asking about anomalies and unusual patterns. Now, identify the data and analyses that will answer these questions.

Diagnostic Analyses Examples

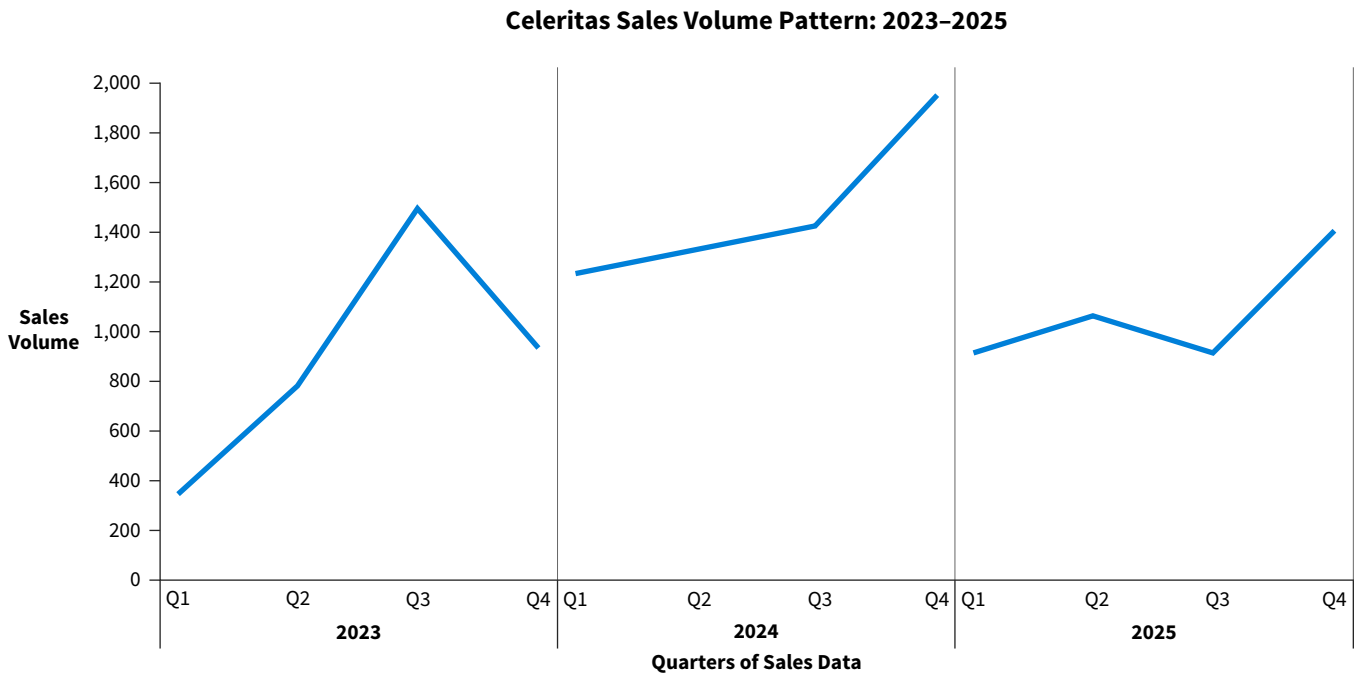
There are four common types of diagnostic analyses: anomaly detection, correlation, pattern detection, and trend analysis.

Using the Super Scooters questions in Illustration 3.11, let’s examine these sub-questions:

- 1. Are there identifiable patterns in sales volume for the Celeritas model?
- 2. Are there unusual patterns in the Seattle location’s Celeritas model sales?

Line charts can identify patterns by revealing repeating patterns in the data. **Illustration 3.12** shows Celeritas sales by month for each year.

ILLUSTRATION 3.12 Celeritas Sales Volume Patterns



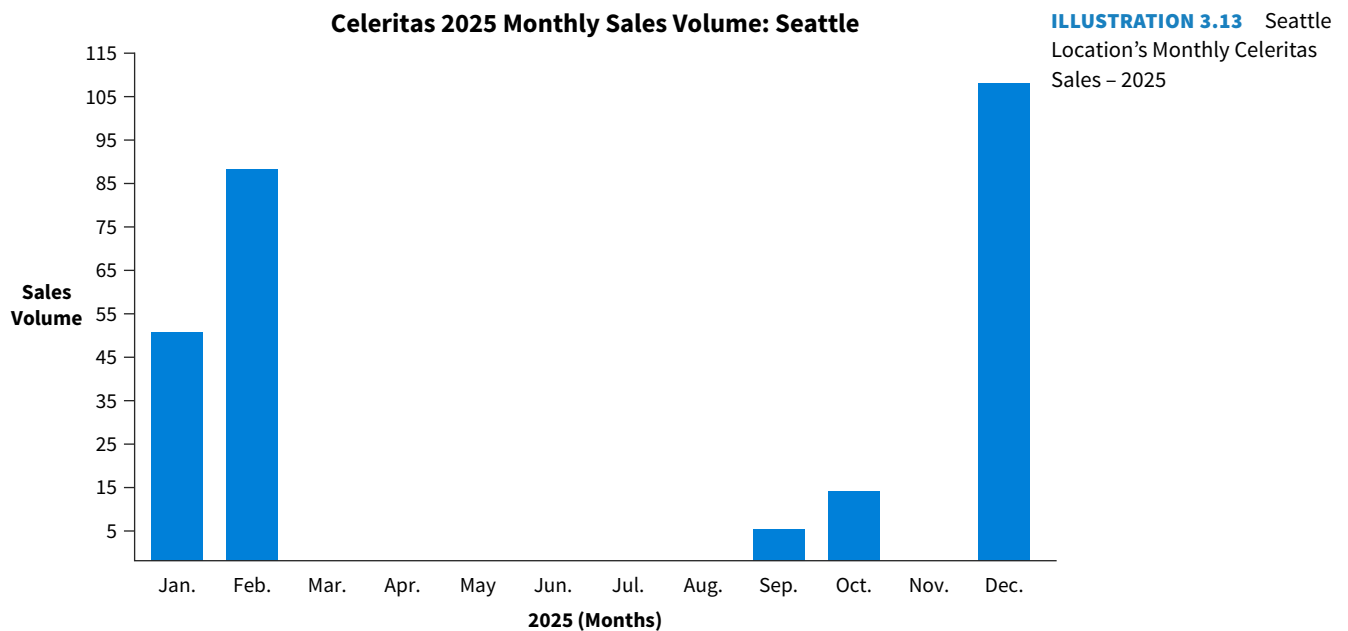
The line charts in Illustration 3.12 do not appear to show a pattern in sales volume for the Celeritas model. However, the line chart shows all locations, and previous analyses revealed the Seattle location had a drop in Celeritas sales.

The sales pattern in years 2023 and 2025 indicate that sales generally increase from the first quarter to the third quarter:

- In 2023 there is a drop in sales in the fourth quarter, but 2024 and 2025 saw an increase in fourth quarter sales.
- However, 2025 sales are below 2024 sales, and there was a large drop in the third quarter.

To understand why sales are lower in 2025 and why there was a drop in the the third quarter of 2025, we must further analyze sales volume by location.

Illustration 3.13 shows an analysis examining monthly sales of the Celeritas at the Seattle location.



The column chart of Celeritas sales over 2025 shows one thing and suggests another:

- In 2025 there are several months with no sales.
- This could be a contributing factor to the decline in overall Celeritas sales in 2025.

The next step would be asking management what happened in March 2025 that led to six months without any sales of the Celeritas model. This also raises additional questions to investigate:

- Did sales drop at the Seattle location only?
- Was the Celeritas model the only product with declining sales?

This investigative process is called exploratory data analysis, which you learn about later in this course.

Data Auditing During the Best Bakes Bakery financial statement audit, you have been asked to determine if there are unusual changes in sales from prior years that might affect risks of material misstatement. You have been given an excerpt of the transactions in an Excel file so that you know the data available.

Apply It 3.3

Determine the Risk of Material Misstatement of Sales

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930



Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Costof GoodsSold	Profit	Profit Margin
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60	0.463768
853300	Poppyseed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40	0.587629
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00	0.529412
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50	0.359375
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60	0.646341
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50	0.160193
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80	0.333333

1. What is the objective of the analysis?
2. Develop three questions relevant to the objective, and state the measures necessary to answer them.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective is to analyze the transactions from 2022 to 2025 for unusual changes. The analysis will provide the auditor with information to determine the nature, timing, and extent of audit procedures.

2. Questions	Measures
1. Are there any unusual changes in total revenue over the years 2022–2025?	Quarterly gross sales
2. Are there any unusual changes in total revenue over the years by location?	Quarterly gross sales
3. Are there any unusual changes in revenue by product over the years?	Sales volume by product

3. Anomaly, pattern, and trend analyses can be prepared to answer the questions.

3.4 What are Predictive Objectives?

LEARNING OBJECTIVE

4

Demonstrate how to develop predictive questions.

So far, you have learned how to develop descriptive questions to find out what happened in the past and diagnostic questions to understand why. What if you want to know what may happen in the future? In this case, you would ask **predictive questions**.

Develop Predictive Questions

When making loan decisions, banks use historical information about whether a potential borrower pays bills consistently and on time to predict if they will continue to do so. In the same way, predictive analytics use past and present data to create models so businesses can make predictions.

The use of predictive analytics is not new in the accounting profession, but as the availability of data and software tools to perform predictive analytics has increased, so has performing these analytics in all areas of accounting:

- Financial accountants can identify trends in sales or expenses.
- Cost accountants can predict costs, create forecasts, and evaluate cost drivers.
- Auditors identify potential material misstatements using predictive analytics.
- Tax accountants may use predictive analytics for tax planning.

Suppose Super Scooters is preparing their budget for the next year:

- They want to predict revenue for 2026 assuming a 10% increase in sales volume.
- They also believe there will be a 10% increase in warranty costs.
- Finally, they are considering discontinuing the Celeritas model and want to know if that will change predicted revenue.

What is the overall objective of the data analysis? What questions should Super Scooters ask to meet that objective?

When determining predictive questions, it is helpful to ask, “What do I want to do with the answer?” **Illustration 3.14** shows the overall objective of predicting revenue for 2026, the initial question, plus specific sub-questions.

Objective	Initial Question	Sub-questions
Predict revenue for 2026.	How much will revenue change with a 10% increase in sales volume?	How much will revenue change for each model? How much will revenue change at each location?
	How will revenue be affected with a 10% increase in warranty costs?	What factors influence warranty expense?
	How will revenue be affected by discontinuing the Celeritas and/or the Kicks model?	How much will revenue at each location be affected by discontinuing the Celeritas and/or the Kicks model?

ILLUSTRATION 3.14 Predictive Questions to Forecast Super Scooters' Revenue

Like all questions, predictive questions should be related to the objective, specific, measurable, and the appropriate data and analytics should be used to answer them.

Predictive Analyses Examples

Two common analyses that answer predictive questions are trendlines and regression analysis.²

Trendlines

Trendlines show the underlying functional relationship of data:

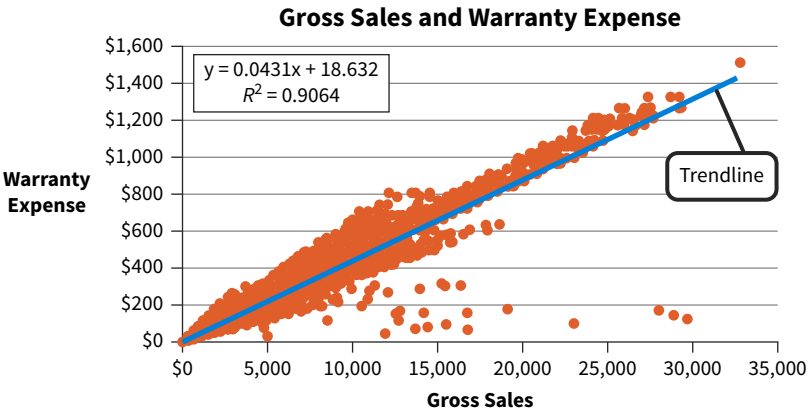
- A **functional relationship** is the effect of an independent variable on a dependent variable.
- A **linear function** shows steady increases or decreases over the range of the independent variable.

The Excel Trendline tool can help determine if the data follow a linear function. Remember that Super Scooters believes warranty costs will rise by 10%. Further, warranty costs are believed to be driven by sales.

²There are many other predictive analytical methods, such as artificial intelligence, which are beyond the scope of this chapter.

Illustration 3.15 shows a trendline created in Excel to determine if there is a linear relationship between sales and warranty expense for Super Scooters.

ILLUSTRATION 3.15 Trendline
Using Sales and Warranty Expense



The trendline supports a linear relationship:

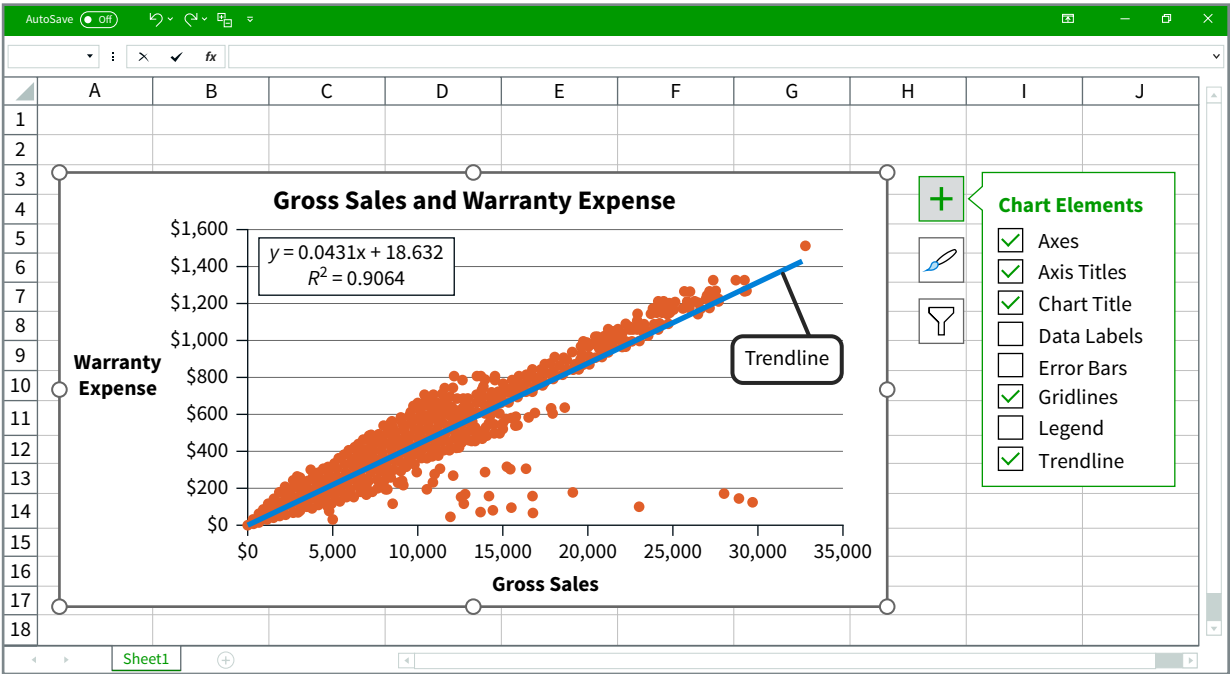
- As gross sale amounts increase, warranty expense also increases.
- The equation for the trendline is shown in the chart. ($y = 0.0431x + 18.632$, $R^2 = 0.9064$).

That equation can be used to predict future warranty expense based on expected sales where y is the amount of warranty expense and x is the amount of the gross sale. For example, if a gross sale was \$2,000, then the expected warranty expense would be:

$$(0.0431 \times 2,000) + 18.632 = \$104.83$$

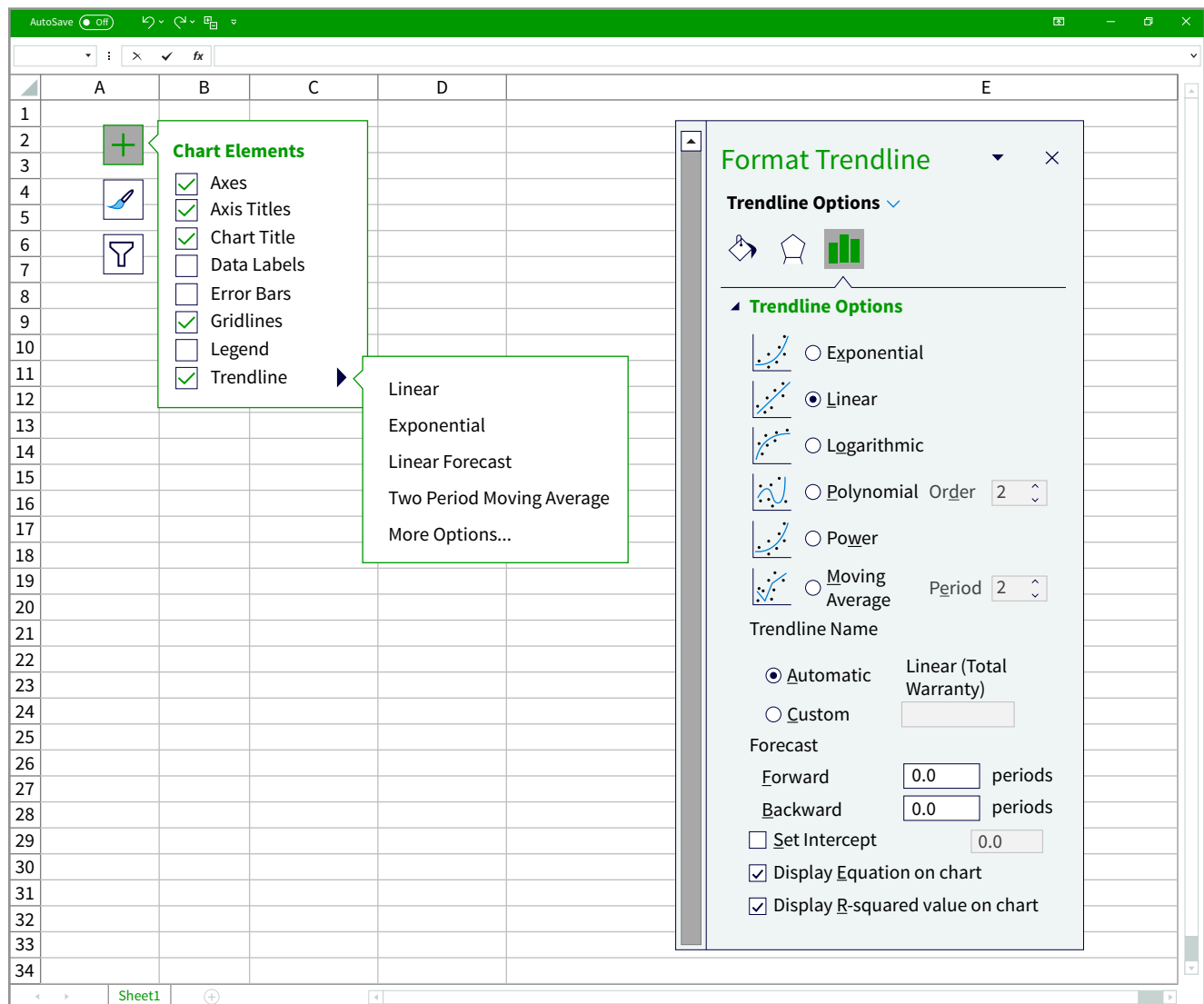
If warranty costs increase by 10% as Super Scooters expects, the prediction model can be adjusted by increasing 0.0431 by 10%. Find the trendline tool **Chart Elements** in Excel by clicking on the graph and then the plus sign (**Illustration 3.16**).

ILLUSTRATION 3.16 Trendline Choice in Chart Elements



Note that clicking on the plus sign at the top left of the chart includes the option to select **Trendline**, and then **More Options**. (Illustration 3.17).

ILLUSTRATION 3.17 Trendline Options in Microsoft Excel



The **Format Trendline** box that opens lets the user choose other functions (exponential, logarithmic, polynomial, power, and moving average). This can be useful if the pattern of the data does not appear linear. Also, click on the boxes to display the equation and the *R*-square on the chart. The *R*-square is a measure of how well the line fits the data. The closer to 1, the better the fit. The trendline in Illustration 3.16 has a very strong *R*-square of 0.9064. We will discuss *R*-square in more detail in the next section.

Linear Regression

While you may not prepare a predictive model during your career, you will likely use predictive models or need to interpret their output. **Linear regression** is a tool for building mathematical and statistical models to explain the relationship between a dependent variable and one or more independent variables.

Predictive analytics build models to predict or better understand a phenomenon. To find out what factors influence warranty expense, we would build a model that predicts warranty expense.

Building a model requires identifying the variables that will be included in it:

- A **variable** is a data field used for analysis.
- A **dependent variable** is the outcome measure (warranty expense).
- **Independent variables** are the variables that influence the dependent variable (specific variables that we believe influence warranty expense, such as sales or number of warranty claims).

Simple linear regression involves a single independent variable, while multiple regression involves two or more independent variables. The goal of a regression model is finding the equation of the line that best fits the data.

Let's examine the output of a regression model to understand how they help answer questions. As an accountant at Super Scooters, you are trying to estimate next year's maintenance expense for the machinery equipment. You believe the cost drivers for maintenance expense are the number of hours the equipment is used and the number of repair requests. The regression model variables would be:

- **Dependent variable:** Maintenance expense
- **Independent variables:** Machine hours and repair requests

Historical data from the previous 36 months will be used to create a regression model. The output from the model is shown in **Illustration 3.18**.

ILLUSTRATION 3.18 Regression Model Output

SUMMARY OUTPUT					
<i>Regression Statistics</i>					
Multiple R	0.97868787				
R Square	0.95782995				
Adjusted R Square	0.95527419				
Standard Error	1283.82134				
Observations	36				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	5252.85845	897.8427419	5.850533	1.5E-06	
Machine Hours	3.56920298	0.43769194	8.154601	2.05E-09	
Repair Requests	759.836675	168.7763028	4.502034	7.94E-05	

The regression in Illustration 3.18 was performed using Microsoft Excel. The summary output is split into three sections.

Regression Statistics First are the **regression statistics**, or the statistical measures used to evaluate the model. **Illustration 3.19** shows the regression statistics from Illustration 3.18, along with the definition of the statistic and an interpretation for the Super Scooters regression model.

ILLUSTRATION 3.19 Regression Statistics

Regression Statistics	Definition	Output	Interpretation
Multiple R	- Also called the correlation coefficient, it measures the strength of the relationship between the dependent and independent variables. - It is a measure from -1 to 1. A positive number equals a positive correlation, so the variables move in the same direction. A negative number equals a negative correlation, where the variables move in opposite directions.	0.97868787	- Multiple R is 0.978. This is a positive correlation between total expenses and the independent variables in the model (machine hours, repair requests). - As the variables in the model increase, total maintenance expenses will also increase.
R Square (R^2)	- Also called the coefficient of determination, it is a measure of how well the regression line fits the data. R^2 gives the proportion of the variation in the dependent variable that is explained by the independent variables. - The closer the R^2 is to 1, the better the regression line fits to the data.	0.95782995	This shows that 95.8% (0.9578) of the change in total maintenance expense is explained by how many hours the equipment is used (machine hours) and how many repair requests are made.
Adjusted R^2	- Explains how well the regression line fits the data. - Modifies the value of R^2 by incorporating the sample size and the number of independent variables. - Generally used to evaluate a multiple regression model. - The closer the adjusted R^2 is to 1, the better the fit of the regression line to the data.	0.95527419	Adjusted R^2 is interpreted in the same way as R^2 . This shows that 95.5% of total expenses can be explained by how many hours the equipment is used and the number of repair requests made.
Standard Error	- Represents the variability of the observed dependent variable values from the values predicted by the model. - If the data are clustered close to the regression line, then the standard error is small (optimal). - If the data are more scattered, then it is larger.	1,283.82134	- The standard error is \$1,283.82. - To determine if standard error is large or small, compare it to the standard deviation of the dependent variable. - In this example, compare the standard error to the standard deviation of total expenses. If the standard error is smaller than the standard deviation, the standard error is acceptable. - The mean and standard deviation for total expenses is: - Mean: \$27,450 - Standard deviation: \$6,070
Observations	The number of observations included in the data set.	36	In the Super Scooters example, it is 36 months of data.

Analysis of Variance (ANOVA) The next section of the regression output is the ANOVA (analysis of variance). **Illustration 3.20** is the ANOVA section from the regression model.

ILLUSTRATION 3.20 ANOVA Regression Statistics

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			

In a multiple linear regression such as this, significance is a hypothesis test of whether the regression model is better than a model with no independent variables. In other words, is the model better than no model at all?

Generally, a model is considered significant if the F statistic (*Significance F* in Illustration 3.20) is less than 0.05:

- If F is significant, then the model can explain some of the variation of the dependent variable. In other words, it is better than no model at all.
- The ANOVA in Illustration 3.20 has a significance F of 2.05385E-23. The notation “E-23” after 2.05385 represents scientific notation, also known as exponential notation. 2.05385E-23 is the same as 0.000000000000000000000000205385. It is clearly a number well below 0.05, so the model is significant.

In other words, the independent variables can explain some of the variation of total expenses, so the model is better than no model at all.

Regression Equation The last section of the regression summary output provides the information to create the equation to predict the dependent variable:

- The intercept and coefficients of the model represent the equation of the line that best fits the data.
- The key statistic to analyze in this section is the p -value for each of the independent variables. Like the F statistic, the p -value provides a test of significance. In the case of p -value, it is a test as to whether the independent variable improves the ability of the model to better predict the dependent variable. A p -value of 0.05 or less is generally considered significant.

Let’s use the output in [Illustration 3.21](#) to identify the prediction model for machine maintenance costs and interpret the coefficients.

ILLUSTRATION 3.21 Regression Model

	Coefficients	Standard Error	t Stat	P -value
Intercept	5252.85845	897.8427419	5.850533	1.5E-06
Machine Hours	3.56920298	0.43769194	8.154601	2.05E-09
Repair Requests	759.836675	168.7763028	4.502034	7.94E-05

Notice that the p -values for the independent variables meet the test of being less than 0.05 and are therefore significant. The prediction model will be equal to the intercept plus (or minus if the number is negative) the coefficients of the independent variables times the predicted values for those variables. Based on the regression model in Illustration 3.21, the equation to predict total purchasing department costs is:

$$\text{\$5,252.86} + \text{\$3.57 (number of Machine Hours)} + \text{\$759.84 (number of repair requests)}$$

The calculation of predicted total expenses if there are 2,250 machine hours in one month and 8 repair requests is shown in [Illustration 3.22](#). Start with the intercept and add the product of each independent variable coefficient and variable predicted value for a prediction of \$19,364.08 of total maintenance costs for the year.

ILLUSTRATION 3.22 Prediction Model Example

	Model Coefficient	Variable Values	Prediction
Intercept	\$ 5,252.86	1	\$ 5,252.86
Machine Hours	\$ 3.57	2250	\$ 8,032.50
Repair Requests	\$ 759.84	8	\$ 6,078.72
			\$ 19,364.08

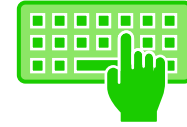
The model could be interpreted as follows:

- **Intercept:** The intercept does not always have a practical interpretation. It is a result of the model that represents the mean for the response when all the independent variables are zero. It is where the equation function crosses the y-axis. However, here the intercept represents the amount of fixed costs that exist regardless of machine hours and repair requests
- **Machine hours:** Each hour machine equipment is used adds \$3.57 to total costs.
- **Repair requests:** Each request adds \$759.84 to total costs.

Using a model like the one in Illustration 3.22 helps businesses predict future outcomes.

(Data) See How To 3.2 to learn how to perform this regression in Microsoft Excel.)

How To



Data Managerial Accounting As a managerial accountant for Best Bakes Bakery, you are preparing an analysis of sales trends to help create the 2026 operating budget. You have been given sales transactions for the years 2022–2025. Following is an excerpt from the file.

Apply It 3.4 Plan a Sales Trend Analysis

SalesOrder Number	Customer Number	Customer Name	Customer Address	Customer City	Customer ZipCode	Customer State	Customer Phone
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
101	521121	Bluebird Cafe	524 W Laurel St	Fort Collins	80521	CO	(720) 296-2323
102	521425	Walrus Ice Cream	125 W Mountain Ave	Fort Collins	80521	CO	(303) 674-0930



Inventory Code	Inventory Description	SalesOrder Date	SalesOrder Quantity	Inventory Price	Gross Sales	Inventory Cost	Cost of Goods Sold	Profit	Profit Margin
853200	Caramel Apple	2/19/2022	36	\$ 3.45	\$ 124.20	\$ 1.85	\$ 66.60	\$ 57.60	0.463768
853300	Poppysed Bagel	2/19/2022	24	\$ 4.85	\$ 116.40	\$ 2.00	\$ 48.00	\$ 68.40	0.587629
853500	Cheesecake Bite	2/19/2022	25	\$ 3.40	\$ 85.00	\$ 1.60	\$ 40.00	\$ 45.00	0.529412
853600	Chocolate Chip Cookie	2/19/2022	30	\$ 3.20	\$ 96.00	\$ 2.05	\$ 61.50	\$ 34.50	0.359375
853800	Blueberry Scone	2/19/2022	24	\$ 4.10	\$ 98.40	\$ 1.45	\$ 34.80	\$ 63.60	0.646341
853900	Raspberry Scone	2/19/2022	10	\$ 4.05	\$ 40.50	\$ 3.40	\$ 34.00	\$ 6.50	0.160193
853100	Cinnamon Bun	3/10/2022	24	\$ 4.35	\$ 104.40	\$ 2.90	\$ 69.60	\$ 34.80	0.333333

1. What is the objective of the analysis?
2. Develop three questions relevant to the objective, and state the measures necessary to answer them.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective of the analysis is to predict sales for the next year's operating budget.

Questions	Measures
How are sales trending from 2022 to 2025?	Sales volume, average sales price
How are sales trending by product from 2022 to 2025?	Sales volume, average sales price
How are sales trending by location from 2022 to 2025?	Sales volume, average sales price

3. Trendline analysis will provide an estimate of sales trends that can then be applied to the budget for 2026.

3.5 What are Prescriptive Objectives?

LEARNING OBJECTIVE
5

Demonstrate how to develop prescriptive questions.

You have learned about questions that help describe what happened, investigate why, and forecast what will happen next. Next, let’s examine the questions we ask when we want to know what *should* happen.

Develop Prescriptive Questions

Prescriptive objectives build on descriptions of the present and predictions about the future to determine the best action. **Prescriptive questions** investigate how to take advantage of future opportunities or mitigate a future risk outcome. The analyses specify the actions necessary to achieve the desired outcomes.

In the Super Scooters example, we answered the previous predictive question regarding warranty expenses for 2026. Now, we can address the third analysis requested by Super Scooters—deciding how many units of each model should be produced to meet the 2026 revenue goal. The first step is to clearly articulate the questions.

Illustration 3.23 starts with the objective of determining the most profitable product mix:

- The initial question is more specific. How many scooters should be produced and sold to maximize contribution margin?
- Next, dig deeper by asking about potential constraints that should be considered and how many units of each model should be produced.

ILLUSTRATION 3.23
 Prescriptive Question to Determine Super Scooters’ Product Mix

Objective	Initial Question	Sub-questions
Identify the most profitable product mix.	How many scooters should we produce and sell to maximize contribution margin?	Are there constraints that should be included in the optimization model? How many units of each model scooter should be produced to maximize contribution margin and satisfy any constraints?

Prescriptive analyses are performed with specific analysis methods and data.

Prescriptive Analyses Examples

The two most common methods of analysis used to answer prescriptive questions are optimization models and what-if analyses.

Linear Optimization

Optimization is the process of selecting values of variables that minimize or maximize some quantity of interest. Optimization modeling helps managers allocate resources more

efficiently and make cost/profit decisions. The most common optimization model used in accounting is **linear optimization**. In linear optimization, the model is comprised of:

- **Decision variables:** The unknown values the model seeks to determine.
- **Objective function:** The mathematical equation that describes the output target to minimize or maximize.
- **Constraints:** The limitations, requirements, or other restrictions that must be imposed on any solution, such as demand, material, or labor constraints.

The output from a linear optimization model will show the optimal solution.

Super Scooters' management has decided to continue producing both the Celeritas and the Kicks models for at least one more year. They want to know how many of each model should be produced to maximize contribution margin. They have forecasted demand for each model:

- Captain: 18,000 units.
- Celeritas: 10,000 units.
- Kicks: 7,000 units.
- Lazer: 24,000 units.

Because they want to avoid excess inventory, they do not want to produce more than they expect to sell. There is also a limit to the number of machine hours available for the year.

Illustration 3.24 shows information needed to create an optimization model for Super Scooters.

ILLUSTRATION 3.24 Contribution Margin and Resource Requirements

	Captain	Celeritas	Kicks	Lazer	Total
Number Produced	1	1	1	1	
CM per Unit	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	
Total CM	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	\$ 840.10
Machine Hours Per Unit	20	25	10	18	
Total Machine Hours	20	25	10	18	73
Constraints:					
Machine Hours					1,000,000
Demand	18,000	10,000	7,000	24,000	

The number of units produced are the decision variables.

The objective function is the formula for total contribution margin. Contribution margin is sales minus variable expenses.

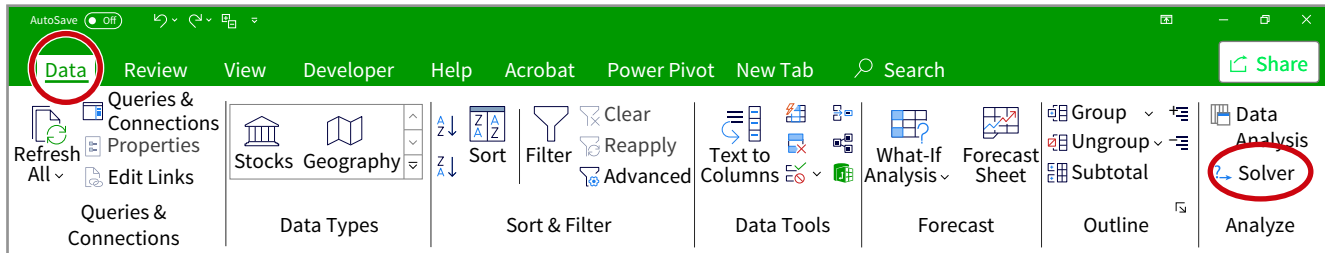
The model also shows how many machine hours are required to produce each unit.

The model shows the constraints of predicted demand for each scooter and the total machine hours available for the year.

The linear optimization program will use this data to solve for the optimal number of each model that should be produced where contribution margin (objective function) is maximized subject to the constraints. Note that we started with an arbitrary number of one in the number of units produced cells. This could also be zero to start; however, using the number one makes it possible to confirm the formulas. The linear optimization available in Microsoft

Excel Solver can illustrate how optimization models work. The Solver program is accessed via the **Data** tab in the toolbar ribbon ([Illustration 3.25](#)).

ILLUSTRATION 3.25 Accessing Microsoft Excel Solver



Clicking on **Solver** opens a dialog box to input the objective function cell, decision variable cell, and create any relevant constraints. [Illustration 3.26](#) is the Solver dialog box used to create the optimization program for Super Scooters.

ILLUSTRATION 3.26 Solver Dialog Box Inputs for Super Scooters Contribution Margin Optimization

The image shows the 'Solver Parameters' dialog box with the following inputs and annotations:

- Set Objective:** \$M\$6 (Annotated with 'Total CM')
- To:** ☒ Max, ☐ Min, ☐ Value Of: 0
- By Changing Variable Cells:** \$I\$4:\$L\$4 (Annotated with 'Cell range for Number Produced')
- Subject to the Constraints:**
 - \$I\$4 <= 18000
 - \$J\$4 <= 10000
 - \$K\$4 <= 7000
 - \$L\$4 <= 24000
 - \$M\$9 <= 1000000
(Annotated with 'Constraints for demand and machine hours')
- ☒ **Make Unconstrained Variables Non-Negative**
- Select a Solving Method:** Simplex LP
- Solving Method:** Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.
- Buttons:** Add, Change, Delete, Reset All, Load/Save, Help, Solve, Close.

Once all the constraints are input, check the box to ensure the Solver results are not negative (we cannot “unproduce” a product) and select the **Solving Method Simplex LP**, since this is a linear optimization. Clicking **Solve** produces the box in [Illustration 3.27](#).

ILLUSTRATION 3.27 Solver Results Choice Box

Solver Results

Solver found a solution. All Constraints and optimality conditions are satisfied.

☒ Keep Solver Solution
☐ Restore Original Values

☐ Return to Solver Parameters Dialog
 ☐ Outline Reports

Reports

☐ Answer
☐ Sensitivity
☐ Limits

OK Cancel Save Scenario...

Solver found a solution. All Constraints and optimality conditions are satisfied.

When the GRG engine is used, Solver has found at least a local optimal solution. When Simplex LP is used, this means Solver has found a global optimal solution.

This shows that Solver found an optimal solution that satisfied the constraints. The default is to **Keep Solver Solution**. If this radio button is checked, the spreadsheet will reflect the new decision variable amounts and the optimal contribution margin amount ([Illustration 3.28](#)). There is also the choice to create three reports. Choose the **Answer** report and click **OK**.

ILLUSTRATION 3.28 Optimization Model after Running Solver

	Captain	Celeritas	Kicks	Lazer	Total
Number Produced	18,000	5,520	7,000	24,000	
CM per Unit	\$ 371.78	\$ 201.87	\$ 104.27	\$ 162.18	
Total CM	\$ 6,692,106.26	\$ 1,114,328.39	\$ 729,872.21	\$ 3,892,219.65	\$ 12,428,526.51
Machine Hours Per Unit	20	25	10	18	
Total Machine Hours	360,000	138,000	70,000	432,000	1,000,000
Constraints:					
Machine Hours					1,000,000
Demand	18,000	10,000	7,000	24,000	

The number of units produced are the decision variables.

The objective function is the formula for total contribution margin. Contribution margin is sales minus variable expenses.

The model also shows how many machine hours are required to produce each unit.

The model shows the constraints of predicted demand for each scooter and the total machine hours available for the year.

Illustration 3.29 is the answer report. The first section of the report shows the objective function's original value and then the final value when the optimal solution is reached. In this case, the optimal production will be 18,000 Captain, 5,520 Celeritas, 7,000 Kicks, and 24,000 Lazer scooters.

ILLUSTRATION 3.29 Solver
Results Output

Objective Cell (Max)			
Cell	Name	Original Value	Final Value
\$M\$6	Total CM Total	\$ 840.10	\$ 12,428,526.51

Variable Cells				
Cell	Name	Original Value	Final Value	Integer
\$I\$4	Number produced Captain	1	18,000	Contin
\$J\$4	Number produced Celeritas	1	5,520	Contin
\$K\$4	Number produced Kicks	1	7,000	Contin
\$L\$4	Number produced Lazer	1	24,000	Contin

Constraints					
Cell	Name	Cell Value	Formula	Status	Slack
\$M\$9	Total Machine Hours Total	1,000,000	\$M\$9<=\$M\$12	Binding	0
\$I\$4	Number produced Captain	18,000	\$I\$4<=\$I\$13	Binding	0
\$J\$4	Number produced Celeritas	5,520	\$J\$4<=\$J\$13	Not Binding	4,480
\$K\$4	Number produced Kicks	7,000	\$K\$4<=\$K\$13	Binding	0
\$L\$4	Number produced Lazer	24,000	\$L\$4<=\$L\$13	Binding	0

The middle section of the report (Variable Cells) shows the final value for the decision variables (**Illustration 3.30**). It shows the number of each model scooter that Super Scooters should sell to achieve maximum contribution margin.

ILLUSTRATION 3.30 Solver
Results Output – Variable Cells
Report

Variable Cells				
Cell	Name	Original Value	Final Value	Integer
\$I\$4	Number produced Captain	1	18,000	Contin
\$J\$4	Number produced Celeritas	1	5,520	Contin
\$K\$4	Number produced Kicks	1	7,000	Contin
\$L\$4	Number produced Lazer	1	24,000	Contin

Finally, the last section of the report shows how much of the constraints were used in the optimal solution (**Illustration 3.31**). The Status column indicates whether the constraint is binding or not binding. In other words, additional production is not possible without an increase in the constraint. The amount shown in the Slack column represents the amount of the constraint that is remaining after the optimal solution.

ILLUSTRATION 3.31 Solver
Results Output – Constraints
Report

Constraints				
Name	Cell Value	Formula	Status	Slack
Total Machine Hours Total	1,000,000	\$M\$9<=\$M\$12	Binding	0
Number produced Captain	18,000	\$I\$4<=\$I\$13	Binding	0
Number produced Celeritas	5,520	\$J\$4<=\$J\$13	Not Binding	4,480
Number produced Kicks	7,000	\$K\$4<=\$K\$13	Binding	0
Number produced Lazer	24,000	\$L\$4<=\$L\$13	Binding	0

Since machine hours are limited, the optimal model prescribes the best use of those hours to maximize contribution margin is to produce all Super Scooters can sell of the Captain, Kicks, and Lazer models and 4,480 less than demand of the Celeritas model. Any other mix will result in less contribution margin than the optimal model.

What-if Analyses

A spreadsheet model that evaluates changes and specific combinations of model inputs and assumptions is called a **what-if analysis**. A what-if analysis is an easy way to change values in a spreadsheet and recalculate the outputs. Microsoft Excel has three tools built into the Data tab under What-if Analyses. Two of these tools – Scenario Manager and Goal Seek – are

useful tools to facilitate what-if analyses. We will discuss each tool in a later chapter, but here is a short explanation:

- Scenario Manager in Excel allows changing or substituting input values for multiple cells (maximum 32). Therefore, the results of different input values or scenarios can be viewed at the same time.
- Goal Seek is used when the desired result is already known but the input value to achieve that result is not. Goal Seek is limited because it can only use one input variable. If the analysis being performed requires more than one variable to change, then an optimization model using Excel Solver is necessary. For example, the Super Scooters optimization model had more than one variable because it was necessary to consider the demand and machine hours constraints.

Data Managerial Accounting You are a managerial accountant for Best Bakes Bakery asked to prepare an analysis to determine the optimal mix of products to maximize profit. You have been given sales transactions for the years 2022–2025. Besides prior sales data, you know there are some resource constraints (such as supplies, or labor hours) that should be included in the analysis.

1. What is the objective of the analysis?
2. Develop three questions that will be relevant to the objective.
3. What analyses will you use to answer these three questions?

SOLUTION

1. The objective is to determine the optimal sales mix of products given the resources available.
2. Three questions:
 - What resource constraints should be included in the decision?
 - What are the resource requirements for each product?
 - What is the expected profit per product?
3. Linear optimization can be used to determine the best mix of products for maximum profit.

Apply It 3.5 Prescribe Optimal Sales Mix



3.6 What are Data Analytics Motivations and Objectives in Professional Practice?

LEARNING OBJECTIVE 6

Describe motivations and objectives for data analytics in professional practice.

While the methods of analysis – descriptive, diagnostic, predictive, and prescriptive – are the same across accounting areas, project objectives and what motivates them can be different given the variety of purposes and stakeholders. In auditing, stakeholders are largely external (e.g. shareholders, regulatory agencies), whereas stakeholders in managerial accounting are primarily internal (e.g. management, employees). Their perspectives help identify the objective and develop good questions.

Accounting Information Systems

Motivations for data analyses in AIS can range widely in scope. **Illustration 3.32** provides a summary of typical AIS data analysis motivations, objectives, and questions.

ILLUSTRATION 3.32 Motivations, Objectives, and Questions for AIS Data Analytics

Motivation	Objective	Question
Investing in new technologies.	Determine the technology with the highest return on investment.	What are the labor cost savings if we adopt a new technology?
Increasing security of AIS assets, including data.	Determine how many security breaches or attempted breaches have occurred.	What is the total number of attempted security breaches per day?
Innovations in operational processes, either human or computer, to improve processes.	Identify inefficient processes.	Which processes take the longest?
Improving system performance, such as processing time, reliability, and availability.	Evaluate processing time for monthly financial statement reporting.	How many hours does it take to process financial statements each month?
Improvements in AIS system maintenance.	Evaluate how long it takes to address system maintenance issues.	How long does it take to resolve maintenance requests?
Improvements in data integrity.	Determine how often data must be corrected.	How many corrections were made to data in the last month?

Although the list of questions in Illustration 3.32 is not exhaustive, the questions are a starting point for conducting analyses to address them.

APPLYING CRITICAL THINKING 3.1: Understand AIS Stakeholders

Understanding the stakeholders’ motivations helps accurately identify the objectives of analyses. Stakeholders in AIS data analysis projects can be internal or external to the organization (**Stakeholders**):

- Internal stakeholders include the chief financial officer, internal auditors, managers, chief information officer, and employees. They want to understand and improve processes, so objectives focus on specific aspects of internal processes and how to improve them.
- External stakeholders may include investors, external auditors, customers, and vendors. These stakeholders are motivated by increased return on investment, so the projects with the greatest potential to offer a return on any investment are usually prioritized.

Auditing

Auditors perform data analyses to verify financial statement information is not materially misstated. Their purpose is providing a professional opinion about whether the financial statements provide reliable, accurate information for the client’s primary stakeholders, which include company owners and investors and creditors in the capital markets and financial institutions.

Illustration 3.33 includes examples of motivations, objectives, and questions in auditing data analytics.

ILLUSTRATION 3.33 Motivations, Objectives, and Questions for Auditing Data Analytics

Motivation	Objective	Question
Evaluate the risk of material misstatement of revenue.	Determine if there are any unusual changes in revenue compared to prior years.	Are there any unusual trends or changes in sales from prior years?
Evaluate the risk of material misstatement in the company's general ledger accounts.	Analyze balances in the company's general ledger to identify unusual changes from prior year.	Are any changes in general ledger balances higher than our specified material amount?
Evaluate the risk of fraudulent payments.	Verify employees are not receiving unauthorized payments.	Are there any vendors in the vendor file with the same address as an employee?
Verify physical inventory and fixed asset counts.	Test for existence of assets.	Do inventory counts match amounts on the balance sheet?

Note that the motivation identifies a specific aspect of the audit. The objective states the goal of the analysis. The third column in the illustration provides an example of a specific question the auditor can address with the analysis. Audit analytics are used to perform risk assessment, substantive analytical procedures, and to perform tests of detail.

APPLYING CRITICAL THINKING 3.2: Acquire Auditing Knowledge and Avoid Biases

Auditors acquire and apply certain types of information when performing audit data analytics (**Knowledge**):

- Client's industry, governance, policies, and procedures.
- Relevant accounting and auditing standards, and SEC regulations.
- Risk assessment and statistical and sampling techniques.

Auditors can be biased when they make assumptions about what they expect to find during an audit engagement. They must be vigilantly skeptical of all information they are given (**Risks**).

Financial Accounting

There are a few typical motivations for financial accountants to perform data analyses:

- Ensuring economic transactions, value changes, and period closing entries have been properly captured by the accounting system, valued in the appropriate account, and in the correct accounting period.
- Predicting future net income and cash flows for top management.
- Identifying, evaluating, and securing alternative capital sources.

Illustration 3.34 shows examples of objectives for these motivations and possible questions to address them.

ILLUSTRATION 3.34 Motivations, Objectives, and Questions for Financial Accounting Data Analytics

Motivation	Objective	Question
Ensure the financial statements reflect all transactions.	Verify transactions are not missing.	Does the sales transaction file agree to the financial statements?
Predict cashflows.	Determine expected cash inflow from sales.	Based on prior year sales and expected growth, what are predicted sales for the next year?
Evaluate alternative capital sources.	Decide whether to issue stock or bonds to raise capital.	What is the expected value of issuing more stock based on current market prices?

Managerial Accounting

Managerial accountants perform data analyses to improve management decision-making and operational performance.

APPLYING CRITICAL THINKING 3.3: Consider Alternative Methods

If the objective of an analysis is to predict future income, then managerial accountants may use regression, trendlines, or a what-if analysis. Always choose the method that can answer the questions most effectively and efficiently (**Alternatives**).

Illustration 3.35 shows some typical motivations, objectives, and questions for the projects managerial accountants work on.

ILLUSTRATION 3.35 Motivations, Objectives, and Questions for Managerial Accounting Data Analytics

Motivation	Objective	Question
Identify and evaluate new market, product, service, and business process opportunities.	Determine whether to expand an existing product line.	What are the projected sales for the new product line?
Efficiently allocate resources for successful operations.	Evaluate the best use of materials for production.	Which products use the most materials?
Maximize revenue generation and cash flows.	Determine the most profitable product mix.	How many units of each product should we produce to optimize contribution margin?
Increase organizational efficiency by managing costs.	Determine potential cost cutting.	Which production line has the highest amount of waste?

Tax Accounting

Tax accountants perform data analytics to enhance the quality of their professional advice and decisions, as well as to meet compliance requirements for their organization or clients. Typical motivations for tax analytics include:

- Performing tax research.
- Designing tax plans.
- Correctly calculating tax liability and completing applicable tax returns.

Illustration 3.36 summarizes examples of the objectives related to these motivations and some specific questions.

ILLUSTRATION 3.36 Motivations, Objectives, and Questions for Tax Accounting Data Analytics

Motivation	Objective	Question
A new tax law has been passed that potentially changes a client's tax liability.	Identify relevant tax law changes.	Based on last year's tax return, how will the change in tax law affect the client's tax liability?
A client requested tax planning for the next three years.	Prepare a tax plan to minimize tax liability.	Based on the client's past tax returns and current business projections, how much tax will they owe next year?
Analyzing a client's tax liability after preparing their tax return.	Identify unusual deductions.	How do the client's deductions compare to their prior year return and to national averages?

Although general motivations in tax analytics may be similar, objectives and questions will be specific to individual clients. For example, tax planning is a common motivation, but a client may have differing objectives for their tax plan, such as saving for retirement or estate planning. The objective of the client will determine the type of analysis.

As this chapter has emphasized, accounting professionals are increasingly performing data analysis to provide value to their organizations and clients. Wherever your career takes you, thinking critically about what is motivating a project will help you ask the best objective-based data questions to achieve your goals. These are the first two steps of successful data analysis projects across professional practice areas.

Match at least one professional practice area's acronym to the following motivations to perform data analyses.

- | | |
|-----------------------------------|--------------------------|
| a. Accounting information systems | d. Managerial Accounting |
| b. Auditing | e. Tax Accounting |
| c. Financial Accounting | |

Apply It 3.6
Match Motivations
to Professional
Practice Areas

Motivation	Practice Area
Performing tax research.	
Investing in new technologies.	
Performing analytical procedures on variances in account balances and on classes of transactions.	
Ensuring all economic transactions have been recorded.	
Investing in business intelligence tools.	
Increasing security of AIS assets, including data.	
Innovations in operational processes, either human or computer, to increase process effectiveness.	
Designing defensible tax plans.	
Learning if all adjusting entries at the end of the period have been recorded.	
Improving system performance, such as processing time and availability.	
Reviewing and evaluating the internal control system.	
Increasing security of AIS assets, including data.	
Computing tax liability and completing tax forms.	
Performing physical inventory and fixed asset counts at the fiscal year end.	
Testing documentation to determine if account balances are supported.	

SOLUTION

Motivation	Practice Area
Performing tax research.	Tax accounting
Investing in new technologies.	AIS
Performing analytical procedures on variances in account balances and on classes of transactions.	Auditing
Ensuring all economic transactions have been recorded.	Financial accounting
Investing in business intelligence tools.	AIS
Increasing security of AIS assets, including data.	AIS
Innovations in operational processes, either human or computer, to increase process effectiveness.	Managerial accounting
Designing defensible tax plans.	Tax accounting
Learning if all adjusting entries at the end of the period have been recorded.	Financial accounting
Improving system performance, such as processing time and availability.	AIS
Reviewing and evaluating the internal control system.	Auditing
Increasing security of AIS assets, including data.	AIS
Computing tax liability and completing tax forms.	Tax accounting
Performing physical inventory and fixed asset counts at the fiscal year end.	Financial accounting
Testing documentation to determine if account balances are supported.	Auditing

Chapter Review and Practice

Learning Objectives Review

1 Summarize the relationship between motivations, objectives, and data analysis questions.

Accountants generally perform data analyses for four reasons:

- Opportunities within their organizations and in the marketplace.
- Professional issues and requirements from regulations and laws.
- Solving problems within their organizations or for clients.
- Process and performance evaluation of their own work, or work performed within their organizations.

First, identify the objective, which is the goal of the data analysis. Then, develop questions designed to achieve the project's goal. Data analysis questions can be evaluated for clarity, conciseness, and measurability:

- Will the answer address the objective of the analysis?
- Does the question address a single topic? If not, break it down into sub-questions.
- Is the question measurable?
- Are the data needed to answer it available?

If the answer to any of these questions is no, then either revise the question or drop it.

The six elements of critical thinking can help us think about what is motivating data analyses and how to develop objectives and questions:

- Understanding the stakeholders and their perspectives helps identify the project's motivation and develop specific objectives and questions.
- Critically thinking through the purpose of data analyses supports moving from objective to specific questions.
- Considering several alternative methods of analyses increases the likelihood of selecting the best option for the purpose.
- Being aware of potential risks makes it possible to mitigate them when evaluating motivation, determining objectives, and developing questions.
- Considering what knowledge is necessary to successfully complete the project involves either confirming we already have that knowledge or doing additional research.
- Understanding why we are performing data analyses lets us leverage the experience to future data analytics contexts and tasks. Considering past experiences can speed up the process of identifying objectives and developing questions.

2 Demonstrate how to develop descriptive questions.

Descriptive questions focus on understanding what is currently happening or what has happened in the past. They are developed by identifying the objective of the analysis and then formulating

sub-questions that narrow the focus. The next step is identifying the data and the method of analysis to answer those questions. Common descriptive methods:

- Frequency measures help understand categories of data.
- Measures of location (mean, median, mode) reveal average observations in a data set.
- Measures of dispersion (minimum, maximum, range, variance, and standard deviation) show how much variance there is among the observations in the data set.
- Measures of percentage change show percentage increases and decreases as compared to prior periods and the percent of total.

3 Demonstrate how to develop diagnostic questions.

Diagnostic questions build on what was learned in descriptive analysis and explore the data to find the cause of the outcome. Diagnostic analytics determine why an outcome occurred by looking for anomalies, correlations, patterns, or trends. Common diagnostic analyses:

- Anomaly detection: Scatterplots, bar charts.
- Correlation (linear): Scatterplots, correlation.
- Pattern detection: Line graphs, bar charts.
- Trend analysis: Bar charts, line charts, trendlines.

4 Demonstrate how to develop predictive questions.

Predictive analytics use past and present data to forecast and create models so businesses can make predictions about the future. The increased availability of data and software tools to perform predictive analytics means it is now used in all areas of accounting. To determine a predictive question, ask, "What do I want to do with the answer?" Common analyses used to answer predictive questions:

- Trendlines: Show the underlying functional relationships of the data.
- Regression analysis: Linear regression builds mathematical and statistical models to explain the relationship between a dependent variable and one or more independent variables.

5 Demonstrate how to develop prescriptive questions.

Prescriptive objectives build on predictions about the future and descriptions of the present to determine the best course of action. The objectives for prescriptive analytics focus on what should happen. There are two common methods of prescriptive analysis:

- Optimization is the process of selecting values of variables that *minimize* or *maximize* some quantity of interest. Optimization modeling helps managers allocate resources more efficiently and make cost/profit decisions.

- A spreadsheet model that evaluates how changes and specific combinations of model inputs and assumptions is called a what-if analysis. A what-if analysis is an easy way to change values in a spreadsheet and recalculate the outputs.

6 Describe motivations and objectives for data analytics in professional practice.

The primary professional accounting practice areas have both common and unique motivations to perform data analyses:

- Accounting information systems professionals have governance, strategic, operational, and compliance motivations.
- Auditors have professional compliance, quality, firm, and market motivations.
- Financial accountants have both internal and external stakeholder motivations.
- Managerial accountants have internal governance, strategic, and operational motivations.
- Tax accountants have professional compliance, client, firm, and market motivations.

Key Terms Review

Coefficient of determination 3-21
Constraints 3-25
Correlation coefficient 3-21
Decision variables 3-25
Dependent variable 3-20
Descriptive questions 3-8
Diagnostic questions 3-13

Functional relationship 3-17
Independent variables 3-20
Linear function 3-17
Linear optimization 3-25
Linear regression 3-19
Motivation 3-2
Objective 3-3

Objective function 3-25
Optimization 3-24
Predictive questions 3-16
Prescriptive questions 3-24
Regression statistics 3-20
Variable 3-20
What-if analysis 3-28

How To Walk-Throughs



HOW TO 3.1 Create a Highlight Table in Tableau

The table shown in Illustration 3.10 was created in Tableau version 2021.1.2. Create it yourself by following these steps.

What You Need: **Data** The How To 3.1 data file.

STEP 1: Open a new worksheet and drag the **Sold Date** to **Columns** and **Location** to **Rows**. Drag **Gross Sales** to the open area under years, or to the Text symbol under **Marks** ([Illustration 3.37](#)).

ILLUSTRATION 3.37 Tableau Worksheet

Pages

Filters

Marks

T Automatic

Color

Size

Text

Detail

Tooltip

T

SUM(Gross Sale)

Columns

Year(Sold Date)

Rows

Model

Sheet 22

		Sold Date		
Location	2023	2024	2025	
Boston	686,611	1,211,197	1,433,578	
Charlotte	618,987	1,254,221	1,653,506	
Chicago	711,852	1,239,689	1,624,719	
Dallas	679,008	1,417,584	1,113,397	
Miami	582,623	1,410,659	1,606,576	
Phoenix	640,681	1,252,712	1,376,503	
Salt Lake City	544,630	1,205,880	1,593,184	
Seattle	460,424	1,514,392	977,384	

STEP 2: Create a calculation that will calculate the change in sales from 2024 to 2025. Because data from 2023 is unnecessary and we are only interested in the Celeritas model, create two filters. First, drag **Model** to the **Filters** box. This opens an input box ([Illustration 3.38](#)).

Filter [Model]

General Wildcard Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

☐ Captain
☐ Celeritas
☐ Kicks
☐ Lazer

All None ☐ Exclude

Summary

Field: [Model]
 Selection: Selected 0 of 4 values
 Wildcard: All
 Condition: None
 Limit: None

Reset OK Cancel Apply

ILLUSTRATION 3.38 Filter Input Box

Check the box for Celeritas and click **OK**. Next, drag **Sold Date** to **Filter** and select **Years** from the input box ([Illustration 3.39](#)).

ILLUSTRATION 3.39 Filter Field Choices

Filter Field [Sold Date]

How do you want to filter on [Sold Date]?

☒ Relative Date
☒ **Range of Dates**
☐ Individual Dates

Years
 # Quarters
 # Months
 # Days
 # Week numbers
 # Weekdays
 # Month / Year
 # Month / Day / Year

Count
 # Count (Distinct)
☐ Minimum
☐ Maximum
☐ Attribute

Next > Cancel

Click **Next** to see the following input box choices ([Illustration 3.40](#)).

ILLUSTRATION 3.40 Filter Choices

Filter [Year of Sold Date]

General

Condition

Top

☒ Select from list

☐ Custom value list

☐ Use all

Enter search text

☐ 2023

☒ 2024

☒ 2025

All

None

☐ Exclude

Summary

Field: [Year of Sold Date]

Selection: Selected 2 of 3 values

Wildcard: All

Condition: None

Limit: None

☐ Filter to latest date value when workbook is opened

Reset

OK

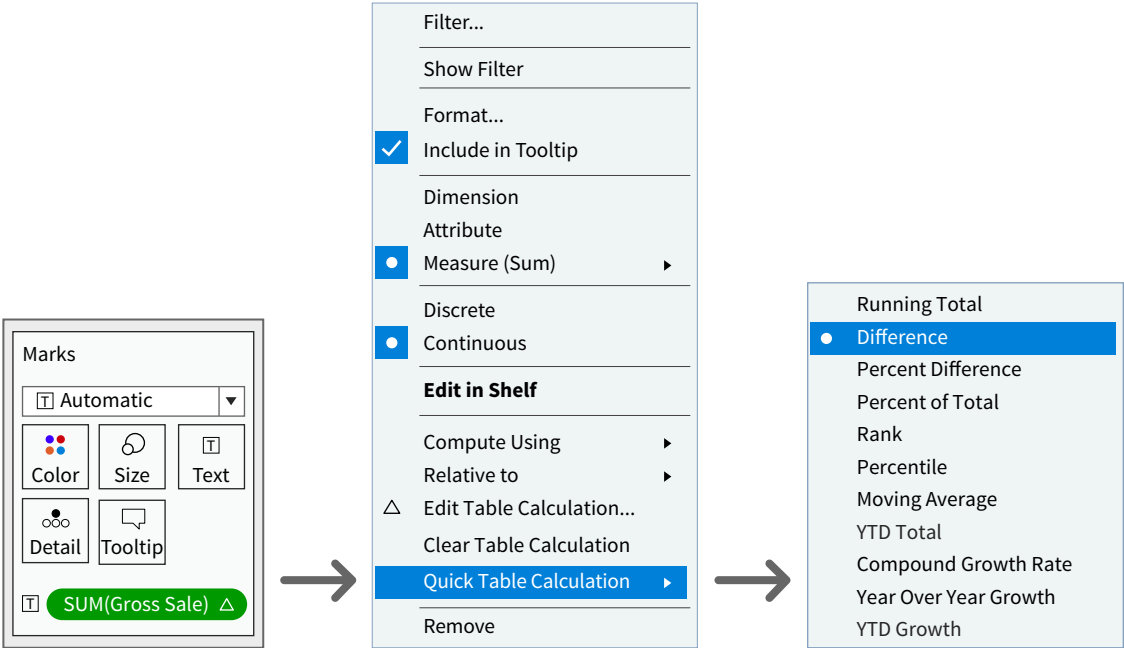
Cancel

Apply

Select 2024 and 2025 and click **OK**.

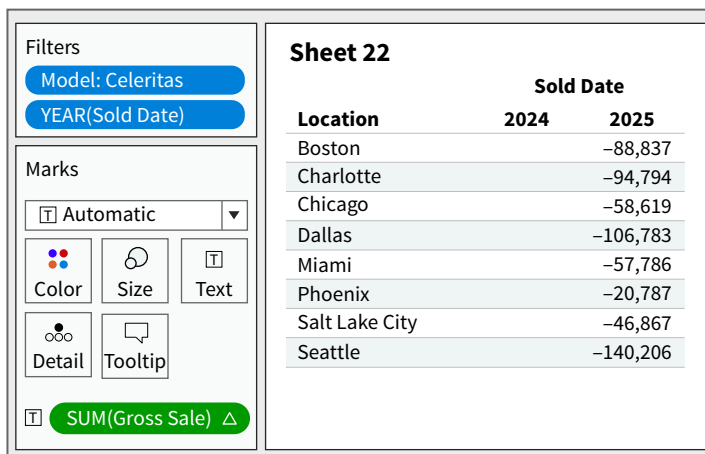
STEP 3: Create the calculation. Clicking the down arrow in the green pill for **Sum(Gross Sales)** in the **Marks** area generates several options ([Illustration 3.41](#)).

ILLUSTRATION 3.41 Creating a Quick Table Calculation



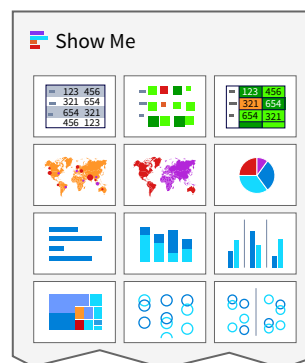
Select the **Quick Table Calculation** option, then select **Difference**. This will change the columns in the table to differences ([Illustration 3.42](#)).

ILLUSTRATION 3.42 Quick Table Calculation Results



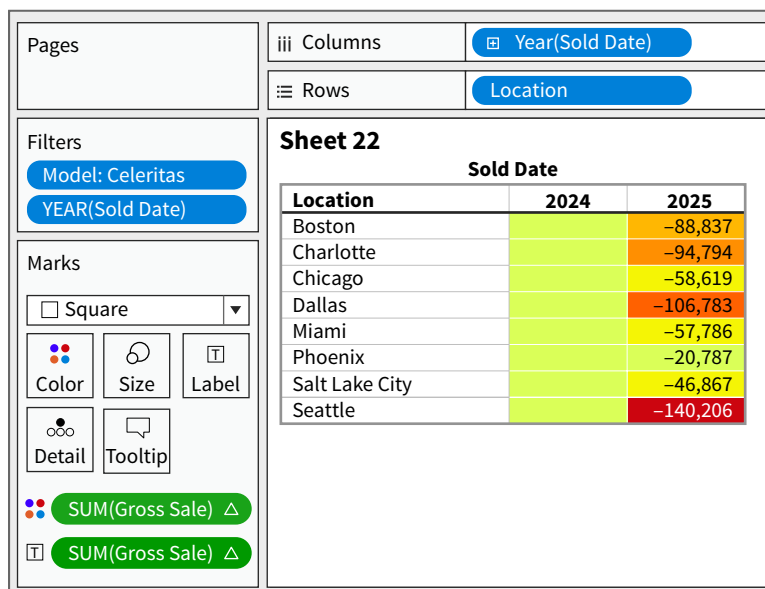
STEP 4: To create the highlight table in which darker colors represent larger numbers, choose that chart type from the **Show Me** options in the top right corner ([Illustration 3.43](#)).

ILLUSTRATION 3.43 Choosing a Highlight Table



Now the table looks like the table in [Illustration 3.44](#).

ILLUSTRATION 3.44 Highlight Table Results



STEP 5: To only show the column for 2025, right click on the 2024 in the column and choose **Hide** ([Illustration 3.45](#)).

ILLUSTRATION 3.45 Hide Column Feature

Sheet 22

Location	Sold Date	
	2024	2025
Boston		
Charlotte		
Chicago		
Dallas		
Miami		
Phoenix		
Salt Lake City		
Seattle		

✓ Keep only

✕ Exclude

Hide

+ Drill down

Format...

Rotate Label

✓ Show Header

Show Missing Values

Click on **Hide** and *not* **Exclude**. (Selecting **Exclude** means that the calculation will no longer work because the 2024 data would be removed.)

STEP 6: Order the locations by the largest loss to the smallest loss. To do that, click 2025 at the top of the column. The box will appear as shown in [Illustration 3.46](#).

ILLUSTRATION 3.46 Sorting Data from Smallest to Largest

Sold Date

Location	2025
Boston	-88,337
Charlotte	-94,794
Chicago	-58,619
Dallas	-106,783
Miami	-57,786
Phoenix	-20,787
Salt Lake City	-46,867
Seattle	-140,206

✓ Keep Only ✕ Exclude

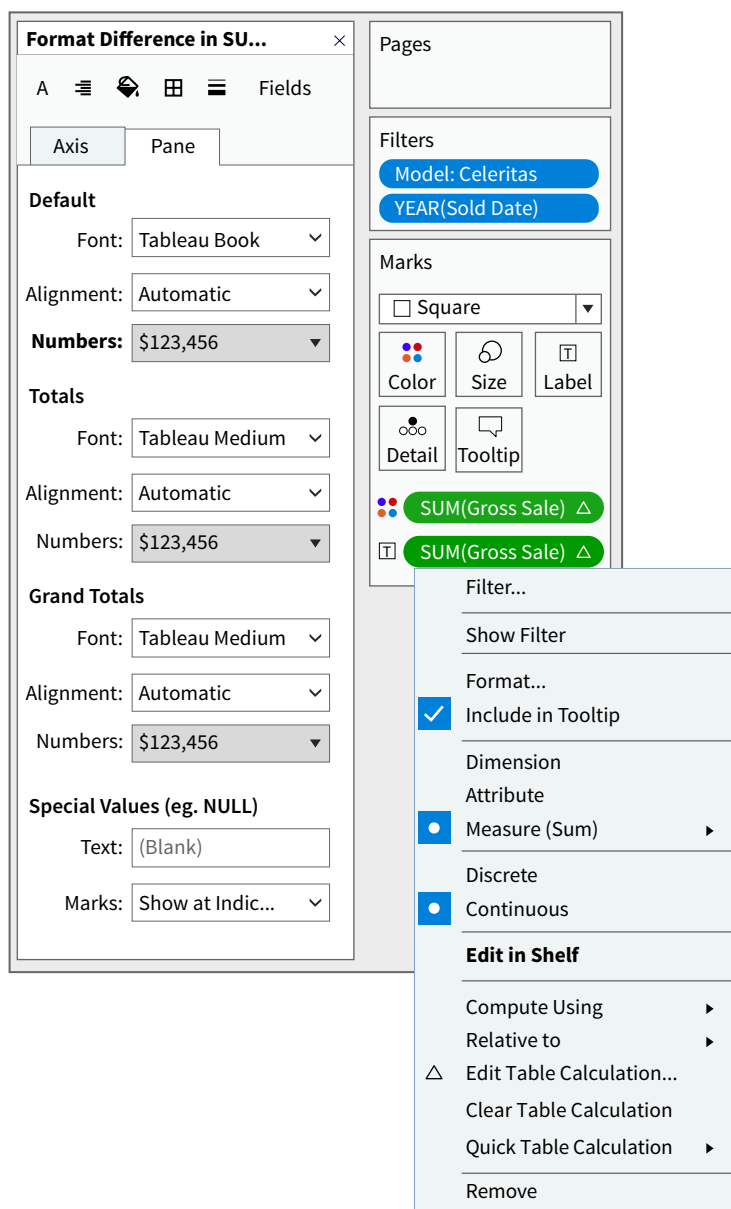
8 items selected Difference in SUM(Gross Sales): -614,679

Click the sort icon that shows the smallest bar on the top. Data will then sort from the smallest number to the largest ([Illustration 3.47](#)).

ILLUSTRATION 3.47 Sorted Highlight Table

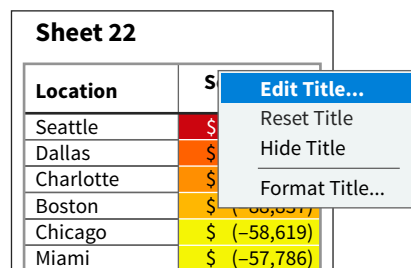
Sold Date	
Location	2025
Seattle	-140,206
Dallas	-106,783
Charlotte	-94,794
Boston	-88,837
Chicago	-58,619
Miami	-57,786
Salt Lake City	-46,867
Phoenix	-20,787

STEP 7: Format the difference column to whole dollars by clicking the green **Sum(Gross Sale)** pill in the **Marks** area and choosing **Format** ([Illustration 3.48](#)).

ILLUSTRATION 3.48 Format Options

Here, the default format can be changed to currency.

STEP 8: Create a title for the visualization. Tableau will name the visualization by the sheet number by default. To change that, right click in the sheet title area and then choose **Edit Title** (Illustration 3.49).

ILLUSTRATION 3.49 Editing a Worksheet Title

Enter a title in the box that appears ([Illustration 3.50](#)).

ILLUSTRATION 3.50 Editing a Worksheet Title

Sheet 22

Location	Sold Date 2025
Seattle	\$ (140,206)
Dallas	\$ (106,783)
Charlotte	\$ (94,794)
Boston	\$ (88,837)
Chicago	\$ (58,619)
Miami	\$ (57,786)
Salt Lake City	\$ (46,867)
Phoenix	\$ (20,787)

Edit Title

Tableau Light 15 B I U Insert

Change in Gross Sales from 2024-Celeritas Model

Reset OK Cancel Apply



HOW TO 3.2 Perform a Regression in Microsoft Excel

Illustration 3.18 is regression performed in Excel. You can create it by following these steps.

What You Need: **Data** The How To 3.2 data file.

STEP 1: The Data Analysis tool in the **Data** ribbon has an option for regression. [Illustration 3.51](#) shows the dialog box that opens when **Data Analysis** is selected.

ILLUSTRATION 3.51 Excel Data Analysis Tool - Regression

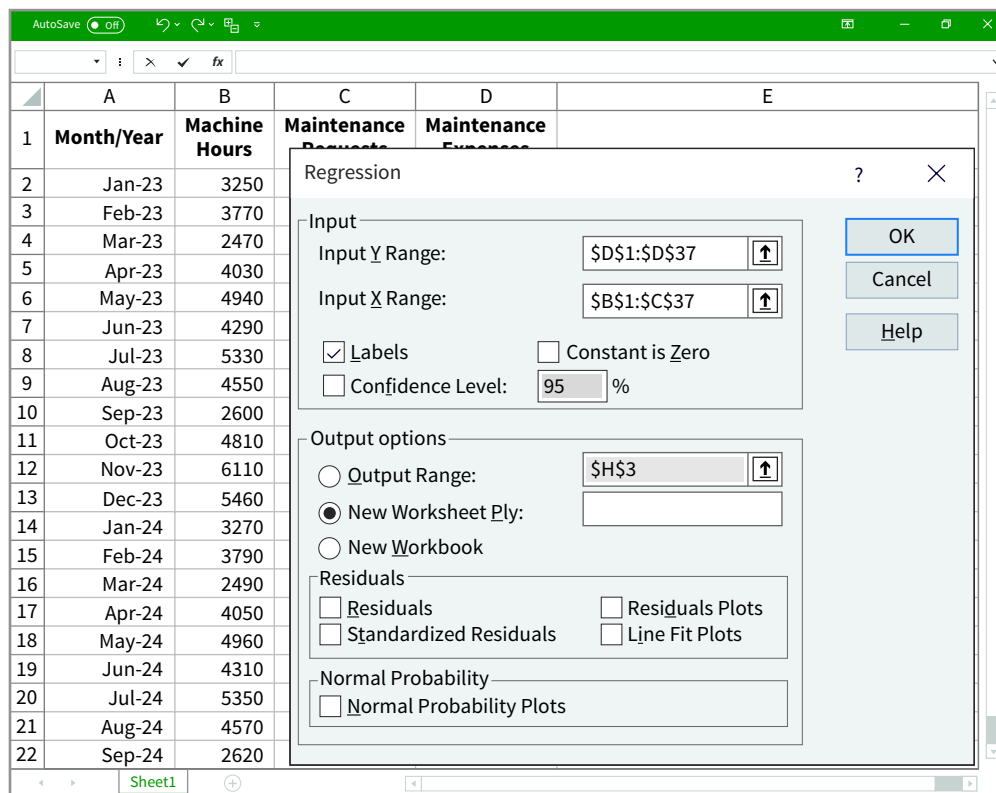
Data Analysis

Analysis Tools

F-Test Two-Sample for Variances
Fourier Analysis
Histogram
Moving Average
Random Number Generation
Rank and Percentile
Regression
Sampling
t-Test: Paired Two Sample for Means
t-Test: Two-Sample Assuming Equal Variances

OK
Cancel
Help

STEP 2: Click on **Regression**, then **OK**. A Regression dialog box will appear ([Illustration 3.52](#)).

ILLUSTRATION 3.52 Regression Dialog Box

STEP 3: The Input Y Range is for the dependent variable. The dependent variable is what will be predicted. In this example, it is column D, Maintenance Expenses (\$D\$1:\$D\$37). The Input X Range is for the independent variables. In this example, there are two independent variables—Machine Hours and Maintenance Requests (\$B\$1:\$C\$37).

Select **Labels** if the first row of the data is the column title (row 1). In this example, the output will be in a new worksheet. Finally, **Residuals**, **Standardized Residuals**, **Residual Plots**, and **Normal Probability Plots** are selected. These residuals options will be used to evaluate the regression assumptions.

STEP 4: Click **OK** for the regression results ([Illustration 3.53](#)).

ILLUSTRATION 3.53 Regression Output

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.978687872				
R Square	0.95782995				
Adjusted R Square	0.955274189				
Standard Error	1283.821342				
Observations	36				
ANOVA					
	df	SS	MS	F	Significance F
Regression	2	1235399491	6.18E+08	374.773	2.05385E-23
Residual	33	54390508.88	1648197		
Total	35	1289790000			
	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	5252.858452	897.8427419	5.850533	1.5E-06	3426.183659
Machine Hours	3.569202976	0.43769194	8.154601	2.05E-09	2.678712028
Maintenance Request	759.8366754	168.7763028	4.502034	7.94E-05	416.4587054

Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Motivation in data analytics in accounting is
 - a. the reason why the analysis is being performed.
 - b. the interpretation of results of analysis.
 - c. based only on factors outside analysis.
 - d. based only on whether your career will benefit from performing the analysis.
2. **(LO 1)** One source of motivation for data analyses in accounting relates to opportunity. Which of the following is an example of the motivation source of opportunity?
 - a. Desiring future success in your career path.
 - b. Considering whether to expand production of a newly launched product.
 - c. Ensuring compliance with the newest FASB pronouncement.
 - d. Considering why the company's sales volume decreased over a period of time.
3. **(LO 1)** Accountants' motivations to perform data analyses can originate from
 - a. opportunities in the marketplace or organization.
 - b. changes in laws, regulations, and professional rules.
 - c. indications that a problem needs to be solved.
 - d. needing to perform evaluation on processes or performance.
 - e. Any of these may trigger the motivation to perform data analyses.
4. **(LO 1)** Critically thinking through the motivation to perform data analyses includes considering
 - a. stakeholders, purpose, alternatives, risks, knowledge, and self-reflection.
 - b. stakeholders and purpose.
 - c. stakeholders, purpose, risk management, and alternatives.
 - d. stakeholders, acquiring knowledge, alternatives, and self-reflection.
 - e. stakeholders, risk management, alternatives, and self-reflection.
5. **(LO 1)** Analyses of the stakeholders and their perspectives helps to critically think through motivation to perform data analyses because
 - a. it reminds you who pays your salary.
 - b. some may be external to your organization.
 - c. considering the perspectives of stakeholders helps identify the objective and develop good questions.
 - d. internal stakeholders should never be considered.
 - e. All of these are reasons for considering stakeholders in your critical thinking.
6. **(LO 1)** Critically considering the purpose for performing data analyses
 - a. keeps us focused on the reason why something is done or created.
 - b. provides free data for analyses.
 - c. reveals the stakeholders' perspectives.
 - d. helps avoid biases and assumptions.
 - e. None of these list the critical thinking benefits of considering the purpose of data analyses.
7. **(LO 1)** Performing the most appropriate data analyses is more likely if you critically think through issues by
 - a. using bar charts from last year's operational results.
 - b. starting with the results you want to achieve.
 - c. paying others to help you.
 - d. acquiring and applying new knowledge.
 - e. None of these help you perform better data analyses.
8. **(LO 1)** Critically thinking through the risks to your perspective as you determine your motivation to perform data analyses helps you minimize
 - a. errors from biases.
 - b. errors from lack of experience.
 - c. errors from your assumptions.
 - d. faulty conclusions from errors or omissions in your data.
 - e. All of these are benefits of considering the risks to your data analyses choices.
9. **(LO 1)** Considering several alternative methods of analysis helps you
 - a. go with the first alternative that comes to mind.
 - b. select the option that is best for the project's purpose.
 - c. pick the alternative with the highest incremental cost.
 - d. pick the alternative with the lowest likelihood of informative results.
10. **(LO 2)** If the objective of data analysis is to understand what is currently happening or has happened in the past, then what type of question are you asking?

a. Descriptive	c. Predictive
b. Diagnostic	d. Prescriptive
11. **(LO 2)** Once you have articulated a descriptive question, what type of analysis is necessary to understand the dispersion of the data?

a. Median	c. Standard deviation
b. Frequency	d. Percent change
12. **(LO 2)** Assume you plan to perform descriptive analysis of your entity's maintenance expenses. Which of the following analyses would help understand categories of maintenance expenses?
 - a. Create a frequency distribution table based on maintenance expense categories.
 - b. Calculate the mean and median maintenance expense for the period.
 - c. Build a regression model using labor hours as a predictor of maintenance expenses.
 - d. Calculate the percentage change in maintenance expenses since prior year.
 - e. None of these are considered descriptive analysis used to understand categories of maintenance expenses.

13. (LO 3) What type of question is “What is driving the increase in labor expenses?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

14. (LO 3) The sub-question “Are there any unusual sales transactions in 2023?” is best associated with which of the following objectives of data analysis?

- a. What is the average dollar amount of sales in 2023?
- b. What is the best predictor of increased sales for future periods?
- c. Why are sales decreasing during 2023?
- d. What are the minimum and maximum dollar amount of sales during 2023?

15. (LO 4) What type of question is “How much will revenue be next year if we increase sales by 10%?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

16. (LO 4) Which of the following is a question asked in predictive analysis?

- a. Are there any unusual changes in total revenue from 2021 to 2024?
- b. What is the average sales amount for online customers each year?
- c. How will revenue be affected if we have a 5% decrease in variable costs?
- d. How much revenue is categorized as business-to-business sales?

17. (LO 5) What type of question is “How many units of product should we buy each month to minimize inventory holding costs but still meet demand?”

- a. Descriptive
- b. Diagnostic
- c. Predictive
- d. Prescriptive

18. (LO 5) In an optimization model, what is being determined?

- a. Constraints
- b. Decision variables
- c. Objective function
- d. How much to produce

19. (LO 6) AIS accountants are most likely to be motivated to perform data analyses for which of the following opportunities?

- a. New technologies that can provide cost savings for operational processes.

b. New customers in the marketplace.

c. Turnover in the regulatory body leadership.

d. Changes to the footnote disclosures required by generally accepted accounting rules.

e. Changes in tax rules.

20. (LO 6) Auditors’ motivation to perform data analyses would be high for which of the following situations?

a. Evaluating the effectiveness of internal controls in their client’s accounting system.

b. Evaluating the completeness of an account balance on their client’s balance sheet or income statement.

c. Considering the risk of material misstatement in engagement planning.

d. Aggregating and analyzing key variables in a smart contract for evaluation regarding whether a client recorded their lease liabilities in accordance with GAAP.

e. All these are examples of situations where auditors are motivated to perform data analyses.

21. (LO 6) Financial accountants would be most likely motivated to perform data analyses for which of the following?

a. Changes to the management team.

b. New competitors in the marketplace.

c. Identifying and supporting assumptions and estimates for end of period adjusting entries.

d. Changes to the tax rates applicable to their organization.

e. None of these would motivate data analyses by financial accountants.

22. (LO 6) Managerial accountants would most likely be motivated to perform data analyses for which of the following?

a. Changes to the tax code.

b. Changes to the financial statement footnote disclosure rules.

c. Changes to reporting rules by the Securities and Exchange Commission.

d. Changes to their organization’s strategies, performance, and resource costs.

e. None of these would motivate data analyses by managerial accountants.

Review Questions

1. (LO 1) Discuss the following motivations for data analyses in accounting by providing an example for each:

1. Opportunities
2. Professional issues and requirements
3. Problem solving
4. Process and performance assessment

2. (LO 1, 6) Compare and contrast the professional considerations and requirements that motivate financial and managerial accountants to perform data analyses.

3. (LO 2, 4) Assume you are a financial analyst at a non-public entity. Your supervisor has asked you to calculate and analyze debt solvency ratios over the last five years. You note that there are several stakeholders associated with this analysis, including your manager

and the bank that holds your entity’s debt. Provide an example of how these two stakeholders may be impacted differently by the results of your data analyses.

4. (LO 1, 4) As an auditor assigned to a public company audit in the manufacturing industry, you have been asked to analyze the company’s inventory to understand the types of products in inventory and the mix of raw materials, work-in-process, and finished goods. You receive the inventory on hand file from the information technology department. Identify the knowledge you might need to acquire or apply to this analysis that may contribute to your motivation to perform it.

5. (LO 2) Assume you are performing a descriptive analysis of the employee expense reimbursement data for your company’s current fiscal year compared to the prior fiscal year. Give an example of an initial question related to this analysis.

6. (LO 2) As an auditor working in the internal audit group, you are asked to perform a descriptive analysis of the employee login attempts during business hours and non-business hours. Your objective is to understand whether the quantity of failed login attempts has increased or decreased. Your initial question is “Have login attempts increased?” Provide an example of a sub-question that would address that objective and initial question for descriptive analysis.
7. (LO 3) Explain how accountants can use diagnostic analytics to determine why an outcome occurred.
8. (LO 3) Your company controller asked you to perform diagnostic analysis on the company’s product sales to understand why sales increased compared to the prior year. Your initial question is “What is driving sales volume increase?” Identify two sub-questions to consider as you plan your analysis.
9. (LO 2, 4) Describe the difference between predictive objectives and descriptive objectives in accounting analytics.
10. (LO 4) Define the terms dependent variable and independent variable. How are these variables used in predictive analytics?
11. (LO 5) Define and explain optimization and what-if analyses. Why are these two types of analyses considered prescriptive analytics?
12. (LO 5) Assume you are a financial analyst working for a manufacturing company. Your company manufactures and sells a variety of

component parts for high-end remote-controlled cars. Your product mix includes three popular products:

- Metal Gear
- High-Torque 330
- High-Torque 400

Your controller has asked you to identify the most profitable product mix. Give an example of an initial question and sub-question that may drive your analysis plan.

13. (LO 6) As an accounting information systems accountant, identify the primary stakeholders you prioritize when performing data analyses.
14. (LO 6) What are external auditors’ primary motivations to perform data analyses?
15. (LO 6) Your controller has asked you to prepare adjusting journal entries for the month-end close. Discuss why a financial reporting accountant may engage in data analysis as they prepare adjusting journal entries.
16. (LO 6) Tax accountants often consider alternatives when deciding whether they are motivated to perform data analyses. Discuss some alternatives that could improve their decisions.

Brief Exercises

BE 3.1 (LO 1) Match the following events with the type of motivation for performing data analyses.

- a. Significant cost increases

b. Annual accomplishments
- c. New industry expertise

d. New standard for leases

Motivation	Event
1. Opportunity	
2. Regulation or law	
3. Problem	
4. Process or performance evaluation	

BE 3.2 (LO 1-6) **Accounting Information Systems** Assume you are an accounting information system professional tasked with performing several analyses related to your company’s information security controls. Match the analytics area to the motivation scenario (analytics areas can be used more than once).

- a. Descriptive analysis

b. Diagnostic analysis
- c. Predictive analysis

d. Prescriptive analysis

Motivation to Perform Analyses	Analytics Area
1. Analyze the mean, median, and mode number of failed login attempts after the company changed password requirements to understand if the benefits of enhanced password requirements outweigh the costs.	
2. Analyze the correlation between the quantity of phishing attempts reported and employee attendance at cybersecurity training to determine if training is related to reporting of phishing attempts.	
3. Analyze trends of failed login attempts over time to determine if employees are complying with company policy regarding login credentials.	
4. Perform a linear regression analysis to consider the likelihood of future information security breaches based on the independent variables of dollars spent on cybersecurity training and encryption software.	
5. Perform a what-if analysis to identify the appropriate dollar amount of spending that is necessary to meet the entity’s goals for information security.	

BE 3.3 (LO 1, 6) Match the motivation source for data analytics in professional accounting to the appropriate scenario. Each source may be used once, more than once, or not at all.

- a. Opportunities
b. Process and performance assessment
c. Regulatory changes
d. Problem solving

Scenario	Motivation Source
1. You are a tax accountant working at a large multi-national company. Executive management is trying to decide the best country in which to expand operations. You have been tasked with identifying independent variables and performing regression analysis to predict potential revenue from this expansion.	
2. Your company has several key performance indicators (KPIs) associated with its manufacturing process – especially quality control. You have been tasked with performing an analysis to identify whether quality control benchmarks are being accurately recorded in the information system.	
3. You are an external auditor working on a public company manufacturing client. You have been asked to test the operating effectiveness of internal controls associated with the purchasing groups' compliance with the internal control that all purchase orders above \$10,000 are approved by a purchasing supervisor.	
4. You are a financial analyst working for a company that distributes consumer products to retailers across the U.S. You must find out why sales of your historically most popular product, the electric grill, have been declining.	

BE 3.4 (LO 2, 6) Auditing As an audit senior, you have asked your intern to perform descriptive analysis on the company's maintenance expenses for the period under audit. You remind the intern that they must first identify the purpose of the analysis and then break that down into questions. The intern is concerned and wonders how they would know if they developed a good descriptive question. Fill in the blank to formulate a response to the intern's concern. Each term can be used once, more than once, or not at all.

Word Bank: objective, specific, available, aspirational, accurate, considerable, exclusive, ambiguous

A good question is one that relates to the _____, is _____, measurable, and can be answered with the _____ data.

BE 3.5 (LO 2) Financial Accounting You are a financial accountant working for the city. Your controller would like to better understand vendor payments made by the city. You have downloaded the vendor payment file for fiscal years 2024 and 2025 from the city's database. An excerpt of the data set is provided.

FY	Activity	Department	Check Date	Vendor	Check Total
2024	Landfill Management	Sanitation Svcs	8/22/2024	Smith Temporaries, Inc	\$ 570.24
2025	Service Maintenance Centers	Public Works and Transportation	8/14/2025	Ferrell Gas, LP	\$ 32.22
2023	Water Treatment	Water Utilities	10/30/2023	Custom-Crete Redi-Mix, LLC	\$ 100.00
2023	Inventory	Inventory Purchase	11/21/2023	Installer Sales and Service	\$ 8.76
2024	Fleet Replacement	Equipment and Fleet Management	6/28/2024	Sam Pack's Five Star Ford	\$ 22,155.00
2024	Homeless Housing Services	Housing/Community Services	6/26/2024	Saint Augustine Estates	\$ 614.00
2024	Complete Street	Public Works and Transportation	3/11/2024	HERC Rentals Inc	\$ 4,236.00
2023	Non-Departmental	Office of Budget	12/24/2023	William H Bancroft Jr.	\$ 2,000.34

- What is the objective of the analysis?
- Develop three questions relevant to the objective, and determine what measures you will use to answer the questions.
- What analyses will you use to answer these three questions?

BE 3.6 (LO 2, 6) Managerial Accounting Match the appropriate analysis that can be used to answer descriptive questions or sub-questions. Each analysis choice can be used once, more than once, or not at all.

- Filter the data to only analyze the machine of interest and use the maximum function.
- Use the minimum and maximum functions to identify the smallest number of units produced and the largest number of units produced.
- Filter the data to only analyze the unit of interest and machine of interest and calculate the mean number of units produced for each shift.
- Create a linear equation to estimate the number of machine hours needed to produce a unit.
- Use the optimization model function in Excel to identify the maximum number of units that can be produced given the constraint of time during shift.
- Create a frequency distribution table to categorize product category.

Initial Descriptive Analysis Questions or Sub-Questions	Analysis Choice
1. How many units were produced for each product category in the facility?	
2. What is the average number of units produced by machine #1,065 during first shift, second shift, and third shift?	
3. What is the greatest number of units that machine #1,810 produced?	
4. What is the range of units produced during the period?	

BE 3.7 (LO 3) Auditing You are an internal auditor at a remote-controlled car manufacturer and sales company. You have been asked to perform an analysis to see if there have been unusual purchases from vendors during the period. Your information technology group provided you with a transaction file that contains all purchases made during the period. An excerpt of the file is provided.

Purchase Order No	Purchase Order Date	Vendor ID	Vendor Name	Item Code	Item Description	Item Quantity	Item Price	Total PO Amount	PO Approved?
2001	1/18/2025	13789	ABC Parts	V473	Cylinder Head	150	\$ 56.24	\$ 8,436.00	
2002	1/18/2025	13356	L&M Manufacturing	V302	Camshaft	250	\$ 57.56	\$ 14,390.00	Y
2003	1/18/2025	13780	TGIF Works	V292	Crankshaft	155	\$ 58.45	\$ 9,059.75	
2004	1/18/2025	13852	LP, LLP	K385	Oil Pan	153	\$ 36.24	\$ 5,544.72	
2005	1/18/2025	13513	GAPISH, CO	K395	Oil Valve	157	\$ 36.47	\$ 5,725.79	
2006	1/19/2025	13253	PTO Park, Co	M390	Nickel Metal Hydride Battery	201	\$ 18.46	\$ 3,710.46	
2007	1/19/2025	13850	STY Manufacturing	M392	Lithium Ion Battery	200	\$ 19.50	\$ 3,900.00	
2008	1/19/2025	13861	Bing Blitz, Inc.	Z389	Struts	268	\$ 37.89	\$ 10,154.52	Y

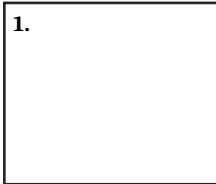
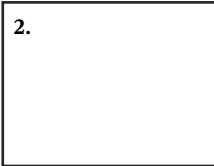
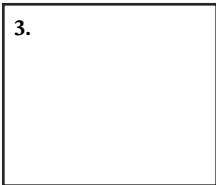
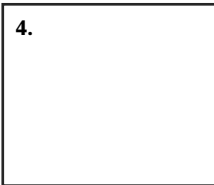
- What is the objective of the analysis?
- Develop three questions that will be relevant to the objective.
- What analyses will you use to answer these three questions?

BE 3.8 (LO 3) Auditing As an auditor for a private entity, your senior has asked you to examine your client's sales returns in the first 30 days after year end. The descriptive analysis revealed sales returns dramatically decreased compared to the prior year. Further, you noted that the largest decrease related to sales returns was in the northwest region. You must perform diagnostic analyses to understand why there was a decrease in sales returns.

Next is an outline of the data analysis objective, initial question, sub-questions, and possible measures. Complete the outline by matching the appropriate statement to the corresponding blank square. Statement choices may be used once, more than once, or not at all.

- What is driving the sales return decrease?
- Are there unusual patterns in the northwest region sales returns for certain product lines?

- c. Create a trend analysis showing returns by date for each product line for the current year compared to the prior year.
- d. Filter data to isolate the northwest region sales returns and examine total sales returns, average sales returns, and sales return quantity.
- e. Create an optimization model to identify the most appropriate number of sales returns given the constraint of sales and quantity manufactured.
- f. Calculate the standard deviation of sales and compare to the standard deviation of employee expenses for the period.
- g. Calculate the overall spending by the company in customer services to determine if sales returns are related to manufacturing quality.

Objective	Initial Question	Sub-questions	Possible Measures
Why did sales returns decrease from prior year?	1. 	Are there anomalies in sales returns for any specific product the company sells? Are there identifiable trends in sales returns for any specific product?	Create a scatterplot comparing quantity returned to sales price for items to identify anomalies in returns. Create a scatterplot comparing sales dollars to sales returns for items to identify anomalies in returns. 2. 
Why is there a large decrease in sales returns in the northwest region?	What factors are driving the decrease in sales returns in the northwest region?	3. 	4. 

BE 3.9 (LO 4) Match the analysis to its description. Each term may be used once, more than once, or not at all.

Analyses:

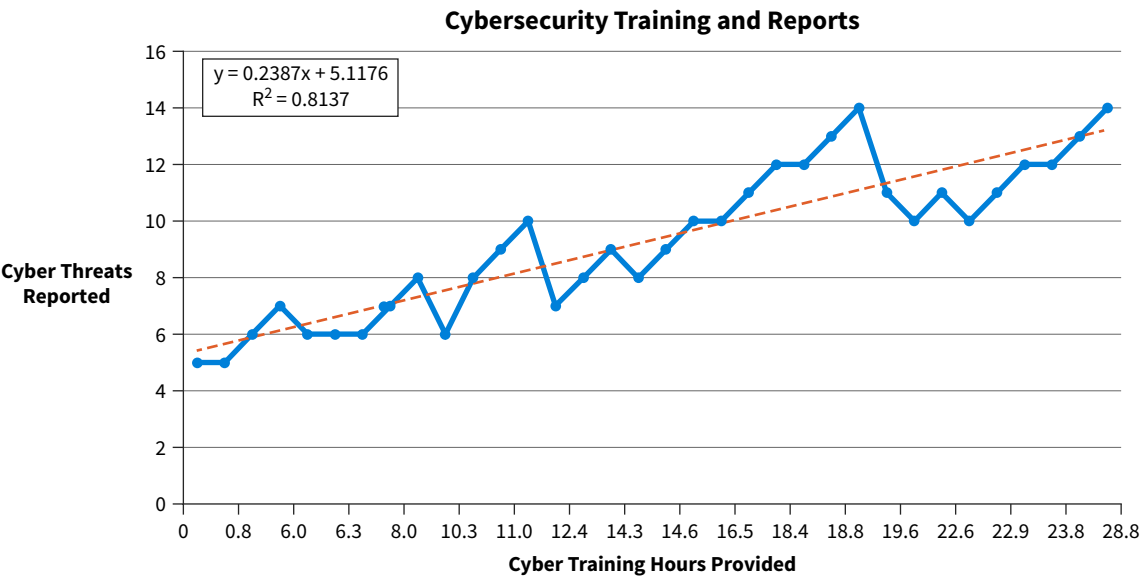
- a. Independent variable
- b. Linear function
- c. Regression statistics
- d. Correlation coefficient
- e. Dependent variable
- f. Adjusted R^2

Description	Analysis
1. The statistical measures used to evaluate a regression model.	
2. The outcome variable in a regression model.	
3. The variable or variables that influence the outcome variable.	
4. This type of relationship shows steady increases or decreases over the range of the independent variable.	
5. This statistic measures the strength of the relationship between the dependent and independent variables.	
6. This statistic explains how well the regression line fits the data.	

BE 3.10 (LO 4) The objective is to engage in predictive analyses. For each of the following initial or sub-question, identify the independent variable and the dependent variable.

- 1. How much will revenue change with a 15% increase in sales volume?
- 2. How much will revenue increase with a 10% increase in production volume?
- 3. How much will maintenance expenses change if we increase production volume by 5%?
- 4. As unemployment increases, how much will sales volume decrease?
- 5. As the number of employees increase, how much will information security costs increase?

BE 3.11 (LO 4, 6) Accounting Information Systems Assume you are an accounting information systems accountant asked to consider the success of your company’s cyber training program. You collected data on the monthly number of cybersecurity training hours provided to employees and the number of cyberthreats reported. You created the following trendline line graph.



- Use the information provided in the line graph and the equation to answer the following questions:
- 1. If the company offered 9 hours of cyber training during a period, what is the expected number of cybersecurity threats reported by employees?
 - 2. If the company offered 35 hours of cyber training during a period, what is the expected number of cybersecurity threats reported by employees?

BE 3.12 (LO 5) Match the appropriate term to each definition. Terms can be used once, more than once, or not at all.

- a. Constraints
 - b. Objective function
 - c. Decision variable
 - d. Optimization
- e. Linear regression
 - f. Dependent variable
 - g. Independent variable

Definition	Term
1. The process of selecting values of variables that minimize or maximize some quantity of interest.	
2. The unknown values a model seeks to determine.	
3. The mathematical equation that describes the output target that we seek to minimize or maximize.	
4. The limitations, requirements, or other restrictions that must be imposed on any solution.	

BE 3.13 (LO 5, 6) Managerial Accounting As a managerial accountant for a manufacturing company, you are asked to identify the optimal number of units that must be produced in order to maximize the company's contribution margin (CM). Your company manufactures four different types of units:

- Standard widgets
- Blue flying widgets
- Red swimming widgets
- Yellow hopping widgets

To perform prescriptive analysis, your manager has built a spreadsheet for you to execute a Microsoft Excel Solver function.

	Standard	Blue Flying	Red Swimming	Yellow Hopping	Total
Number Produced	1	1	1	1	
CM per unit	17.87	15.20	21.00	19.87	
Total CM	17.87	15.20	21.00	19.87	73.94
Machine Hours Per Unit	6	8	12	11	0
Total Machine Hours	6	8	12	11	37
Constraints					
Machine Hours					50,000
Demand	1,200	1,750	1,670	1,250	

1. What is the decision variable in the model?
2. What are the constraints in the model?
3. What is the objective function?

BE 3.14 (LO 5, 6) Financial Accounting Your private company wants to expand and grow into a new product line; however, that growth requires increased capital and investment in physical assets, people, and materials. You are asked to consider the most appropriate strategy to acquire the needed capital. You volunteered to engage in data analysis to provide insight to the discussion.

1. What is the objective of the analysis?
2. Develop two questions that will be relevant to the analysis.
3. What analyses will you use to answer each question?

BE 3.15 (LO 6) Identify whether the professional has developed a measurable and specific question appropriate for the practice area.

1. A financial statement auditor asks, "What is the expected value of the client issuing stock based on current market prices?"
2. An accounting information systems accountant asks, "How many units should the company produce to optimize the company's contribution margin?"
3. A tax accountant poses the question, "Based on last year's tax return, how will the current year changes in tax law impact the client's tax liability?"
4. A managerial accountant considers "Which products use the highest amount of raw materials in production?"
5. A financial accountant ponders "What dollar amount of accounts receivable are classified in the over 90 days outstanding category?"

BE 3.16 (LO 6) Match each statement to the most appropriate accounting professional practice area. Each accounting professional practice can be used once, more than once, or not at all.

- a. Accounting information systems

b. Financial accounting

c. Managerial accounting
- d. Auditing

e. Tax accounting

Statement	Practice Area
1. Simone and the information technology team are investigating the possibility of investing in new technologies.	
2. Chung would like to improve the company’s information system performance and is evaluating processing time to prepare month-end management reports.	
3. Christine wants to identify unusual sales transactions that may have contributed to an unexpected increase in the client’s revenue compared to last year.	
4. Daniel is currently using the Scenario Manager function in Excel to identify interest rates, payment terms, and length of a loan to evaluate various capital requirement strategies.	
5. Ellis is analyzing a client’s tax liability calculation to identify if there are expenses that are not properly included as deductions.	
6. Pierre is currently examining the relationship between machine hours and maintenance expenses to predict future manufacturing modifications.	
7. Mica is performing a regression analysis to estimate the predicted monthly revenue for a retail client. He will then compare the predicted monthly revenue to the client’s recorded amount.	

Exercises

EX 3.1 (LO 2, 6) Data Financial Accounting Analysis of Measures As the company controller for a yarn manufacturing supply company, you are asked to investigate sales by customer for the month of June 2025. You have critically assessed the objective and questions of your analysis and documented them as follows:

Objective	Initial Question	Sub-questions	Possible Measures
Understand sales by customer for the month of June.	What happened to sales in the month of June?	What was total sales? What was the average sales amount? What was total sales by customer? Which customers had the highest sales?	1. Sum of sales amount (total measure). 2. Average of sales amount (measure of location). 3. Total sales separated by customer. 4. Total sales separated by customer using a highlight table.

Use the available data to perform the analyses proposed in the possible measures section of the analysis plan.

EX 3.2 (LO 2, 6) Data Managerial Accounting Analysis of Measures Your company manufactures high-quality yarn for sale at craft stores and yarn and specialty shops across the country. Specifically, your facility manufactures merino wool yarn in a variety of colors and in two weights: chunky and DK weight. The product manager wants to understand differences in production quantities by product between 2025 and 2024 in the July manufacturing period, and has provided you with production data for select products. Use the spreadsheet to identify the available data and complete the following:

1. What is the objective of the analysis?

2. What are two initial questions you can ask about the data?

3. Create a column chart depicting the production quantity by ProductDescription and Year.

4. Calculate and identify several descriptive measures:

a. Calculate the Average ProductionQuantity by ProductDescription in 2024 compared to 2025.

b. Identify the lowest number of units produced by ProductionDescription in 2024 compared to 2025.

c. Identify the highest number of units produced by ProductionDescription in 2024 compared to 2025.

EX 3.3 (LO 2, 6) Data Financial Accounting Tableau Highlight Table Your controller would like to better understand changes in specific vendor payments between 2025 and 2024 that were made by the city you work for. The controller is specifically interested in understanding the payments made to the following vendors:

- 4-Star Hose & Supply
- Aecom Technical Services, Inc.
- WRG, LLC
- Winston Water Cooler Ltd.
- Zoetis Inc.

You have downloaded the vendor payment file and uploaded it into Tableau for analysis. Create a highlight table in Tableau to identify changes in payments to each of these five vendors between 2024 and 2025.

EX 3.4 (LO 2, 6) Data Accounting Information Systems Analysis of Measures As part of your company's new cybersecurity program, each employee must participate in an annual training. The trainings are conducted in a variety of modalities (in-person, online synchronous, and online asynchronous). You and your team have tracked the training expenses each month as well as the number of cybersecurity threats reported by employees. Employees can report cybersecurity threats such as phishing attempts, or IT specialists can report threats through examination of the monthly login report. You have been asked to investigate cybersecurity training costs over the period from January 2023 through December 2025.

Review and complete the chart by identifying sub-questions and possible measures used to address your identified sub-questions.

Objective	Initial Question	Sub-questions	Possible Measures
Understand cybersecurity training expenses.	What is the total amount of cybersecurity training expense for 2023, 2024 and 2025?	1.	2.
	What is the percentage change of cybersecurity training expenses between 2023 and 2024 and between 2024 and 2025?	3.	4.

5. Then, conduct your descriptive analyses using the provided data file.

EX 3.5 (LO 3, 6) Data Auditing Diagnostic Measures Assume you are an audit staff assigned to audit the financial statements of a private entity distribution client for the year ended December 31, 2025. The client does not manufacture items, but acquires finished products and sells to a variety of retailers. The senior on the engagement team asked you to perform analytical procedures on sales returns. Your descriptive analysis revealed that sales returns increased as a percentage of sales. Use the spreadsheet to perform the following:

1. Calculate total sales dollars, returns dollars, sales quantity, and returns quantity for each year 2024 and 2025.
2. Calculate the percentage change in sales dollars, returns dollars, sales quantity, and returns quantity between 2024 and 2025.
3. Create a scatterplot whereby sales dollars are on the x-axis and the sales returns are on the y-axis.
4. Which months in 2025 have sales returns that you may consider as anomalies?

EX 3.6 (LO 3, 6) Data Managerial Accounting Trend Analysis with Line Graphs You are an intern for a manufacturing company, and your company controller has learned you have completed a data analytics course with an emphasis in accounting. The controller has performed basic descriptive analysis of the company's sales for the month of June 2025 but wants to understand why sales increased 5.14%. Use the data set to complete the following:

1. What is the initial question for this analysis?
2. Identify two sub-questions for the analysis.
3. Identify potential measures that you may calculate to answer these two sub-questions.
4. Create a line graph showing the trend in June sales in 2024 compared to 2025.
5. Create a column bar chart showing sales to each customer during June 2024 compared to June 2025.

EX 3.7 (LO 3, 6) Financial Accounting Trend and Correlation Analysis The corporate controller asked the team to analyze accounts receivable and the aging of accounts receivable compared to prior year. Your descriptive analysis indicates accounts receivable increased over prior year, while sales have remained steady. Since you are the intern on the job, your colleagues have asked you to consider the diagnostic analysis questions that you must ask to plan your analysis. You were provided with the data dictionary.

Data Field	Description
SalesOrderNumber	Uniquely assigned sales order number.
CustomerID	Uniquely assigned customer number.
CustomerName	A customers name.
CustomerCredit	Customer’s credit limit.
CustomerAddress	U.S Postal mailing address and street name.
CustomerCity	Mailing address city.
CustomerState	Mailing address state.
CustomerZipcode	Mailing address zip.
CustomerPayTerms	Payment terms negotiated with the customer.
SalesOrderDate	Sales order date.
SalesOrderID	Uniquely identifiable item number.
SalesOrderDescription	Description of item.
ItemPrice	Sales price of the item per negotiated terms with the customer.
ItemQuantity	Total quantity of items sold to the customer.
InvoiceNumber	Uniquely assigned invoice number.
InvoiceAmount	The total amount of the invoice, including shipping charges and tax.
PaidAmount	The total amount of the payment received from the customer.
PaidDate	The date the invoice was paid.

Complete the diagnostic objective questions in the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Understand why accounts receivable increased compared to the prior year.	1.	2.	3.

EX 3.8 (LO 3, 6) Data Accounting Information Systems Trend and Correlation Analysis You and your team have tracked the number of cybersecurity threats reported by employees. Employees can report cybersecurity threats such as phishing attempts, and IT specialists can report threats through examination of the monthly login report.

Your initial descriptive analysis revealed that in the last three years, 313 cyber incidents were reported. There were 73 reports in 2023, 111 in 2024, and 129 in 2025. Your manager would like to understand why cyber incident reports have increased. You have started to identify initial questions, sub-questions, and possible measures for your analysis in the following chart. Complete the chart.

Objective	Initial Questions	Sub-questions	Possible Measures
Determine why cyber incident reports are increasing.	1.	Are there more cyber incidents reporting during a specific period of time?	Create a line chart of cyber incident reports over time and review for a trend or pattern.
	2.	3.	Perform a correlation between training expenses and cyber incidents reported by using the CORREL function in Excel.

4. Perform the possible analyses.

EX 3.9 (LO 4, 6) Data Accounting Information Systems Trendlines Your controller has asked you to demonstrate the relationship between cybersecurity training expenses and the number of valid cybersecurity threats reported by employees. You have collected monthly data reporting the monthly cybersecurity training expense and the number of valid cybersecurity threats reported by employees. Create a line graph showing the relationship between these two variables. Include the trendline, equation, and R-square of your analysis.

EX 3.10 (LO 5, 6) Data Auditing Sales Returns Analytical Procedures You are the audit intern staffed on the fiscal year ended December 31, 2025 financial statement audit of a private company. Your audit senior is thrilled that you have just completed a data analytics in accounting course and wants you to use your knowledge of predictive analytics to develop an expectation of sales returns based on sales quantity to perform analytical procedures. You have engaged in critical thinking and developed the partial chart.

Complete the chart:

Objective	Initial Question	Sub-question	Possible Measures
Predict sales returns.	How much will sales returns change if we have a 10% increase in sales quantity?	1.	Regression analysis

2. Perform the regression analysis to predict sales return dollars using sales quantity.
 3. Write the regression equation.
 4. What is the R^2 for the regression model?

EX 3.11 (LO 5, 6) Data Financial Accounting Estimate Warranty Expense for Accruals As a financial accountant for a public widget manufacturing company, you are given a data file and asked to calculate and record month-end adjusting journal entries. You must calculate the warranty expense amount for January 2025.

1. Perform a regression to predict warranty expense using sales quantity.
 2. Write the regression equation.
 3. You just performed a query of your company's accounting information system and determined that January 2026 sales quantity is 10,385. Write the journal entry to record warranty expense.

EX 3.12 (LO 5, 6) Data Managerial Accounting Trend Analysis You are a financial analyst supporting the control in a private company that manufactures paper plates, trays, and other disposable paper products. The company is owned by a private equity (PE) investment group. The PE board is interested in cost savings and identifying efficiencies within the company's processes. You collected monthly data on the number of die-cut changes and the monthly maintenance expenses. The goal is to predict the cost of maintenance expenses for the next period.

1. Run a regression analysis using die-cut changes to predict maintenance expenses.
 2. Write the regression equation.
 3. If the company plans to have 650 die-cut changes during the month of January 2024, what is the company's expected maintenance expense?

EX 3.13 (LO 4, 6) Data Auditing Predictive Analytics Kahn & Williams CPA firm are the external auditors for Super Scooters. The audit team is performing substantive procedures on revenue and designed a test that uses predictive analytics to evaluate the reasonableness of Super Scooters reported revenue. The objective is to evaluate recorded revenue for reasonableness. The initial question is: Does reported monthly revenue vary significantly with the predicted monthly revenue from the engagement team's regression model? The team's model is based on the prior three years of monthly data. Further, the team believes that revenue is driven by sales volume and the average price of scooters sold during the month. The team has decided that any difference between predicted revenue and the actual revenue that is greater than the standard error (in absolute dollars), should be investigated further.

The summary output is given here:

PREDICTED REVENUE MODEL

SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.997170819							
R Square	0.994349643							
Adjusted R Square	0.994007197							
Standard Error	27302.53042							
Observations	36							
ANOVA								
	df	SS	MS	F	Significance F			
Regression	2	4,328,953,815,697.83	2,164,476,907,848.92	2,903.67	0.00			
Residual	33	24,599,129,513.81	745,428,167.09					
Total	35	4,353,552,945,211.64						
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	-597355.3555	79141.28598	-7.547961195	1.10769E-08	-758369.5125	-436341.1985	-758369.5125	-436341.1985
Volume	424.0763009	7.525920213	56.34876386	2.20655E-34	408.7647011	439.3879007	408.7647011	439.3879007
Average Price	1453.623412	208.2682902	6.97957145	5.57457E-08	1029.89839	1877.348434	1029.89839	1877.348434

1. What is the dependent variable in the model (y range)?
2. What are the independent variables in the model (x range)?
3. What is the regression equation for this model?

EX 3.14 (LO 5, 6) Data Managerial Accounting Optimization Your production manager has asked you to identify the quantity of items that should be produced to maximize the company's contribution margin for the period. Your company has four products:

- Standard widgets
- Blue flying widgets
- Red swimming widgets
- Yellow hopping widgets

You have created a spreadsheet outlining the demand, machine hours per unit, contribution margin per unit, and overall machine hours. Use Microsoft Solver to identify the number of units to be produced for each product and the maximum contribution margin.

EX 3.15 (LO 5) Data Managerial Accounting Use Linear Optimization to Determine Product Mix and Profit You and your college roommates open a business designing and crafting dog and cat leashes with your college logo and colors. You have decided to offer leashes in three lengths: 6, 8, and 12 feet.

Further, you have two types of leash: "standard" or "decorative." While the decorative leashes take longer to create, they have a large profit margin per each. However, the standard leashes are in higher demand. You and your two roommates have put a schedule together so that you can fit everything into your week – this includes classes, social events, and crafting time. Since you are enrolled in the data analytics in accounting course, you have volunteered to calculate how many of each type of leash you should produce weekly based on demand, gross profit for each product, and the constraint of work hours.

1. Perform a linear optimization using Excel Solver.
2. How many of each unit should you and your two roommates produce each week?
3. What is the total gross profit you should expect to earn based on the weekly production of dog leashes?
4. Are there any constraints identified in the answer report that are non-binding?

EX 3.16 (LO 4, 5) Data Build the Spreadsheet and Identify Constraints Along with your college roommates, you are opening a business to make decorative hats, T-shirts, and socks with your college colors. You have decided to start with five core products:

- Decorative ball cap
- Decorative “GO Team” t-shirt
- Decorative “Defense Wins Championships” t-shirt
- Decorative socks
- Decorative knee-high socks

As part of your business plan, you want to calculate the total number of each product to create each week. You plan to use Microsoft Solver to identify the maximum number of units to produce given your constraints. You want to maximize your company’s profit given the constraints of work hours per person. The table outlines the gross profit and crafting hours per unit for each item.

Product	Gross Profit per Item	Crafting Hours per Item
Ball cap	5.00	4
GO Team t-shirt	7.50	2.5
Defense Wins Championships t-shirt	7.50	3
Socks	3.30	6
Knee-high socks	4.50	3.5

In addition, you and your roommates have committed to working the following hours each week:

- You: 20
 - Roommate 1: 20
 - Roommate 2: 15
 - Roommate 3: 10
1. Build a linear optimization model spreadsheet.
 2. Calculate the total number of units for each product that should be produced each week.
 3. What is the total gross profit you should expect to earn based on the weekly production?
 4. Are there any constraints identified in the answer report that are non-binding?

EX 3.17 (LO 5, 6) Data Managerial Accounting Linear Optimization Tiana’s Jewel Design, Inc. designs and manufactures mass produced metal jewelry and custom designed silver-plated jewelry. Currently, the owner designs and produces three lines of custom jewelry: necklaces, bracelets, and earrings (decision variables). The contribution margin and the resource requirements for each product follows.

	Necklaces	Bracelets	Earrings
Contribution Margin per unit	\$ 12.00	\$ 8.00	\$ 14.00
Resource Requirements:			
Direct Materials, ounces per unit	22	8	13
Design Hours per unit	4	1	5
Machining Hours per unit	2	1	2

They would like to maximize the contribution margin (objective function). However, the owner has these constraints:

- A contract to produce at least ten pairs of custom earrings a month.
 - They can sell all the custom necklaces and custom earrings that are produced
 - Demand for custom bracelets is only 30 a month.
 - Each month 800 ounces of silver plate can be purchased.
 - Two designers work a total of 120 hours a month.
 - An estimated 80 hours of machining time are available each month.
1. Use Microsoft Excel Solver to create a linear optimization model.
 2. How many of each product should be produced to maximize the contribution margin?
 3. Are any of the constraints binding?
 4. Based on the output from the answer report, what advice would you give the owner?

Professional Application Case: Healthcare Service Company



Kindred Healthcare is a post-acute healthcare service company that operates long-term acute care and rehabilitation services across the United States. Kindred's hospitals offer the same in-depth care that a patient would receive at a traditional hospital, but for an extended recovery period. The data used in this case is based on actual data from the California Health and Human Services Open Data Portal.

Kindred has 13 long-term acute care hospitals in California:

- Baldwin Park
- Brea
- La Mirada
- Los Angeles
- Ontario
- Paramount
- Rancho
- Riverside
- Sacramento
- San Diego
- San Francisco Bay Area
- South Bay
- Westminster

The hospital locations in California are consolidated into financial statements which are audited by external auditors and provided to the board of directors, the company's primary investors, and the financial institutions who have extended the bank loans. The income statement for the prior three years is provided.

	2019 (unaudited)	2018 (audited)	2017 (audited)
Gross Patient Revenue	\$2,802,211,999	\$2,515,881,547	\$2,423,431,540
Deductions from Revenue	\$2,206,888,182	\$1,940,233,238	\$1,869,518,824
Net Patient Revenue	\$ 595,323,817	\$ 575,648,309	\$ 553,912,716
Other Operating Revenue	\$ 2,262,385	\$ 2,367,007	\$ 2,663,316
Operating Expenses	\$ 579,928,100	\$ 523,090,033	\$ 506,521,633
Net Revenue from Operations	\$ 17,658,102	\$ 54,925,283	\$ 50,054,399
Non-Operating Revenue	\$ 1,237,718	\$ 970,497	\$ 763,174
Non-Operating Expense	\$ 110,477	\$ 200,367	\$ 385,724
Income Tax	\$ 86,442	\$ 73,200	\$ 25,176
Net Income	\$ 18,698,901	\$ 55,622,213	\$ 50,406,673

The financial statements are prepared based on annual amounts from each of the California locations. The consolidated entity has provided a data file that contains annual data by location for a variety of data fields.

PAC 3.1 Auditing: Develop Objectives and Questions for an Audit Plan

Data Auditing You are a second-year staff member at a public accounting firm who is assigned to the Kindred Hospitals engagement, which is a new engagement for the firm. You have been asked to perform descriptive, diagnostic, and predictive analytics for your engagement team.

The audit senior provided you with an Excel file that includes gross patient revenue by hospital location from 2013 to 2019. The senior asked you to perform exploratory data analytics and outline an audit plan, which you recognize is similar to outlining objectives, questions, and sub-questions for data analysis. Before you can fully explore the data, you must understand it, so you begin with descriptive statistics.

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross patient revenue (GrossPatientRevenue) for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross patient revenue for the year 2019 compared to 2018. Specifically, GrossPatientRevenue, GrossInpatientRevenue, and GrossOutpatientRevenue data points.	Which locations had changes in revenue from 2018 to 2019?	3.	4.
Predictive Analysis: Develop an expectation for gross patient revenue in 2019 based on the components patient days, out patient visits, and average length of stay.	What factors drive changes in gross revenue?	5.	6.

7. Perform the possible analytics as described for each measure.

PAC 3.2 Managerial Accounting: Evaluate Labor Costs and Productivity

Data Managerial Accounting Assume you are a managerial accountant for the consolidated California group of Kindred Hospitals. The director of operations has asked that you analyze the annual data associated with hospital labor. Specifically, the director of operations is interested in the following variables.

Variables	Definition per Data Dictionary
HopsitalFTE	Number of hospital full time employees
NurseFTE	Average number of nursing personnel
ProductiveHours	Total productive hours
NonProductiveHours	Total nonproductive hours
PaidHours	Total paid hours
MedicalStaff	Number of active medical staff

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand labor information for 2018 - 2019.	What happened to labor hours during 2018 and 2019?	1.	2.
Diagnostic Analysis: Determine why variables changed between 2018 and 2019.	Which hospitals have increasing salary expenses from 2018 to 2019?	3.	4.
Predictive Analysis: Identify if there is relationship between total productive hours and gross patient revenue.	Can productive hours predict gross patient revenue?	5.	6.

7. Use the data to calculate the measures.

PAC 3.3 Financial Accounting: Understand Patient Revenue

Data Financial Accounting You are a financial analyst working for the controller's group for the California location of Kindred Hospital group. Your controller has highlighted the fact that gross revenue consists of both inpatient revenue and outpatient revenue. However, only one facility – Kindred Hospital-Rancho – has outpatient revenue. Therefore, the controller wants a full understanding of inpatient revenue by facility for 2019. Specifically, you are asked to understand the dollar amount of inpatient revenue contributed by each facility in 2019, and how inpatient revenue has changed since 2018. Finally, the controller is creating a financial forecast for inpatient revenue for 2020 and wants to understand the utilization data that may contribute to the generation of this forecast. To satisfy your company controller, you are planning your analysis and have started the chart which follows. The controller has also provided you with a data dictionary for the data relevant to this analysis.

Variables	Definition per Data Dictionary
DayTotal	Patient Days Total
OccupancyLicensed	Occupancy rate – Licensed beds
OccupancyAvailable	Occupancy rate – Available beds
AverageLengthStayAll	Average Length of Stay – Including Long Term Care
AverageLengthStayNoLTC	Average Length of Stay – Excluding Long Term Care
GrossInpatientRevenue	Gross Inpatient Revenue

Complete the chart for descriptive, diagnostic, and predictive analytics.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand gross inpatient revenue for the year 2019, which is the year under audit.	What happened to gross patient revenue during 2019?	1.	2.
Diagnostic Analysis: Understand what happened to gross inpatient revenue for the year 2019 compared to 2018.	How has revenue changed at the locations?	3.	4.
Predictive Analysis: Identify the utilization measures that are related to in-patient revenue.	Are there any measures correlated to inpatient revenue?	5.	6.

7. Use the data available to calculate the measures.

PAC 3.4 Tax Accounting: Evaluate Non-Profit and For-Profit Tax Return Data

Data Tax Accounting As a tax accountant at Kindred you have been asked to analyze the non-profit and for-profit hospital data. Kindred Hospitals has three hospitals in California that are tax exempt organizations. These three hospitals must be separated from the for-profit hospitals for tax purposes. In addition, you have been asked to do an analysis of a sample of non-profit tax filings and compare those filings to the amounts that Kindred's non-profit hospitals filed for 2019. The variables and data definitions for both files are included in the following tables:

IRS 990 Data:

Variables	Definition per Data Dictionary
Assets	Total asset book value at year end
Income	Net Income
Revenue	Revenue reported on line 12 of Form 990

Kindred 2013 – 2019 Financial Data

Variables	Definition per Data Dictionary
TotalAssets	Total Assets
TotalRevenue	Net patient revenue plus other operating revenue
NetIncome	Net Income

Complete the chart.

Objective	Initial Question	Sub-questions	Possible Measures
Descriptive Analysis: Understand assets, revenue, and income information for Kindred's non-profit and for-profit hospitals	Which hospitals have the highest income?	1.	2.
Descriptive Analysis: Understand how IRS 990 data for non-profit hospitals compares to the Kindred non-profit hospitals	Are the Kindred Hospitals comparable to the sample of IRS 990 filings.	3.	4.
Diagnostic Analysis: Identify trends in assets, revenue, and income for Kindred hospitals.	5.	6.	7.
Diagnostic Analysis: Compare Kindred data to IRS 990 data.	8.	9.	10.

11. Use the data to calculate the measures.

Le Grind Continuing Case: Motivation and Specific Questions for Analyzing Gross Profit

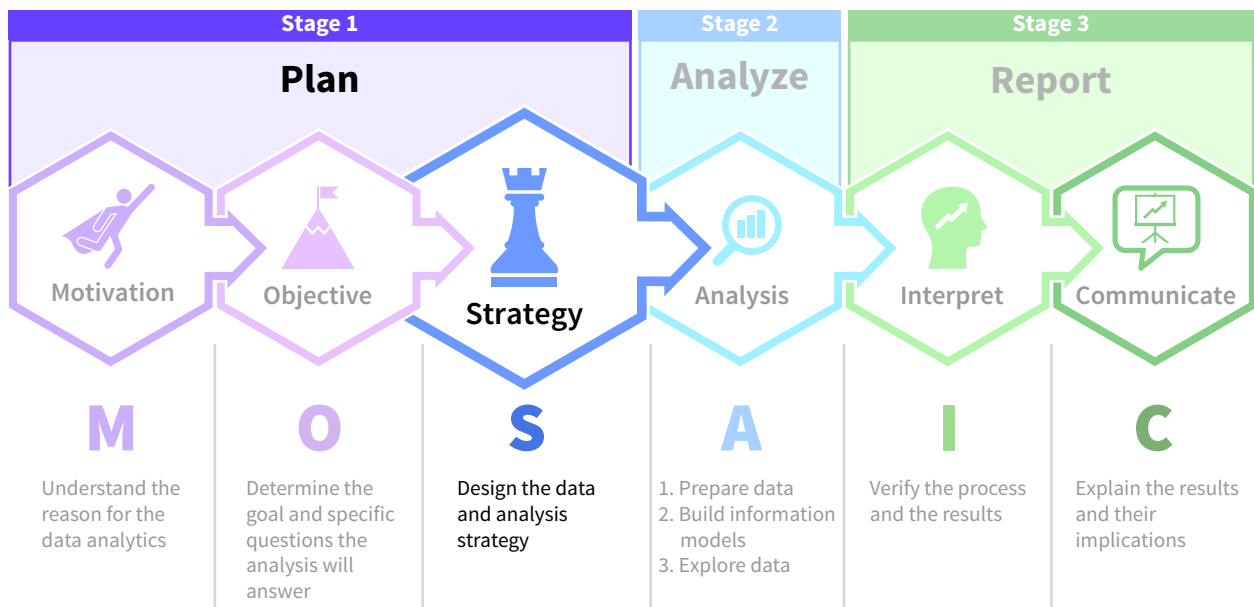
Data Visit Wiley's online learning platform for the case background, additional questions, data, and more details about the continuing case.



Planning Data and Analysis Strategies

Chapter Preview

Armed with an understanding of the motivations and objectives for the data analysis project and after formulating the necessary objective questions, it is time for the final step in the planning stage—designing the data and analysis strategies. You will soon be expected to apply your knowledge, common sense, and problem-solving skills to design a variety of accounting, audit, and tax related data and analyses strategies. This chapter will help you develop the skills necessary to create effective data analysis project plans.



Professional Insight: Why Should You Plan Data and Analysis Strategies?

Taylor was excited to begin her career in auditing after interning at a Big Four accounting firm during the last busy season.

My audit team had me design my own data and analysis strategies and document each step I performed. While I lacked confidence at first, I felt very supported. I learned so much, even when I made mistakes and had to pivot my strategies.

I was more successful when I understood the motivation for and the objective of my tasks. I was able to formulate better questions, which resulted in better

data choices. The measurement scales of my selected data also pointed me toward the appropriate analyses.

My first task was a descriptive analysis. I had to determine if the year-end accounts receivable subsidiary ledger and general ledger balances reconciled. After checking if any credit customer's activities were excluded from the general ledger accounts receivable balance, I created an accounts receivable aging report listing each credit customer's outstanding balances from the subsidiary ledger. I compared the aging reports' items to the general ledger accounts receivable postings and noted each unmatched item. If there were no unmatched items, the accounts receivable balance in the general ledger was successfully reconciled and verified.

I then designed a diagnostic analysis strategy to determine which receivable activities were causing any unmatched items I had found. My strategy began with gathering the client's accounts receivable process documentation. For each unmatched item, I identified which process caused the items to be included in both ledgers. Documenting my work helped me explain what, how, and why I performed each step. This record will be helpful when we repeat this verification in next year's audit. **Designing my own data and analysis strategies helped me gain relevant skills and the confidence to start my auditing career.**

Chapter Roadmap

LEARNING OBJECTIVES	TOPICS	APPLY IT
LO 4.1 Identify the components of a data analysis project plan.	- Create a Data Analysis Project Plan - Sample Data Analysis Project Plan	Build a Project Plan for Inventory Costs (Example: Financial Accounting)
LO 4.2 Describe how to develop a data strategy.	- Identify Appropriate Data - Evaluate Data Fields and Sources - Consider Data Strategy Risks and Implement Controls	Identify Data Characteristics (Example: Accounting Information Systems)
LO 4.3 Explain how an analysis strategy is designed.	- Designing Analyses to Describe and Diagnose - Designing Analyses to Predict and Prescribe	Create a Predictive Data Analysis Project Plan (Example: Financial Accounting)
LO 4.4 Summarize data and analyses strategies in professional practice areas.	- Accounting Information Systems - Auditing - Financial Accounting - Managerial Accounting - Tax Accounting	Match Strategies with Professional Practice Areas

Data The Data tag appears in the chapter when the data for an example, illustration, or application are available on Wiley's online learning platform.

Data analytics software is continuously changing, and there may be more recent versions of the software referenced in this chapter. For more information access the accompanying video on Wiley's online learning platform.

4.1 How do Accountants Design Data Analysis Projects?

LEARNING OBJECTIVE 1

Identify the components of a data analysis project plan.

Recall that the first step in planning a data analysis project is understanding the motivations for the project, such as identifying problems, evaluating patterns and opportunities, measuring performance, and regulatory compliance. Considering the perspectives of the project's primary stakeholders is an important part of that understanding. Next, thinking critically about a project's purpose helps define the objective and articulate the project's specific questions. Common objectives and questions can be grouped by whether the analysis results describe, diagnose, or predict an outcome, or if the analysis results should prescribe future organizational strategy.

The final step of the data analysis planning stage is designing a professional plan for the data analysis project. Critical thinking continues to play a vital role, from selecting a strategy for the data and the analysis, to identifying inherent risks to both and embedding internal controls to reduce them. A data and analysis strategy plan is an organized and deliberate blueprint for the project. It also allows someone other than its creator to perform the analysis, which can free accountants to perform other valuable services with their business and regulatory expertise.

Create a Data Analysis Project Plan

Many of us use planning tools every day, such as relying on navigation applications to plan travel time and how to get to new destinations. Planning tools make tasks more efficient and the results more effective. They also often provide logical explanations for the necessity of certain steps or why they should be performed in a particular order.

This chapter summarizes a tool for data analysis project planning that provides similar benefits for accountants ([Illustration 4.1](#)). The order of the components is intentional:

Step 1 Focus on the objective:

- Keep the project's objective and specific questions in mind to select the best data and analysis strategies to fulfill the objective and answer those questions. Simply remembering to ask how the plan's proposed data and analysis strategy decisions relate to the objective helps us make better choices.

Step 2 Select a data strategy:

- Use critical thinking to develop and rank a few data alternatives. This ensures that we choose the data option most appropriate for the objective.

Step 3 Select an analysis strategy:

- Use what was learned from selecting the data strategy and apply that same development and ranking process to analyses alternatives. Following this step increases the likelihood of selecting the best analysis option.

Step 4 Consider risks:

- Considering and prioritizing critical risks to both the data and the analysis strategies reveals how these risks can create misleading and invalid results.

Step 5 Embed controls:

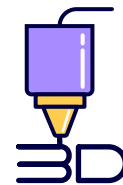
- Designing and implementing preventative and detective controls into the analysis process leads to results that are accurate, valid, and reliable.

ILLUSTRATION 4.1 A Data Analysis Project Plan

A data analysis plan increases the likelihood that analysis results accurately describe what happened, diagnose why it happened, predict what is likely to happen, or prescribe the best course forward. When the project plan is complete, so is the data analysis process planning stage, and it is time to move to the analysis stage. The next chapter describes how to properly prepare data for analysis. But first, practice using this project planning tool in an accounting context.

Sample Data Analysis Project Plan

Imagine you are a financial accountant for WeMakeIt, Inc., which is an employee-owned 3D printing corporation in Phoenix, Arizona. They sell superhero figurines, mechanical shop tools, and medical prostheses that are manufactured with 3D printing technologies. Founded five years ago by a local college student, Marisabel Cordoba, WeMakeIt, Inc. grew quickly to fill their production capacities and has maintained a stable size for the last three years.



Step 1: Focus on the Objective

The project in this example is to estimate the bad debts in the 2025 year-end accounts receivable balance. The result will be the value estimate for the 2025 year-end adjusting journal entry that debits bad debts expense and credits the allowance for uncollectible accounts contra-asset.

APPLYING CRITICAL THINKING 4.1: Estimate Bad Debts

The U.S. GAAP ending accounts receivable method requires the following:

- The year-end accounts receivable balance can be considered in total or in age group subtotals. Age grouping rules for manufacturing companies are typically less than 30 days, 30 to 60 days, and greater than 60 days outstanding, but these vary by industry.
- Reviewing regulatory guidance, industry practices, business policies and processes, and past bad debt estimation percentages is the professional approach for selecting bad debt estimation percentages. Typically, as the age of the accounts receivable increases, the percentage uncollectible applied to that age also increases (**Knowledge**).

Assume the following for this example:

- The company offers its credit customers payment terms of 30 days.
- The 2025 year-end accounts receivable balance in the general ledger is \$83,734.30.
- The 2025 year-end unadjusted allowance for uncollectible accounts has a credit balance of \$3,000.
- WeMakeIt, Inc. decided to use the percentage of year-end accounts receivable method to estimate bad debts expense each year. They have authorized a 2% bad debts assumption for invoices that are 30 to 60 days outstanding and 30% for invoices that are outstanding for more than 60 days.

This data analysis project has the following objective and related questions:

- **Objective:** Accurately estimate bad debts within the year-end accounts receivable balance in accordance with U.S. GAAP.
- **Specific questions:**
 1. What is a reasonable estimate of the uncollectible receivables in the 2025 year-end outstanding accounts receivables?
 2. What amount should be used in the adjusting journal entry for bad debts expense?

With the objective defined and the questions articulated, it is time to develop the data strategy.

Step 2: Select the Data Strategy

Develop several data alternatives that could help answer the objective question. Then, to select the most useful data alternative for the project plan, identify the factors you want to use to rank these alternatives, and assign values to each alternative's factors. The best data strategy alternative is the one with the highest overall factor ratings.

APPLYING CRITICAL THINKING 4.2: Consider Different Perspectives

The net realizable value of accounts receivable (accounts receivable minus the allowance for uncollectible accounts) is reported on the balance sheet as a current asset. The bad debts expense reduces net income reported on the income statement. Keep in mind that parties with very different perspectives use this information for decision-making (**Stakeholders**):

- Creditors, who want to be repaid, are interested in the liquidity of the assets.
- Shareholders, who predict future stock values as well as dividend expectations, must understand the expected future cash inflows from operations.
- Managers want to earn bonuses that are determined by growth in net income and total assets. The change in the allowance for uncollectible accounts will influence both amounts.

There are a few data strategy alternatives that could be considered either individually or in combination for this project:

- All credit sales invoices and collections data that pertain to the accounting period.
- All outstanding invoices included in the year-end accounts receivable balance.
- The prior year-end's adjusted account balance and the current year-end's unadjusted account balance in the allowance for uncollectible accounts.
- Current year's total accounts receivable write-offs.
- Authorized bad debt estimation percentages for accounts receivable age categories.

Recall that the objective question for this analysis project asks for a reasonable estimate of the year-end uncollectible accounts receivables. Because WeMakeIt, Inc. uses the GAAP percentage of ending accounts receivables method with the three age categories typical for manufacturing companies, the data that would be most useful considering the objectives would include:

- The outstanding invoices in the year-end accounts receivable balance.
- This year's unadjusted allowance for uncollectible account balance.
- Authorized bad debt estimation percentages for each accounts receivable age group.

APPLYING CRITICAL THINKING 4.3: Choose the Data Strategy that Fits Your Objective

Both data and analysis strategies are improved when they are developed, evaluated, and ranked by the factor values placed on each potential data strategy:

- Factors that value data relevance, such as expected future collections, prior estimation errors, and expected risk of uncollectibility should be used to select the best data strategy for WeMakeIt, Inc.'s analysis plan (**Alternatives**).
- Using the data from outstanding invoices, the unadjusted allowance for uncollectible accounts receivable, and authorized bad debt estimation percentages was the best choice for WeMakeIt, Inc.'s analysis. These data choices best match the objective of the analysis, which is to calculate a reasonable estimate for the year-end uncollectible accounts receivable (**Purpose**).
- For a personal example, if your objective was to dress successfully for an interview, then you could develop, evaluate, and rank the alternatives of wearing something you already own, borrowing from a friend, or buying something new to wear to the interview. Using clothes you already own would rank higher than buying new clothes if you were ranking options based on lowest costs and time factors.

Step 3: Select the Analysis Strategy

For any project plan, choosing the best analysis strategy involves considering and evaluating several possible alternative analyses given the objective questions and the already selected strategy for the data. The analysis alternatives in this bad debt estimation example involve different methods for determining the age of an outstanding invoice:

- Days outstanding from the invoice date forward.
- Days outstanding from the due date forward.

The first analysis alternative is preferable because WeMakeIt, Inc.'s policy is to allow customers 30 days to pay their invoices and because they estimate bad debts only when an invoice is past its due date, which is equal to or more than 30 days outstanding.

Therefore, the first question, what is a reasonable estimate of the uncollectible receivables in the 2025 year-end outstanding accounts receivables, will be answered by summing



each age group’s estimated bad debts value. This is a measure estimating the uncollectible portion of the year-end accounts receivable balance (**Illustration 4.2**). (**Data** See How To 4.1 at the end of the chapter to learn how to calculate the figures in this illustration.)

ILLUSTRATION 4.2 WeMakelt, Inc.: Estimated Value of Uncollectible Receivables

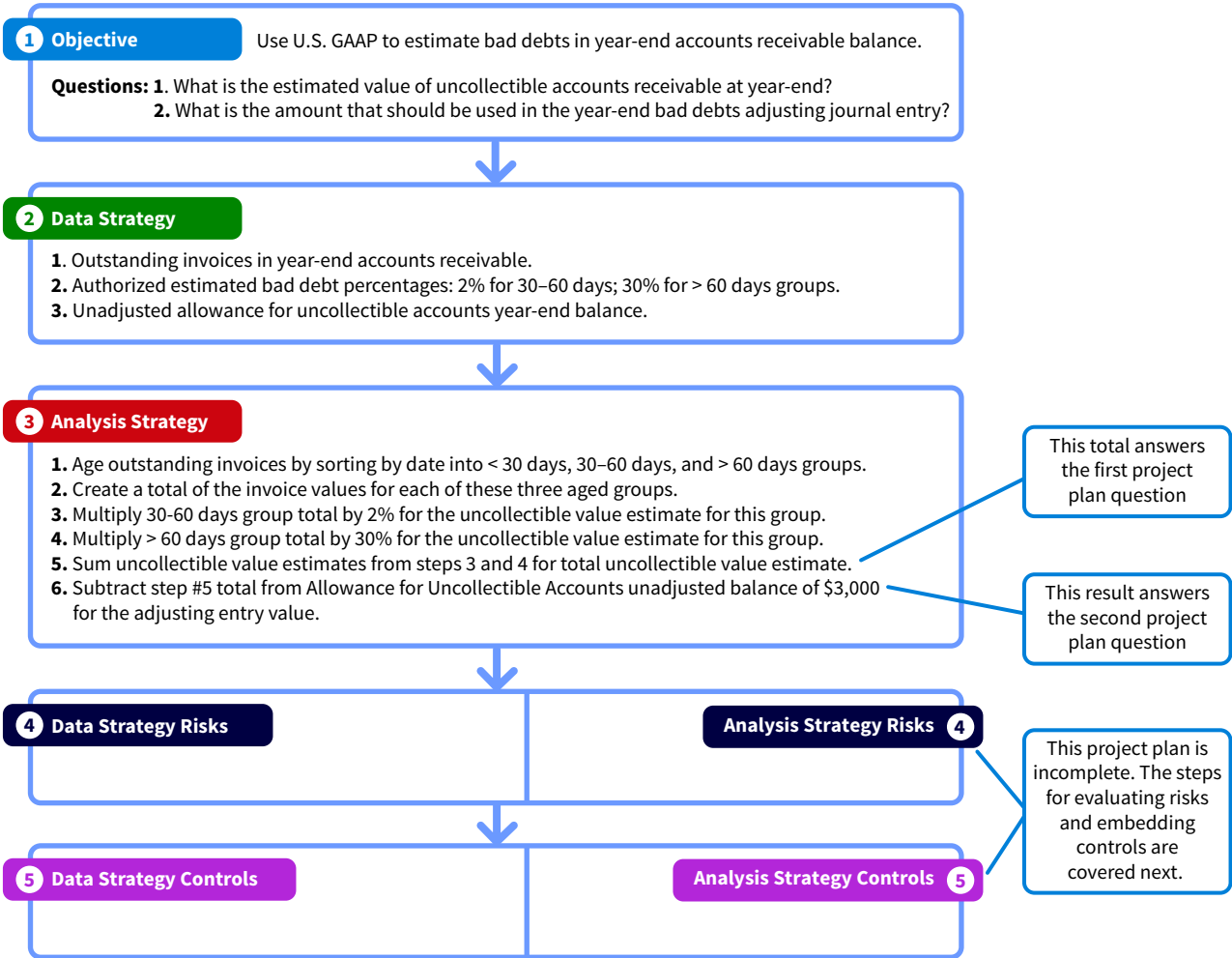
	< 30 days	30–60 days	> 60 days	Total
Total accounts receivable value	\$ 30,099.29	\$ 20,441.46	\$ 33,193.55	\$ 83,734.30
x % uncollectible	0%	2%	30%	
\$ Estimated as uncollectible	\$ 0	\$ 408.83	\$ 9,958.07	\$ 10,366.89

The second question, what value should be used in the adjusting journal entry at year-end for bad debts expense, will be answered when this total of \$ 10,366.89 is reduced by the unadjusted year-end allowance for uncollectible accounts balance (\$ 3,000). The resulting calculated value is the year-end adjusting journal entry debiting bad debt expense and crediting allowance for uncollectible accounts:

	Debit	Credit
Bad debts expense	\$ 7,366.89	
Allowance for uncollectible accounts		\$ 7,366.89

Increasing the allowance account’s \$3,000 unadjusted balance by \$ 7,366.89 will result in an adjusted balance of \$ 10,366.89, equal to the estimated uncollectible amounts in accounts receivable at the 2025 year end. **Illustration 4.3** summarizes the first three steps of this project’s plan.

ILLUSTRATION 4.3 In-Progress Project Plan for Estimating Year-End Bad Debts: Steps 1-3



The last two steps are considering risks and controls for both the data strategy and the analysis strategy. First we will evaluate the data strategy's risks and controls, and then the analysis strategy risks and controls.

Steps 4 and 5: Data Strategy Risks and Controls

While organizations invest in processes and controls designed to capture accounting data correctly, accounting databases inevitably have some data issues. In this example, the data include the 2025 year-end accounts receivable outstanding invoices and the authorized bad debt estimation percentages. There could be some risks to using this accounts receivable data:

- Missing or incorrectly included invoices in the general or the subsidiary ledger.
- Errors in the invoice due date fields or the invoice total amounts.
- Errors created when the data are exported from the AIS system into the analysis tool.

APPLYING CRITICAL THINKING 4.4: Threats to Data Strategy

Data risks can impact the accuracy and validity of analysis results. These risks must be evaluated so appropriate controls can be added to the project plan. Data strategy risks can include **(Risks)**:

- Dirty data such as incomplete data, inaccurate data, and incorrectly formatted data fields.
- Data irrelevant to the objective of the project.
- Insufficient data for the analysis to be reliable.
- Data that is an unrepresentative sample of the underlying population.
- Errors in management estimates and assumptions about the data.

These data risks can be controlled by comparing the extracted data to the source invoice and collection documents. Possible risks involving the management's estimates of bad debt percentages include human bias embedded in the authorized bad debt percentages, changes in customer payment behavior, and business process changes to credit customer approvals and collection practices.



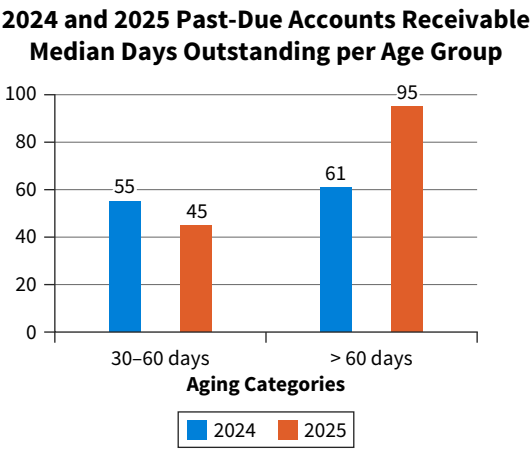
One way to control for these risks is to ask the finance, sales, and accounts receivable managers if there were changes to the customer base, market conditions, or business policies and procedures for credit approval, write-offs, or collection policies during the year. This knowledge could confirm existing bad debt percentages or motivate their adjustment.

Another control for evaluating the risks in management estimates and assumptions is to compare current data to prior year data. Evaluating increases in the median number of days outstanding and the number of new to returning customers in each age group can offer insights into the reasonableness of bad debt estimation percentages.

To illustrate, imagine you were wondering if the 30% bad debt percentage for invoices greater than 60 days outstanding was a reasonable management assumption in 2025.

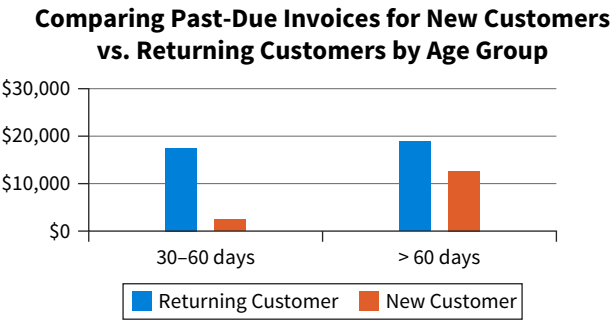
Illustration 4.4 demonstrates that the median number of days outstanding in the greater than 60 days outstanding group increased in 2025.

ILLUSTRATION 4.4 Data Risk in Bad Debt Percentages: Median Days Outstanding



Additionally, you might wonder if this increase was caused by new customers, as the payment habits of new customers are more uncertain than those of continuing customers. **Illustration 4.5** shows that the > 60 days age group has many outstanding invoices from new customers.

ILLUSTRATION 4.5 Data Risk in Bad Debt Percentages: Risks from New Customers



The results of these control tests indicate that if the 2024 bad debt percentage for this age group was used in 2025, it may not accurately measure the bad debts expected in the 2025 accounts receivable balance. These visualizations show that implementing controls, such as data reasonableness tests, before performing an analysis can result in better business and professional intelligence.

Steps 4 and 5: Analysis Strategy Risks and Controls

Common analysis risks include errors in data manipulation, calculation, or using technology, as well as human biases and ethical violations when using sensitive data. Here are some examples of analyses risks when estimating bad debts:

- Errors in age group categorization and age total calculations.
- Incorrect application of bad debt estimation percentages to age group totals.
- Erroneous summing of estimated bad debts across age groups.
- Mistakes in the calculation of the value to use for bad debts expense.

Verifying the accuracy of the age group categorizations and each sum and multiplication performed on age group data items and totals would control for these risks.

Another control is to use common sense when reviewing the results for reasonableness. **Illustration 4.6** uses an example from the 30–60 days age group estimation to show how a common sense check helps catch large errors. The total outstanding accounts receivable for that age group was \$ 20,441.46, and the uncollectible percentage for that group was 2%. The

estimated uncollectible value for that age group was calculated as the product of those two numbers, resulting in \$ 408.83. This control has verified our calculation.

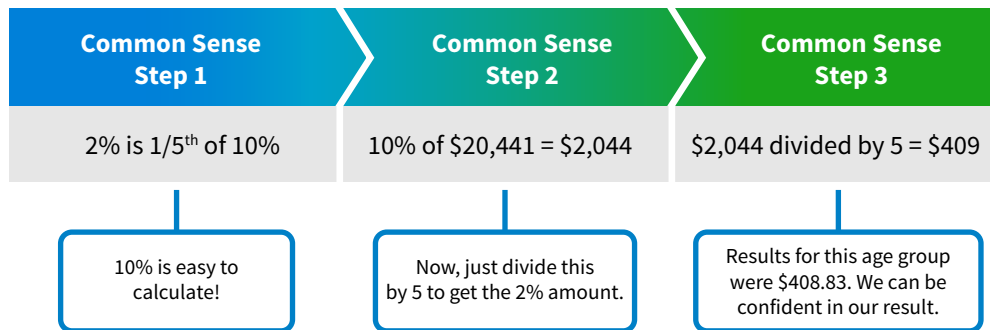
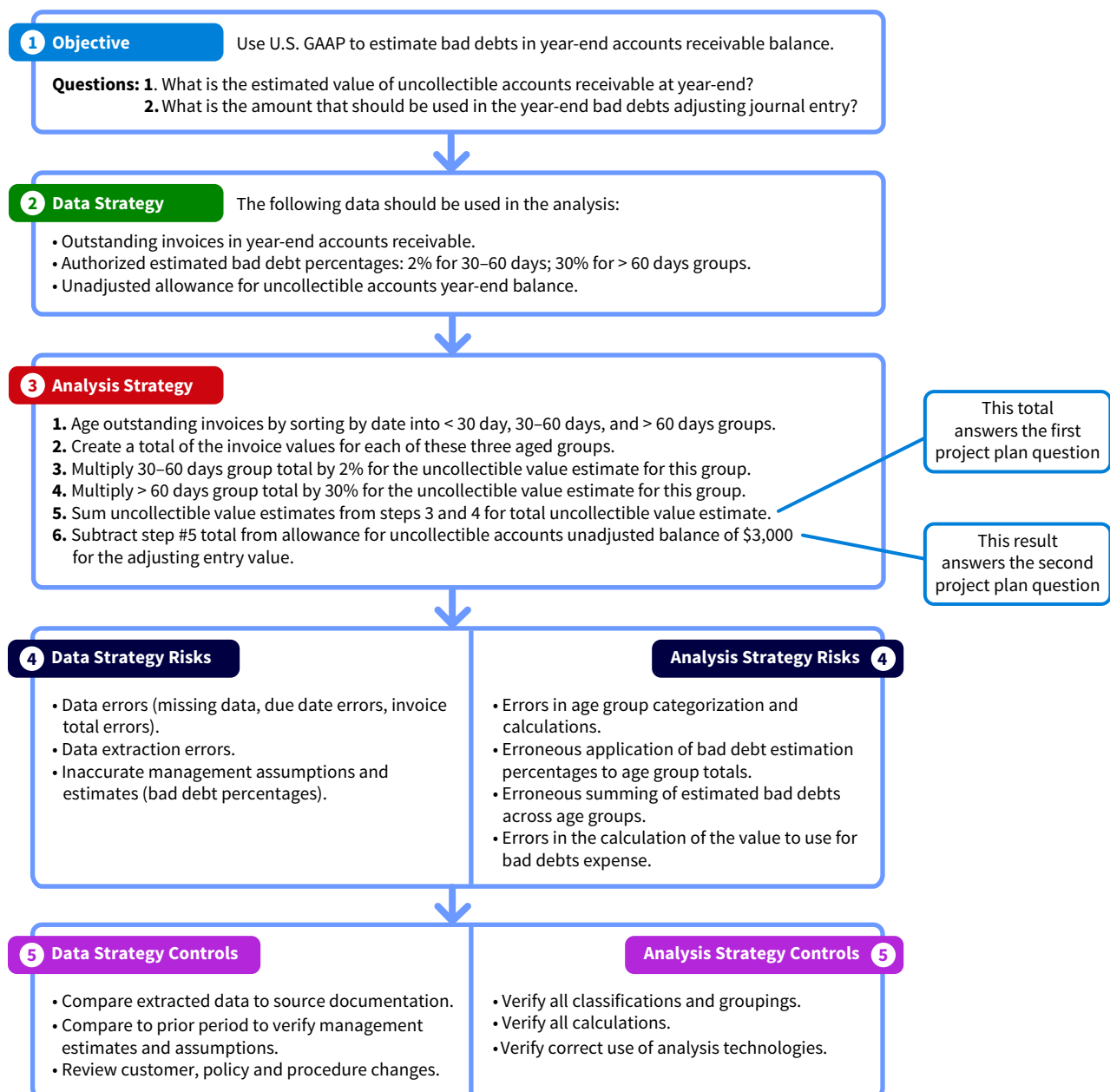


ILLUSTRATION 4.6 Common Sense Check for the 30–60 Days Group Results

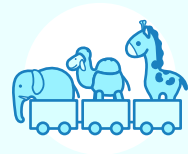
Analysis results that significantly vary from common sense checks should be verified. The project plan design for this example is now complete (**Illustration 4.7**).

ILLUSTRATION 4.7 Complete Project Plan for Estimating Bad Debts



Next, we go into more depth about how to develop both data and analysis strategies.

Apply It 4.1
Build a Project Plan
for Inventory Costs



Financial Accounting Founded by Elisabeth Hess and based in Wisconsin, Tremendous Toys produces and sells unique animal-form toys online. The company has over a million dollars in sales revenues and a strong national and international following. The investment bank underwriting its upcoming initial public offering requires that Elisabeth produce a U.S. GAAP valuation of year-end inventory.

Elisabeth has kept meticulous records, organized in business cost categories, which are listed here. She knows which materials were used to create each toy, which employee built it, and how long each took to create. She has also recorded the number and type of toys produced every month. However, she has never assigned costs to her inventory items.

Materials	Labor	Overhead
Plastic for 3D printers	Production workers' time	3D printer costs
Wood	Production workers' pay rate	Production tools costs
Resin		Computer (peripherals and internet) costs
Fiber and fabric		Shipping costs
Paints, varnishes, and dyes		Factory rent, utilities, and storage
Toy clothing		Production supplies
Glues and epoxies		Tax and legal fees
Bling and other accessories		Donations to charities
Eyes and body parts		Advertising costs

Prepare a data analysis project plan to help Elisabeth develop her inventoriable costs. Use the following structure for your answer.

1. Project objective and specific question
2. Data strategy
3. Analysis strategy
4. Data and analysis strategy risks
5. Data and analysis strategy controls

SOLUTION	
1. Project objective	Prepare a U.S. GAAP compliant full costing valuation for Tremendous Toys' current inventory.
Specific question	What is the U.S. GAAP year-end value of inventory?
2. Data strategy	• Collect all costs and raw measures of consumption of resources related to production (direct materials, direct labor, and manufacturing overhead costs) for the last three years. • Collect records of how many units were produced for each month of cost data received. • Take a physical count of the current inventory by product.

(Continued)

3. Analysis strategy	• Aggregate cost data by monthly production of figurines. • Estimate cost behavior using regression to find variable costs per figurine and total fixed production costs per month. • Apply results to year-end inventory.
4. Risks to data strategy	• Missing data. • Including personal expenses. • Material costs and worker costs may not be representative of true replacement costs of future material and labor costs. • Potential bias for minimizing cost information to make business seem more profitable.
Risks to analysis strategy	• Averaging costs across figurines may not be accurate if true costs of different figurines vary significantly. • Regression may not have strong explanatory power (<i>R</i>-square). • Costs may have seasonality differences (due to scarcity of materials or heating and cooling costs). • Cost formula may need to be recalculated annually to capture cost changes.
5. Data strategy controls	• Interview Elisabeth with a typical list of inventory costs to catch omitted costs. • Inquire about the seasonal or economic trends of materials, labor and production overhead costs, and use the average of the highest and lowest points.
Analysis strategy controls	• Use median tendencies rather than means. • Run sensitivity models with future replacement cost values to note potential risk in final product cost estimations and possible data bias.

4.2 What Should We Consider When Selecting Data for Analysis?

LEARNING OBJECTIVE 2

Describe how to develop a data strategy.

It's easier to make better decisions when we have the necessary information. Each semester, students who know their educational and career goals can select the courses that will prepare them to graduate and begin their careers. For example, accounting majors do not need to take the same courses as education majors. The same idea applies when developing a strategy for selecting data in data analysis. A successful data analysis project hinges on selecting data that are relevant and appropriate for the objective of the project, respecting the data's characteristics and measurement scales, and controlling for inherent data risks.

Identify Appropriate Data

Data can be considered appropriate for analysis when they are relevant, available, and the characteristics match the analysis method requirements. Appropriate data can be internal, external, or a combination of both:

- Internal data are generated within the organization, such as sales data, purchase data, inventory data, customer data, and vendor data. Internal data can typically be more easily controlled and verified by an organization.
- External data are obtained from sources outside of an organization. This data can include weather data, geographic data, and publicly available competitor data. External data are somewhat riskier to use since we often cannot know if the data are accurate or complete. External data can, however, provide insights that internal data alone cannot provide.

After identifying the available and relevant data alternatives, the characteristics of the possible data sets need to be verified as suitable for the planned analysis.

Evaluate Data Fields and Sources

A **data set** is a collection of data columns and rows available for analysis. Understanding the characteristics of a data set is important because, for example, statistical measures and tests often require certain data characteristics or a minimum of data points. Violating the data requirements for these measures and tests can threaten the accuracy, reliability, and significance of the analysis results.

Let's use an example to evaluate the impact data can have on analysis. **Illustration 4.8** shows the inventory data set for the 3D manufacturing company WeMakeIt, Inc.

ILLUSTRATION 4.8 Typical Elements in a Data Set from WeMakeIt, Inc.'s Inventory

The data set is the entire collection of rows and columns.

Column = Data field = Attribute

InventoryCode	InventoryDescription	ProductCategory	UnitCost	NumberOnHand
358500	Medical model foot	1	\$ 30.40	32
358600	Medical model hand	1	\$ 30.20	16
358900	Shop Lamp globe	2	\$ 20.05	43
359000	Cord wraps	2	\$ 5.00	80
361100	Superhero Fantastic Four Mister Fantastic	3	\$ 36.00	30
361200	Superhero Fantastic Four Invisible Woman	3	\$ 36.00	35
361300	Superhero Fantastic Four Thing	3	\$ 36.00	25
361400	Superhero Fantastic Four Human Touch	3	\$ 36.00	30

Row = Record

The individual columns in a data set are called **fields**, and if the source of the data was a database, the columns are called **attributes**. Each column describes and represents one unique characteristic, a description, or aspect of the phenomena captured in the data set. Rows in a data set from a database are **records**, which represent the collection of columns that hold the descriptions of a single occurrence of the data set's purpose. In Illustration 4.8, the highlighted row contains all the information about the inventory item Superhero Fantastic Four Mister Fantastic—inventory code, the description, product category, unit cost, and the number of items on hand. This inventory item belongs to the ProductCategory group #3, indicating that it is a superhero figurine.

The records, or rows, in this inventory data set represent the different products manufactured in each product line. The columns that describe each inventory product for the WeMakeIt, Inc. inventory data set are listed in **Illustration 4.9**.

ILLUSTRATION 4.9 Data Dictionary for the WeMakeIt, Inc. Inventory Data Set

Field Name	Description
InventoryCode	Unique numerical code to identify inventory (primary key)
InventoryDescription	Text that describes the inventory product
ProductCategory	The type of product (1 = medical, 2 = tool, 3 = toy)
UnitCost	Currency value assigned to each unit's cost
NumberOnHand	Quantity of items in inventory

In addition to understanding the content of data fields in accounting databases, considering the source of the data is important because the quality of the data in the fields impact the quality of the analysis. Were the data created by an external party on a web page, entered internally by an employee, or automatically assigned by the computer? Do the data refer to a category, a measurement, or a calculation? For example, in the WeMakeIt, Inc. inventory data in Illustration 4.9, the InventoryCode field's data are probably automatic sequential numbers assigned by the AIS system whenever a new inventory product is added. Since a human is not creating that field's value when a new row is added, we can be confident that the InventoryCode data are consistently correct. **Illustration 4.10** summarizes the typical internal data field sources often used by accountants.

ILLUSTRATION 4.10 Common Data Field Sources and Examples

Source	Description	WeMakeIt Inc. Examples
Raw Data Fields		
Measured Raw Data	- Data created or captured by a controlled process capturing the value of the data. Examples include price, cost, number on hand, weight, date, hours worked, temperature, sensor readings, human observation, or economic value. - Their format can be discrete or continuous data.	InventoryCost NumberOnHand UnitCost
Non-measured Raw Data	- Data often created automatically by the computer or company policy for control. Examples include identification codes, chart of account numbers, standard descriptions, product category codes, or location codes such as city. - These fields are typically formatted as discrete data.	InventoryCode ProductCategory
Calculated Data Fields		
Calculated Data	- Data created when one or more fields in a particular record (row) have any number of mathematical operators (such as +, −, ×, %) applied, and often are derived from using the data in another field or fields within the same record (never across different rows). - These fields can be formatted as discrete or continuous data.	TotalCost (=UnitCost × NumberOnHand)

In the WeMakeIt, Inc. data set example, both the InventoryCode and ProductCategory fields' data are **non-measured raw data** that have been formatted as numeric and discrete, meaning that they are non-continuous and will never have partial unit values. These fields cannot be used in analysis for mathematical calculations because they refer to unique identifiers for inventory items and their product category group. The InventoryDescription field's data source is also a non-measured raw data field formatted as a text field, which cannot be used in numeric calculations during the analysis.

The sources for data fields like UnitCost and NumberOnHand are different because they are **measured raw numeric data**, which can be used in mathematical calculations.

Finally, UnitCost and NumberOnHand can be multiplied in an expression to calculate a new field within a query called TotalCost. This new field source is **calculated data** that is also numeric and likely to be used in a variety of different mathematical calculations in accounting.

A data field’s **measurement scale** refers to the type of information provided by the data. Data measurement scales should be considered when designing the data strategy, as they impact which analyses can be performed on the data. Data measurement scales are grouped into four categories: categorical, ordinal, interval, or ratio (**Illustration 4.11**).

ILLUSTRATION 4.11 Measurement Scale Descriptions and Examples

Measurement Scale Classification	Description	Example
Categorical (nominal) data	Labeled or named data that can be sorted into groups according to specific characteristics. The data do not have a quantitative value.	Regions, product type, account numbers, account category, locations.
Ordinal data	Ordered or ranked categorical data. Distance between the categories does not need to be known or equal.	A survey question inquiring about customer service may ask customers to rate their experience from 1 to 10.
Interval data	Ordinal data that have equal and constant differences between observations and an arbitrary zero point.	Temperature, time, credit scores.
Ratio data	Interval data with a natural zero point. A natural zero point means that it is not arbitrary.	Economic data, such as dollars or euros.

We discuss how data measurement scales influence analysis strategy options later in this chapter. Next, let’s examine the risks and controls related to our data strategy.

Professional Insight 4.2: How Do You Select Financial Data for Analysis?

Jadon is an accounting and finance major, student athlete, and president of the first-generation student organization on campus. During his internship, he realized how much his career would involve data analysis of financial statement information.

On the first day of my summer internship, I was tasked with calculating ten financial ratios for five companies in the same industry for the most recent three years. I was left to figure out how to get the data and perform the calculations on my own. I reached into my sports experience to gather my courage by creating a game plan. **First, I had to select and set up my analysis tool.**

I chose Excel with the Analysis ToolPak and made a raw data template based on the totals I would need for the ratios in the first worksheet. I copied it five times for each company’s raw data. I made another worksheet for all the ratio calculations as the rows, and the companies would be the columns, grouped by each year. I organized the ratios into profitability, liquidity, and solvency groups. I entered ratio formulas that tied back to the raw data worksheet cells and documented my steps on another sheet to remember my choices.

Next, I had to find out where to get the data. I went to the SEC’s xBRL data and downloaded the financial statement information and entered what I needed into Excel. An unexpected benefit of having my ratios all in one sheet was that it was easy to make visualizations of my results. I was amazed by how clear the best and worst performing companies were from these visualizations. I saved my file and emailed it to my boss.

Consider Data Strategy Risks and Implement Controls

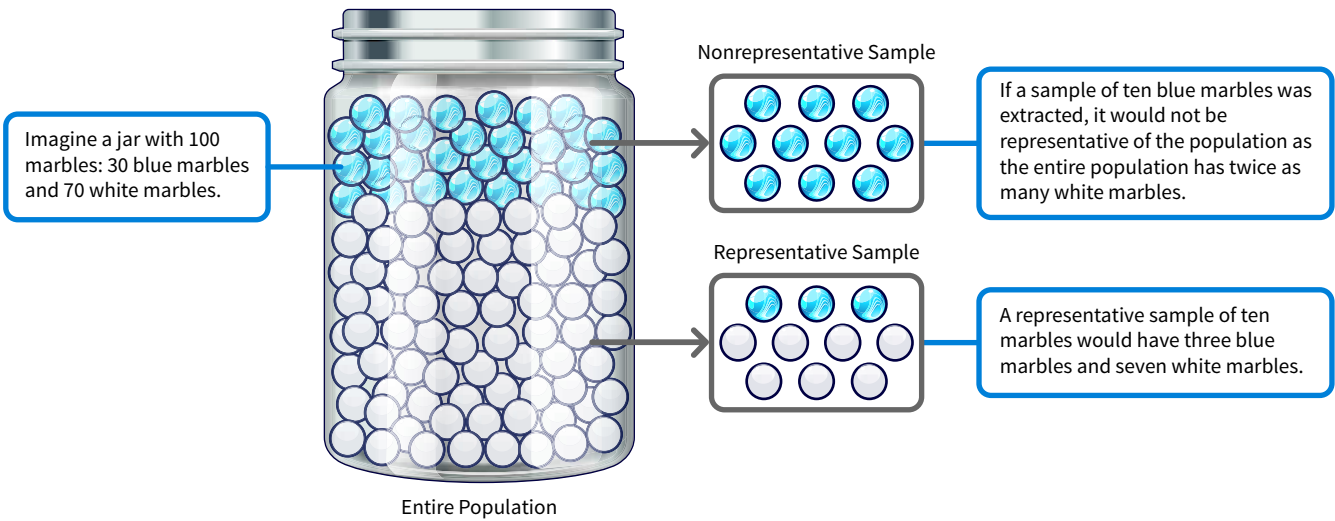
Another benefit of documenting data strategy in a project plan is that examining each choice can help identify three common data risks that might affect an analysis’s validity, accuracy, and reliability (Illustration 4.12).

ILLUSTRATION 4.12 Common Data Risks and Recommended Controls

Data	Risks	Controls
	Non-representative sample selections	Verify representativeness of sample.
	Outlier data points	Perform a histogram or quartile analysis to identify outliers, then explain or justify the rule used for outlier adjustment or removal.
	Dirty data	Verify integrity of data set and clean up dirty data issues.

The first risk is that a sample extracted from a larger population of data are a poor representation of the underlying population. For example, a random selection of different colored marbles might not be a sample that accurately represents the characteristics of the marble population (Illustration 4.13).

ILLUSTRATION 4.13 Representative Sampling Risks



That is why performing tests on the sample’s representativeness can either verify its validity or detect its significant weakness in representing the underlying marble populations.

The second risk is potentially including unusual data points in an analysis. Outlier data points are unusual data points compared to the rest of the data in either an activity point (x-axis or the independent variable), such as number of units produced, or an unusual level of the economic value (y-axis or the dependent variable), such as total costs or total sales. The best control for identifying outliers is to visualize the data in a graph to check if the measure

is very different than the rest of the data. Identified outliers can either be remeasured (if possible) or removed with a logical and documented rule.

The final data risk is that a variety of data errors, or **dirty data**, are problematic for data analysis. Dirty data include missing, invalid, duplicate, and incorrectly formatted data. For example, the purchase orders, invoices, or checks written by the company should be sequentially accounted for, with no missing numbers or two with the same identifying number. Controls should always test for dirty data before the analysis is performed. Data can be compared to its source documents, or if that is not possible, tested for reasonableness. Testing for reasonableness can include checking for missing sequence numbers, duplicate numbers, or verifying the data have the expected format or acceptable characters.

Once data have been selected, evaluated, and any risks identified and controlled, the next step is to develop an analysis strategy appropriate for the objective of the project.

Apply It 4.2

Identify Data Characteristics



Accounting Information Systems Tremendous Toys has recently developed a database to capture inventoriable costs from their production each week.

Data has been extracted from the following data fields:

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber		
ProductionRunDate		
NumberOfToysinRun		
ToyIDNumber		
DirectMaterialsUsed		
DirectLaborUsed		
OverheadCostApplied		
TotalRunCost		
TotalUnitCost		

Complete the three-column table by adding each data field's type and measurement scale.

SOLUTION

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber	Non-measured raw data	Categorical (Nominal)
ProductionRunDate	Non-measured raw data	Interval
NumberOfToysinRun	Measured raw data	Ratio
ToyIDNumber	Non-measured raw data	Categorical (Nominal)
DirectMaterialsUsed	Measured raw data	Ratio
DirectLaborUsed	Measured raw data	Ratio
OverheadCostApplied	Calculated data	Ratio
TotalRunCost	Calculated data	Ratio
TotalUnitCost	Calculated data	Ratio

4.3 What Should We Consider When Selecting an Analysis?

LEARNING OBJECTIVE 3

Explain how an analysis strategy is designed.

Consider these two questions when designing a data analysis strategy:

1. Can the chosen analysis strategy answer the specific objective questions?
2. Is the measurement scale of the data appropriate for the selected analysis strategy?

Earlier in this course you learned there are four types of data analysis objectives: descriptive, diagnostic, predictive, and prescriptive. Each asks different questions of the data. We perform analyses to answer those questions. The specific analysis that can be performed on the data is influenced by the data's measurement scales. Next, we summarize what to consider when designing analyses for projects with descriptive and diagnostic objectives.

Designing Analyses to Describe and Diagnose

Because understanding the phenomena captured by the data is essential, many data analysis projects begin with descriptive analyses. Diagnostic analytics involve many of the same analyses as descriptive, but the focus is on explaining why something happened. Because many types of analyses can be both descriptive and diagnostic, it makes sense to discuss these two analysis strategies together.

The specific type of descriptive and diagnostic analysis used depends on the measurement scale of the data. For example, calculating the median observation in the data requires using either interval or ratio measures. **Illustration 4.14** shows some analysis strategies for descriptive and diagnostic objectives (checkmarks indicate appropriate analyses, and Xs indicate analyses that are inappropriate for the measurement scale).

ILLUSTRATION 4.14 Analyses Strategies for Descriptive and Diagnostic Analytics

Potential Analyses	Data Measurement Scale			
	Categorical	Ordinal	Interval	Ratio
Central Tendency				
Mode	✓	✓	✓	✓
Median	✗	✗	✓	✓
Mean	✗	✗	✓	✓
Frequency Distributions				
Both counts and %s:				
Group frequency	✓	✓	✓	✓
Cumulative frequency	✓	✓	✓	✓
Relative frequency	✓	✓	✓	✓

(Continued)

ILLUSTRATION 4.14 (Continued)

Potential Analyses	Data Measurement Scale			
	Categorical	Ordinal	Interval	Ratio
Data Dispersion				
Minimum, maximum, range, quartiles	✗	✓	✓	✓
Variance and standard deviation	✗	✗	✓	✓
Normality, skewness, and kurtosis tests	✗	✗	✓	✓
Visualizations				
Only bar and pie chart	✓	✓	✓	✓
Many other trends options	✗	✗	✓	✓
Histograms/box plots	✗	✓	✓	✓
Correlation				
Scatterplots, Pearson's R	✗	✓	✓	✓
Cross tabulation analysis	✓	✓	✓	✓
Calculations				
Totals and subtotals	✗	✗	✓	✓
Percent change	✗	✗	✓	✓
Percent of total	✗	✗	✓	✓
Regression				
	✗	✗	✓	✓

Descriptive Analysis Strategies

Recall that accounting data are historic economic transaction and valuation data. Accountants, business managers, and regulators all want to know more about what happened and whether the data signals good or concerning news, so descriptive analyses are the most common analysis strategies using accounting data. It is useful for evaluating strategy performance because it provides more meaning and business intelligence than from just looking, for example, at the totals reported on one year's financial statements.

Let's work through an example using the WeMakeIt Inc. accounts receivable data set to illustrate the explanatory power of descriptive analysis even when using categorical data. Recall that the objective for this analysis was to become more confident in the percentages used for estimating bad debts. The initial descriptive analysis results indicated a large increase in the number of customers with open invoices at the 2025 year end compared to the 2024 year end. This analysis had been designed to answer two descriptive questions:

- **Objective question 1:** How many customers had outstanding invoices at the year end of 2024?
- **Objective question 2:** How many customers had outstanding invoices at the end of 2025?

A data strategy and an analysis strategy were developed:

- **Data strategy:** Use the categorical variable of CustomerNumber to identify customers with outstanding balances.
- **Analysis strategy:** Count and compare the frequency of unique customer numbers with outstanding invoices at the year-end of 2024, and then repeat the analysis for the 2025 year-end. Note that both count and frequency are analyses that can be performed on categorical data.

Illustration 4.15 reports the results of this analysis.

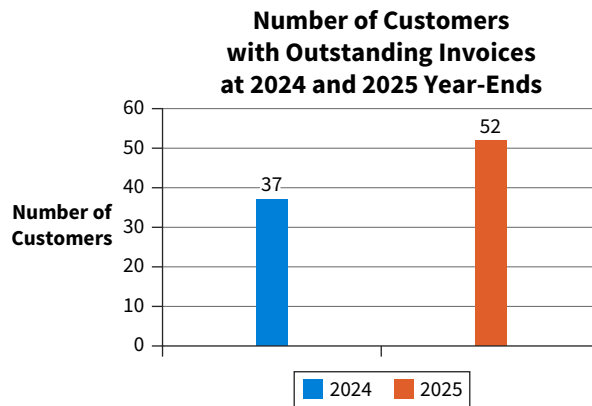


ILLUSTRATION 4.15

Outstanding Invoices by Customer Number

The results indicated that in 2024 there were 37 unique customer numbers with outstanding invoices, and in 2025 there were 52 unique customers numbers with outstanding invoices. Next, we discuss diagnostic analysis strategies.

APPLYING CRITICAL THINKING 4.5: Be Careful Interpreting Initial Results

You might initially conclude that accounts receivable in 2025 has the potential for higher collectability risk simply because there were more customers with outstanding invoices in 2025 than in 2024. However, this initial judgment may be affected by internal assumptions and biases

(Risks):

- The 2025 increase in the number of customers with outstanding balances may not be a negative result, as the company grew and added new customers in 2025.
- These results may motivate you to evaluate alternative explanations and perform new analyses, such as determining if the 2025 customer numbers with outstanding invoices at year end had more invoices owed in 2025 than in 2024, or if the value of their outstanding balances were higher in 2025, possibly signaling a cash liquidity concern. This result is more closely relevant to our company's accounts receivable risks of uncollectability **(Alternatives)**.

Diagnostic Analysis Strategies

Diagnostic analysis strategies can be compared to detective work. Accountants can use these analysis strategies to identify and discover the most likely causes of accounting phenomena. While both descriptive and diagnostic analysis strategies are descriptive techniques, diagnostic analysis strategies also identify what factors have caused the lower sales revenues or the higher costs, for example. Diagnostic strategies always require applying critical thinking skills.

Stakeholders rely on accountants across accounting practice areas to diagnose what is happening and explain, typically in non-accounting terms, why it is happening. For instance, systems accountants often use error dispersion data to identify system performance issues or unusual activity data that may indicate unauthorized party access.

Let's use the WeMakeIt Inc. accounts receivable data set again, this time to illustrate diagnostic analysis:

- **New objective question:** Do the returning credit customers have more outstanding invoices at the 2025 year-end than they had outstanding at the 2024 year-end?
- **Data strategy:** Once again, use the data set for the 2024 and 2025 year-end accounts receivable open invoices.
- **Analysis strategy:** Group the data by CustomerNumber for this question.

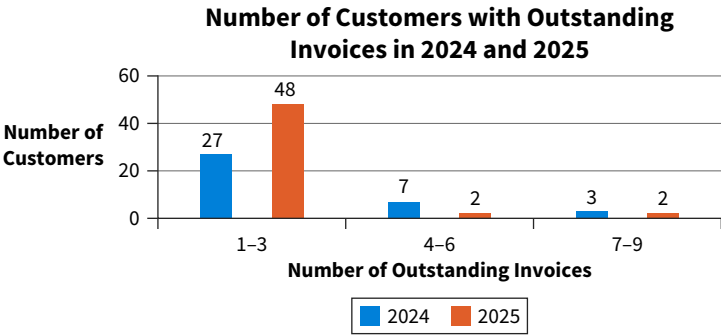
To answer this question, we may choose to create a new, calculated ordinal variable (recall that a **variable** is a data item that will be used in the analysis). This variable will group credit customers into increasing ranges based on the number of invoices they have outstanding at year-end. There are some benefits to this strategy:

- Changing a continuous variable into ordinal ranges results in meaningful groupings rather than a simple count of the number of invoices outstanding per customer.
- The new ordinal variable allows a better understanding of the overall results of the number of outstanding invoices in total.



Let’s create three levels for outstanding invoices: ranges of 1-3 invoices, 4-6 invoices, and 7-9 invoices as the new ordinal variable levels. **Illustration 4.16** provides a visualization of the results of this analysis strategy. (**Data** How To 4.2 describes how to recreate this visualization with a different tool.)

ILLUSTRATION 4.16 Frequency of Customers with Outstanding Balances



The results of this second analysis show that the number of customers with 4-6 and 7-9 outstanding invoices had decreased in 2025 compared to 2024. However, there are almost twice as many customers with 1-3 past due invoices in 2025 compared to 2024. These results are mixed news for concerns about future accounts receivable collectability risks.

This surprise finding is a good reminder about the importance of critical thinking and deliberation about the initial analysis results. Thinking critically when interpreting analysis results involves evaluating both the analysis strategies performed and not performed. Sceptically questioning data analysis strategies can provide valuable insights, as this final analysis did.

APPLYING CRITICAL THINKING 4.6: Controls for Risks to Analysis Strategies

Depending on the objective of the project, analysis strategies can involve a diverse mix of variables with different sources and measurement scales. Simply adding variables and complexities to analysis choices without justification can lead to information overload, which can impair your judgement. For example, sales forecasts might involve fewer variables in industries such as agriculture, as food demand is more stable, than for airlines, as travel is more discretionary. Professional risk-reducing habits include (**Risks**):

- Keeping analysis as simple as possible by including the fewest yet most relevant variables.
- Documenting each step, especially the benefits and drawbacks of each choice as you develop, evaluate, and choose an analysis strategy (**Alternatives**).

The process for analyzing WeMakeIt, Inc.’s accounts receivable, whether descriptive or diagnostic, involves an approach that can be used for most analysis strategies in accounting:

- Start with a clear objective and specific questions that should be answered.

- Develop data alternatives, evaluate each on factors related to the objective, and select the best data strategy. Filter the data set down to the variables and ranges of interest necessary for the objective.
- The best analysis strategies typically create new calculated within-row variables necessary for the analysis. These new calculated variables can be categorical groupings, ordinal rankings, interval measures, and ratio variables. Remember that relevant groupings can make it easier to interpret results.
- Calculate any vertical analysis measures necessary, such as totals, frequencies, averages, and measures of dispersion across the rows in the data set.
- Finally, the analysis variables can be correlated or sliced by dimension variables to add insights from disaggregating the vertical measure total.

Descriptive and diagnostic analyses help to better understand what happened and why it happened. What if the objective is to predict a future outcome or determine what strategies might achieve a specific outcome? In those cases, we need to use predictive or prescriptive analyses, which are explained next.

Designing Analyses to Predict and Prescribe

Predictive analysis strategies use historical data to create models that estimate a future value or outcome. Prescriptive analysis strategies, on the other hand, help determine which option is most likely to produce the best outcome given the objective. Predictive and prescriptive analysis strategies can determine the best use of resources, improve performance, anticipate market reactions and movements, and avoid process breakdowns:

- Accountants commonly use these analysis strategies to evaluate tax planning options, operational and marketing strategy options, financing options, investment options, and exit strategy options.
- Industries with these analysis objectives include lending institutions that want to avoid bad credit decisions and insurance companies that must anticipate natural disasters and disease onset so they can adjust business models before crises that cause their businesses to suffer happen.

Illustration 4.17 summarizes the most common predictive and prescriptive analysis strategies used by accountants that do not involve advanced technology tools. The green checkmarks indicate appropriate analyses, and the red x's indicate inappropriate analyses. As the illustration shows, there are not many predictive or prescriptive models for categorical and ordinal scale variables unless they are used in models in which interval or ratio scale variables are desired for the prediction.

ILLUSTRATION 4.17 Analyses Strategies for Predictive and Prescriptive Analytics

Potential Analyses	Data Measurement Scale			
	Categorical	Ordinal	Interval	Ratio
Trendlines	X	X	✓	✓
Regression analysis (linear)	✓	X	✓	✓
Optimization models	X	X	✓	✓
What-if analyses	X	X	✓	✓

X = inappropriate analysis for the scale

Both predictive and prescriptive analysis strategies often use statistics and sophisticated modeling techniques for their algorithms:

- Common tools for this include statistical regression, time series analyses, process mining and mapping, text analysis, artificial intelligence models, data mining, mathematical modeling, complex simulations, and what-if sensitivity analysis modeling.
- These models train themselves by leveraging patterns, relationships, and structures within the existing data to accurately anticipate future conditions and outcomes.

The key to designing predictive or prescriptive strategies is to select data that will have the best potential to inform the predictive or prescriptive objective questions:

- If the goal is to predict next year's production equipment maintenance expense, data that helps explain the expected amount or increases or decreases in maintenance expenses are necessary. In other words, what variables drive changes in maintenance expense? Once those variables are identified, the related data can be selected and prepared for analysis.
- If we believe production level impacts maintenance expense, then include a data variable for the number of products produced, a measure of the most common levels of production, or a measure of whether production is expected to increase or decrease from current levels.

Suppose the WeMakeIt's auditors are testing the assumptions used to create the balance in the company's allowance for doubtful accounts. The auditors developed a model that included three variables that might impact whether the invoices would be collected on time. The model predicts how many days outstanding an invoice may be based on the invoice amount, if they are a new (1) or existing customer (0), and the credit period allowed on their invoice. The result of the regression model is provided in **Illustration 4.18**.

ILLUSTRATION 4.18 Prediction Model for Accounts Receivable Days Outstanding

Regression Statistics					
Multiple R	0.825790307				
R Square	0.681929631				
Adjusted R Square	0.681542056				
Standard Error	3.549920129				
Observations	2466				
ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	3	66518.36434	22172.78811	1759.475173	0.0000
Residual	2462	31025.95885	12.60193292		
Total	2465	97544.3232			
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	
Intercept	-7.961650225	0.670515711	-11.87392047	1.16228E-31	
InvoiceAmount	0.000829906	0.000903172	0.918879698	0.358248513	
New or Existing Customer	1.535043063	0.189912705	8.082887691	9.81036E-16	
Credit Period in Days	0.395420024	0.006452145	61.28504622	0.000	

This model reveals that these variables (invoice amount, new or existing customer, and credit period) explain about 68% of the change in days outstanding for a customer (adjusted *R*-square). The model is significant with a Significance *F* of less than 0.05. The negative intercept corrects the model's predictions by almost 8 days because customers rarely pay within the first week when they are offered an interest free period of time to pay their invoices. The significance of the intercept indicates that the mere existence of a credit period is a variable that impacts the number of days customers take to pay their invoices. With this model, the auditors can now predict that a new customer with a \$900 invoice and 60 days to settle will take about 16.5 days to make their payment:

$$-7.9617 + \$900 (0.0008) + 1(1.5350) + 60(0.3954) = 16.48 \text{ days}$$

Analysis Strategy Risks and Suggested Controls

The last two steps of a data analysis plan are to consider risks inherent to the data and analysis strategies and to put in place controls to reduce those risks. **Illustration 4.19** captures the most common risks that accountants face when analyzing data.

ILLUSTRATION 4.19 Common Analysis Risks and Recommended Controls

Analysis	Risk	Control
	The analysis is not appropriate for the variable's measurement scales.	Review whether the analysis is appropriate for variables' measure scales. Change analysis strategy if needed.
	The sample size is too small to allow for accurate inferences.	Gather more data to meet the model's minimum data points and assumptions, or select data analyses that do not require large samples.
	The sample is not representative.	Test the reasonableness of sample with descriptive analyses.
	Confounding variables are interfering with the reliability of the information models.	Perform the analysis with and without the confounding variable to test for explanatory power differences.
	The omission of relevant variables, or variables that are unknown and not controlled for, resulting in less reliable modeling.	Add available proxies for the missing variables to see if the model's explanatory power increases.
	Uncorrected and uncontrolled dirty data issues, such as errors, empty fields, outlier data points, and data from different levels of quality sources.	Properly prepare the data before starting the analysis. (Data preparation is explained in the next chapter.)
	Insufficient knowledge about or experience in performing the analysis strategy or interpreting the results.	Acquire proper training or work with a team that has more collective knowledge.
	Poorly made or misleading visualizations.	Perform descriptive analyses to compare and augment the visualization. Use visualization best practices.
	Failing to address the underlying uncertainties in the context.	Perform sensitivity analysis on all assumptions used in the analysis decisions.
	Insufficiently trained prediction and classification models.	Include more data to train the models and fully verify their accuracy before using them on the test data.

It is critical to consider these risks during analysis strategy planning and to add controls to help mitigate them. Without those controls we risk preparing, interpreting, and reporting incorrect analyses results, possibly causing ourselves or our stakeholders to make harmful decisions. Now that we have worked through the process of choosing the best data and analysis strategies, let's examine how accountants do this in professional practice.

Financial Accounting Tremendous Toys has implemented a database to capture inventoriable costs from their production each week. Data from the following data fields have been extracted, noting each field's data type and measurement scale.

Field Name	Field Type	Field Measurement Scale
ProductionRunNumber	Non-measured raw data	Categorical (Nominal)
ProductionRunDate	Non-measured raw data	Interval
NumberOfToysinRun	Measured raw data	Ratio
ToyIDNumber	Non-measured raw data	Categorical (Nominal)
DirectMaterialsUsed	Measured raw data	Ratio
DirectLaborUsed	Measured raw data	Ratio
OverheadCostApplied	Calculated data	Ratio
TotalRunCost	Calculated data	Ratio
TotalUnitCost	Calculated data	Ratio

Apply It 4.3

Create a Predictive Data Analysis Project Plan



Using this production data, as well as the sales data for how many toys of each type were sold each month for the last three years, create a predictive data analysis project plan that will estimate Tremendous Toy's year-end inventory valuation at the end of the next fiscal year. Your project plan should address the following:

1. Project objective and specific questions
2. Data strategy
3. Analysis strategy
4. Risks in data and analysis strategies
5. Controls for data and analysis strategies

SOLUTION

1. Project objective	• Accurately predict the next year-end's inventory value.
Specific questions	• How many units are predicted to be in inventory at the next year end? What is the predicted unit cost for each toy type at the end of next year?
2. Data strategy	• Data from the last three years of Tremendous Toys' production records will be extracted and prepared for analysis regarding materials costs, labor costs, and overhead charges. • The sales units data by toy type for the last three years will be extracted, cleaned for missing, duplicate, and outlier data errors, and aggregated by month. This results in 36 data points for the monthly unit sales of each toy type. • A calculated variable for average monthly cost for each toy type will need to be created using the NumberOfToysinRun and TotalRunCost fields. First, this data must be aggregated by month. Then, for each toy in each month, the total monthly production cost should be divided by the total number of toys produced to calculate the average toy cost per month for each toy type. The result will be 36 data points for the monthly average cost for each type of toy.
3. Analysis strategy	• The December sales in units can be estimated by running a regression on each month's unit sales. • A calculated variable must be created for each month's ending inventory units for each toy type, resulting in 36 data points for each toy. This variable's formula is: beginning inventory units + monthly toys produced – monthly units sold. (You can assume that this business started three years ago with no beginning inventory.) • The December year-end ending inventory in units can be estimated by running a regression on the monthly ending inventory units for each toy type. • The December year-end average cost per toy can be estimated by running a regression on the average monthly costs per toy. • To calculate the total December ending inventory valuation, the December ending inventory units for each toy should be multiplied by its corresponding estimated December average cost. Then these total costs per toy must be added for the total December year end inventory value.

(Continued)

4. Risks to data strategy	• There may be missing, duplicated, or erroneous data in the fields selected for analysis.
Risks to analysis strategy	• Costs may have changed over the three years due to seasonality, changes in vendors, and inflation. • Calculations may contain errors.
5. Data strategy controls	• Perform data cleaning tests to find and correct missing data, duplicates, and errors that result in outlier data points.
Analysis strategy controls	• Compare the unit cost and sales per month to determine if an inflation factor, seasonality, or other changes have occurred. If data shifts are noticed, their cause and permanence should be verified by speaking with the employees who would know the answers. • Each of the calculations performed should be verified, as well as the averaging calculations. • Regression analyses can be verified by making visualizations of the three years of monthly data points and visually inspected for agreement with the regression estimations.

4.4 How Do Data and Analysis Strategies Differ Across Practice Areas?

LEARNING OBJECTIVE 4

Summarize data and analyses strategies in professional practice areas.

As the last step of the data analysis planning stage, accounting professionals across practice areas design data and analysis strategies based on their project objectives. Here, we offer examples of common data and analytics strategies for the accounting information systems (AIS), auditing, financial accounting, managerial accounting, and tax accounting professional practice areas.

Accounting Information Systems

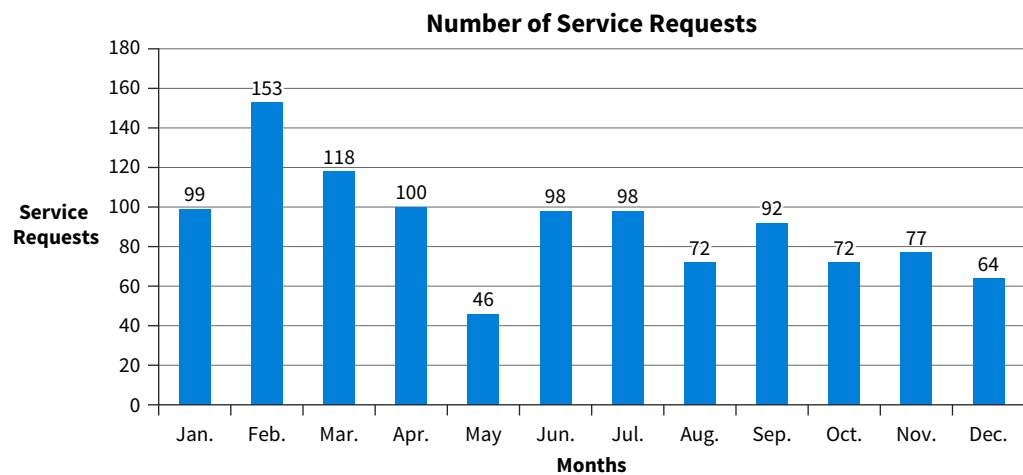
Because a company's accounting information system is involved in planning, executing, controlling, and reporting the business' operations, data analysis projects in this area often involve interdisciplinary data and knowledge. These project objectives can range in scope from impacting just one or two employees to involving most of the organization. All four types of analytics objectives (descriptive, diagnostic, predictive, and prescriptive) are commonly used.

The data analyzed in these projects can include a variety of IT operational data with different levels of data integrity and control documentation. More traditional accounting data are often analyzed with qualitative categorical and ordinal data:

- Counts of help tickets and incidents.
- Error counts and delay times.
- Login issues.
- System satisfaction and IT service satisfaction.

For example, **Illustration 4.20** is an example of a visualization of the number of monthly service requests related to the accounting information system of a company. The objective was to identify how many and which months experienced more than the IT staff's capacity of 90 service requests.

ILLUSTRATION 4.20 Number of Service Requests Each Month



AIS data analyses projects may also choose data strategies which use quantitative interval and ratio data:

- Costs such as equipment and software installations and updates, and IT staff labor.
- Budget variances for any of these costs.

It is important to select data that captures the AIS system's performance, vulnerabilities, and errors. For example, to analyze performance we could analyze the number of spam emails (ratio data) before and after investment in a new firewall.

Accountants designing analysis strategies for AIS projects are focused on increasing competitive advantage and improving the organization's operations. They often use statistics to understand which element of the accounting system they should focus on. The mathematical formality of statistical tools can help to persuade management to make the necessary investments.

There are risks involved with AIS data choice and analysis choices. Some common strategic and critical risks and suggested controls for AIS data and analysis are listed in **Illustration 4.21**.

ILLUSTRATION 4.21 Risks and Controls for AIS Data and Analysis Strategies

	Risks	Controls
Data	• Human behavior is not captured. • Data manipulation from unauthorized access or untrained employees. • Loss of data due to security breaches, ransomware, or natural disasters. • Difficulties capturing high velocity data. • Difficulties measuring interdepartmental dependencies on process software. • Dirty data and outliers.	• Strong logs and segregation of duties in IT. • Strong access controls. • Effective back ups and graceful degradation. • Clear procedures. • Thorough data cleaning and triangulation to verify data capture. • Strong audit trails. • Data and change management controls. • Data retention and destruction policies.
Analysis	• Errors in variable transformations. • Errors in calculations. • Data availability windows are too short to capture more strategic trends and issues.	• Verifying analysis steps and calculations. • Performing sensitivity analyses for critical results. • Creating effective visualizations and explanations.

Auditing

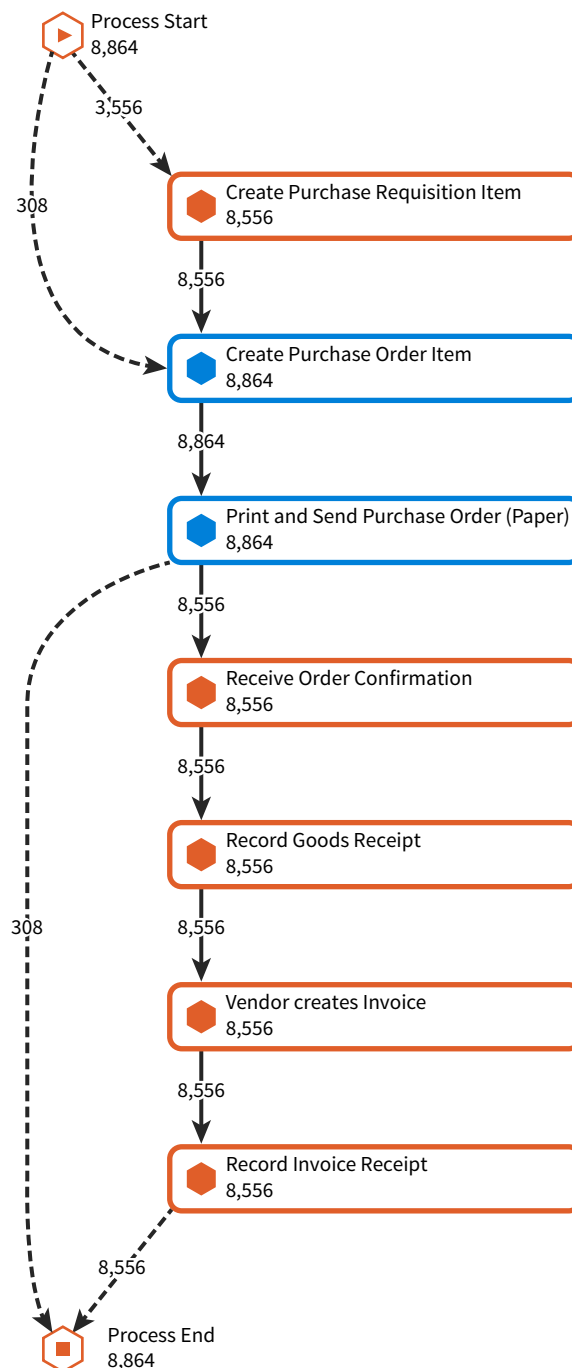
Auditors' historic resistance to change, due to high litigation risks and Public Accounting Oversight Board (PCAOB) review of audit processes, is disappearing as new technologies help them perform higher-quality, lower-risk, and more efficient audits. Audit firms are increasingly investing in technology and new ways to use it. They are reorganizing to offer new services to provide more business intelligence and economic value to their clients.

Auditors use statistics to determine risks to account balances and unusual transactions, especially in journal entries that involve estimates and assumptions. They commonly work with different kinds of data sources, from process documentation, journal entries, general and subsidiary ledgers, trial balances, to financial ratios. Auditors use analysis strategies for:

- Continuous auditing modules to test large data populations rather than testing samples with inference risks.
- Automated identification of dirty data, unusual transactions, and pattern anomalies to better reduce audit risk and fraud risks.
- Testing entire transaction cycles' internal controls with process mining and tracing purchasing to payment flows, P-cards (purchasing credit cards) usage and payroll documentation. Testing the revenue cycle from order to collection.
- Using robotic process automation (RPA) to remove the human (and often inconsistent) element of repetitive audit tasks, freeing auditors to focus on areas that require thoughtful critical thinking and judgments. (RPA applications and benefits to accountants are explained in the chapter that covers data and analysis developments in accounting.)
- Testing hypotheses about inventory and fixed assets with sampling, statistical tests, and inferences for the population of these assets.

Illustration 4.22 is an example of auditors using a process mining analysis strategy to learn whether all purchases followed the correct process from requisition to the recording of the invoice for the purchase. The visualization indicates that more than expected purchases (308 purchases) skip the authorized purchase requisition process step as well as the authorized order confirmation, goods receipt, and vendor invoice steps. These deviations from the authorized process are concerning to an auditor. The blue boxes indicate where the process is letting in purchase orders without purchase requisitions and letting out purchase orders without the subsequent required steps. An auditor will then want to follow-up on these exceptions to the authorized process.

ILLUSTRATION 4.22 Process Mining Results for Purchases Analysis



Regardless of the data and analysis strategies used, auditors must identify the risks and controls necessary to assure they can rely on the results of their analyses (**Illustration 4.23**).

ILLUSTRATION 4.23 Risks and Controls for Auditing Data and Analysis Strategies

	Risks	Controls
Data	• Missing data. • Dirty data. • Fake transaction data.	• Strong audit trail tests of internal controls for data entry, storage, processing, and reporting. • Testing authorization and access controls.
Analysis	• Errors in assumptions. • Errors in calculations and conclusions.	• Verification of all audit steps and calculations. • Checks on authentication and approvals. • Sensitivity analysis for critical results.

Financial Accounting

Financial accountants are primarily responsible for capturing, recording, processing, storing, and reporting accounting information. Accuracy, thoroughness, and documentation are essential. The accounting data they use for analyses are both guided and restrained by accounting regulations and government agency compliance. Due to the measurement scales of the variables they need for analysis, financial accountants may have to transform data depending on their objective questions. These data most often include categorical, interval, and ratio data.

The purpose of a data analysis strategy might be describing and diagnosing based on corresponding financial accounting rules and regulations:

- Nature, timing, and authorization of transactions (and diagnose issues) charged to each account.
- Internal control effectiveness and weaknesses.
- The completeness and reasonableness of adjusting entries at the end of the period.

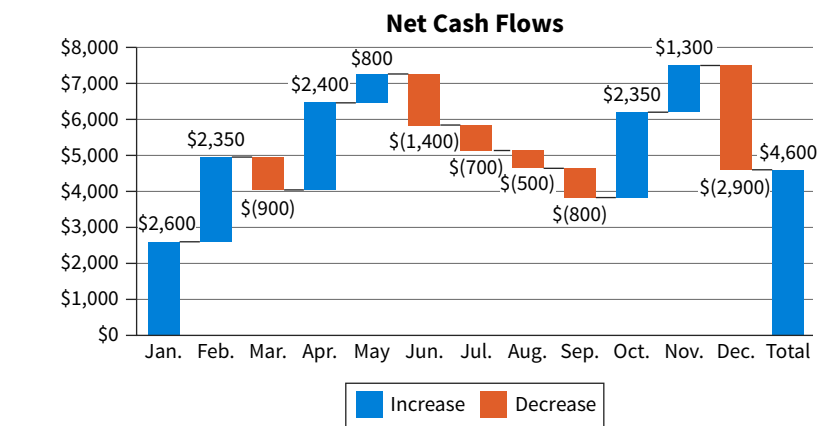
Financial accountants also use analysis strategies designed for their predictive and prescriptive objectives regarding financial outcomes for managers and board members. One example is estimating additional information to be presented along with their financial statements, such as:

- Ranking alternative capital sources and costs of capital in terms of favorability.
- Future net income and cash flows from operations, investing, and financing activities.
- Impacts of new strategies on the financial statement.
- Expected future costs associated with contingent liabilities, pension costs, new business units or the discontinuation of business units.

Financial accountants use statistics to identify opportunities and problems with profitability, liquidity, and business valuation. They analyze categorical industry codes from their client's categorical groups of held to maturity, trading, and available for sale investment assets to ensure desired diversification risks and holding strategies.

Illustration 4.24 is a waterfall chart that shows the analysis results for a project with the objective to better understand the monthly net cash flows for an organization by month. A waterfall chart is a useful visualization to show the positive and negative components of a change.

ILLUSTRATION 4.24 Net Cash Flow Visualization



This waterfall visualization shows the change in cashflow for each month. For instance, in February cashflow increased by \$2,350 but then decreased by \$900 in March. The net of all the monthly changes agrees to the total. Like other accountants, financial accountants must identify potential risks in their data and analysis strategies (**Illustration 4.25**).

ILLUSTRATION 4.25 Risks and Controls for Financial Accounting Data and Analysis Strategies

	Risks	Controls
Data	• Missing operational and adjusting entries. • Dirty data. • Fake transaction data. • Incorrect valuation. • Incorrect account. • Incorrect time period. • Missing data.	• Strong accounting cycle steps. • Strong internal controls for data entry, storage, processing, and reporting. • Authorization and access controls. • Testing for and correcting missing, duplicate, and outlier data.
Analysis	• Calculation errors. • Biases and self-interests. • Incorrect calculations. • Not following GAAP analysis guidance.	• Verifying all cycle steps and calculations. • Checking authentication and approvals. • Sensitivity analysis for critical results. • Verify applicable GAAP guidance.

Managerial Accounting

Managerial accountants add value to their organization with different data analysis strategies. The purpose of most managerial accounting data analyses is to improve planning, operational control, and decision-making to support an organization’s mission and strategies. This type of data analysis is valuable for improving the selection, execution, and evaluation of organizational strategy. These kinds of analyses can also lead to decisions that give employees more access to information, which improves performance and, eventually, organizational culture and operations.

APPLYING CRITICAL THINKING 4.7: Use Multiple Disciplines in Managerial Accounting

The knowledge base needed by managerial accountants is broad, involving multiple disciplines. For example, to reduce factory operating costs, managerial accountants must understand the engineering of their production machines and know how to create reliable statistical models for future production (**Knowledge**):

- Accounting and economic concepts
- Organizational behavior
- Operations management
- Technology tools
- Quantitative analyses disciplines

Managerial accountants prepare analysis strategies, both for routine and ad hoc (one-time) objectives for each functional area of their organization. These strategies use data across measurement scales to describe, diagnose, predict, and prescribe strategy impacts:

- Identifying areas where innovation in business organization, policy and process will increase efficiencies, for example, to reduce non-value-added steps and delays.
- Identifying areas where innovation in business partnerships, operations, and internal controls will increase effectiveness.
- Increasing competitive advantage through many new business intelligence opportunities.
- Improving compliance with all legal and regulatory regulations.

Managerial accountants may perform data analysis to determine the potential business process and internal control benefits of automating the purchase requisition to vendor invoice process (see Illustration 4.22). Automation could force all departments to go through the authorized business process rather than going around it. Identifying risks to the data and analyses used in managerial accounting is important to ensure analysis results are accurate and reliable (**Illustration 4.26**).

ILLUSTRATION 4.26 Risks and Controls for Managerial Accounting Data and Analysis Strategies

	Risks	Control Choices
Data	• Dirty data. • Missing data. • Irrelevant data.	• Independent verification. • Segregation of duties. • Sound personnel governance. • Verify authorization. • Considering the purpose when making data choices.
Analysis	• Errors in calculations. • Errors in attributions. • Biases and self-interests.	• Verifying all steps and calculations.

Tax Accounting

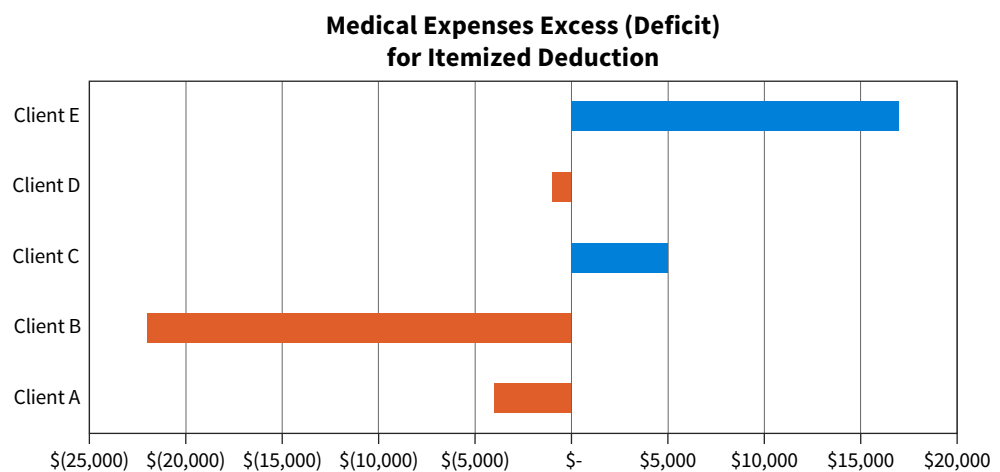
Thanks to data analytics, tax accountants have more information for their tax planning and compliance services, which improves their judgments and how they document defend their

positions to both clients and regulators. Recent surveys show that over 58% of tax accountants are using data and analytics strategies for many areas of their professional practice¹, including:

- Sales and use taxes and local taxes (SALT).
- Value added taxes (VAT).
- Goods and services taxes (GST).
- Customs duties.
- Transfer pricing and intercompany transactions.
- Tax compliance.
- Tax provisions.

Perhaps the biggest effect of data analysis strategies on tax practice is the movement away from a dependency on historic data and toward a forward-looking, value-adding service perspective by using predicting and prescribing analytics strategies. Examples of these predictive and prescriptive analytics include using complex data modeling to evaluate decision and position alternatives, which improves the value and accuracy of advice. **Illustration 4.27** is a live dashboard that shows current year medical expenses compared to the level needed for itemized deductions for five of their individual clients. This would help the tax accountant provide advice to these clients.

ILLUSTRATION 4.27 Medical Expenses Itemized Deduction Threshold Assessment by Client



Another example is sophisticated querying of clients' accounting data to better identify opportunities, such as when to buy or sell assets, and understanding the operational decisions driving tax liabilities.

APPLYING CRITICAL THINKING 4.8: Think Critically in Tax Accounting

- Tax accountants consider their clients, tax authorities, and their tax firm partners when they plan different data analyses. Each of these stakeholders may have different objectives and priorities. For example, clients may prioritize minimizing their tax obligations, so that would be the objective of your tax planning services (**Stakeholders**).
- Strong research and analysis skills are necessary, such as when performing tax research, developing tax strategies, and estimating tax liabilities. For example, state laws are different than the national and international law for every type of tax category (**Knowledge**).
- Clients, both individuals and organizations, may have incentives to omit taxable revenues and include irrelevant and inappropriate expenses. These data risks should be controlled, perhaps with interview questions as well as verifying the records provided through indirect tests of reasonableness (**Risks**).

¹International Tax Review (2019). Preparing for the future of tax. <https://www.internationaltaxreview.com/article/2a6a41udxe4e2kuc8zoqo/preparing-for-the-future-of-tax> (accessed 13 July 2022).

Identifying risks and assuring that proper controls are in place to mitigate risks are also critical (**Illustration 4.28**).

ILLUSTRATION 4.28 Risks and Controls for Tax Accounting Data and Analysis Strategies

	Risks	Control Choices
Data	• Dirty data and outliers. • Missing data.	• Implement backups. • Strong audit trails. • Tax management controls. • Data retention and destruction policies. • Data authenticity and dirty data verification and correction.
Analysis	• Errors in data evaluations and tests. • Errors in calculations and forms. • Lack of tax code knowledge.	• Verification of all tax rules' liability steps and calculations. • Sensitivity analysis for critical results. • Effective visualizations and explanations.

For each of the following data and analyses objectives:

1. Identify the data measurement scale that would be utilized.
2. Identify one practice area that is likely to perform that analysis.

Some of the options have more than one correct answer for both (1) and (2), but only enter one correct answer for each.

Data Measurement Scales: categorical, ordinal, interval, ratio

Practice Areas: Accounting information systems, Auditing, Financial accounting, Managerial accounting, Tax accounting

Objective	Measurement Scale	Practice Area
Aging of accounts receivable's year-end open invoices on days outstanding.		
Predicting cost savings and productivity increases from new AIS investments.		
Creating dollar value stratified sampling plans for investment asset accounts.		
Evaluating web server security incidents by type of incident.		
Prescribing product revenue increases by product from marketing campaigns.		
Predicting future cash flows from operations.		
Calculating before and after process improvements from system investments.		
Testing different tax planning strategies by evaluating future cash savings.		
Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.		
Testing approvals and documentation for transaction authorization.		

Apply It 4.4

Match Strategies with Professional Practice Areas

SOLUTION

Objective	Measurement Scale	Practice Area
Aging of accounts receivable's year-end open invoices on days outstanding.	Interval	Financial and managerial accounting
Predicting cost savings and productivity increases from new AIS investments.	Ratio	Managerial accounting
Creating dollar value stratified sampling plans for investment asset accounts.	Interval	Auditing
Evaluating web server security incidents by type of incident.	Categorical	Accounting information systems
Prescribing product revenue increases by product from marketing campaigns.	Ratio	Managerial accounting
Predicting future cash flows from operations.	Ratio	Managerial accounting
Calculating before and after process improvements from system investments.	Interval	Accounting Information systems
Testing different tax planning strategies by evaluating future cash savings.	Ratio	Tax accounting
Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.	Ratio	Auditing
Testing approvals and documentation for transaction authorization.	Categorical	Financial accounting

Chapter Review and Practice

Learning Objectives Review

1

Identify the components of a data analysis project plan.

A data analysis project plan is a blueprint to follow for the project. Using a planning tool can help choose the best data and analysis strategies for the project objectives. There are five steps:

- Step 1: Focus on the project's objective and the specific questions that must be answered.
- Step 2: Decide which data are needed to answer the questions, develop alternatives, evaluate, and rank the best option.

- Step 3: Consider the methods available to do the analysis and rank the alternatives. Evaluate and choose the best strategy option.
- Steps 4 and 5: Consider critical risks inherent to both the data strategy and analysis strategy.
- Steps 4 and 5: Embed controls to reduce these risks in data strategy and analysis strategy.

2 Describe how to develop a data strategy.

A successful project hinges on using the appropriate data and evaluating its characteristics. Considering the data's potential impact on the analysis and any risks makes it easier to select the most suitable data for a project's goals:

- The first step is selecting the most appropriate data. Appropriate data are data that are available and relevant to the data analysis objective questions.
- The data's measurement scales must also be suitable for use with the chosen data analysis method, as not all analyses are valid for every measure scale. Data can have categorical, interval, ordinal, or ratio scales.
- Data can be either internal (generated within the organization), external (obtained from a source outside of the organization), or a combination of both. Data can be raw measures or assigned non-measures, and can also be calculated fields.
- Consider potential risks related to the data chosen for the analysis and put controls in place to mitigate those risks. Three common data risks (and controls) include: non-representative samples (verify representativeness of the sample), outlier data points (identify outliers and then justify or remove), and dirty data (verify integrity of the data set and clean).

3 Explain how an analysis strategy is designed.

There are two things to consider when designing a data analysis strategy: (1) the objective questions and (2) the measurement scale of the data. There are four data analysis objectives: descriptive, diagnostic, predictive, or prescriptive. The questions we design are based on the objective:

- When designing any type of analysis, the appropriate analysis is dependent on the measurement scale of the data. Typical descriptive and diagnostic analysis methods include measure of central tendency (mean, median), frequency distributions, measures of data dispersion (e.g. range, quartiles, standard

deviation), visualizations, correlation, and calculations (e.g. totals, percent change).

- Typical analyses used for predictive and prescriptive analyses include trendlines, regression analysis, optimization models, and what-if analyses.
- There are many common risks inherent in descriptive, diagnostic, predictive, and prescriptive analyses that can be identified and controlled. Some of these risks include performing an analysis with dirty data, or analysis not appropriate for the variable's measure scales, using too small a sample, or a non-representative sample, or including (or omitting) variables from the analysis.

4 Summarize data and analyses strategies in professional practice areas.

Each professional accounting practice area plans different data and analyses strategies for varying objectives, using data with a range of measurement scales to describe, diagnose, predict, and prescribe results for their stakeholders and objectives. The risks inherent to their data and analysis choices, as well as the controls they implement are a function of their roles, responsibilities, and project objectives:

- Accounting information systems professionals plan data and analysis strategies for AIS governance, strategic planning and evaluation, operational performance, and compliance objectives. Their projects typically involve business process functionality, business intelligence, and cybersecurity purposes.
- Auditors plan data and analyses that have professional audit process compliance and audit quality objectives to verify the representativeness of their clients' financial statement balances or their organization's accounting systems. External auditors have the additional data risks of using data that has been captured and extracted by their clients, oftentimes not able to verify the data completeness or accuracy.
- Financial accountants focus on regulatory compliance regarding transaction capturing, recording, processing, storing, and reporting objectives.
- Managerial accountants have a wide variety of internal governance, strategic, financing, investing, and operational objectives for their data and analysis plans.
- Tax accountants focus on both compliance and more forward-facing tax planning opportunities to serve their clients and organizations with their data analysis strategy plans. Similar to external auditors, tax accountants often must use data sets prepared by their clients, which may have greater risks of inaccuracy and incompleteness.

Key Terms Review

Attributes 4-12
Calculated data 4-14
Data set 4-12
Dirty data 4-16

Fields 4-12
Measured raw numeric data 4-13
Measurement scale 4-14
Non-measured raw data 4-13

Records 4-12
Variable 4-20

How To Walk-Throughs



HOW TO 4.1 Calculate Year-End Bad Debts Estimation Using Excel

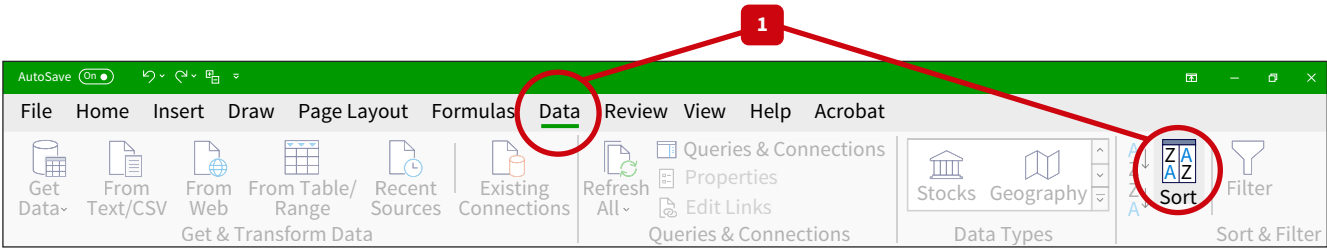
The figures shown in the table in Illustration 4.2 were calculated in Excel. Recall a few facts about this example:

- WeMakeIt, Inc. estimates that 2% of its 30 to 60 days outstanding receivables will be uncollectible, and that 30% of outstanding receivables older than 60 days will be uncollectible.
- The 2025 year-end allowance for uncollectible accounts unadjusted balance is \$3,000.

What You Need: **Data** The How To 4.1 data file.

STEP 1: Select the entire data set, but not the labels, and sort the outstanding invoice data by invoice date. The **Sort** option appears in the **Data** tab. ([Illustration 4.29](#))

ILLUSTRATION 4.29 Sort Outstanding Invoices by Date



STEP 2: Create a new column to the right of the data for the year-end date of 12/31/2025:

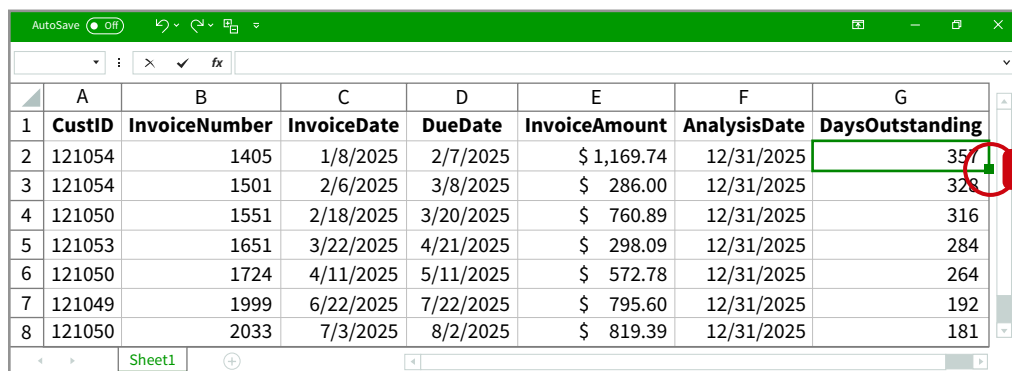
- Add a label name for this column in row 1: “Analysis Date.”
- In row 2, enter “12/31/2025” and copy this date to the bottom row of data. This can be easily done by clicking on the bottom corner of the cell where “12/31/2025” was entered and dragging the cursor down to the last row of data ([Illustration 4.30](#)).

ILLUSTRATION 4.30 Add Analysis Date Column

	A	B	C	D	E	F	G
1	CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	AnalysisDate	
2	121054	1405	1/8/2025	2/7/2025	\$ 1,169.74	12/31/2025	
3	121054	1501	2/6/2025	3/8/2025	\$ 286.00		
4	121050	1551	2/18/2025	3/20/2025	\$ 760.89		
5	121053	1651	3/22/2025	4/21/2025	\$ 298.09		

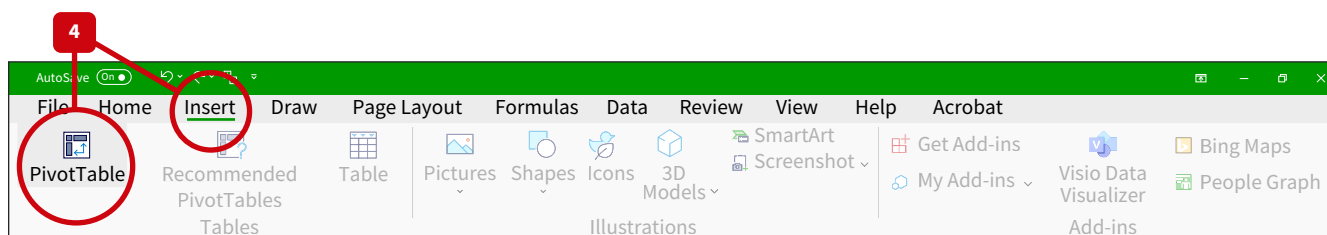
STEP 3: Create a new column to the right of the 12/31/2025 date column:

- Add a label name for this column in row 1: “Days Outstanding.”
- Calculate the days outstanding by creating a formula in cell G2 which subtracts the invoice date (Cell C2) from 12/31/2025 cell (F2) for the Analysis Date. The formula is =F2-C2.
- Copy this formula through all the data rows in Column G. Copying can be done by clicking on the bottom corner of the cell where the formula was entered and dragging in down to the last row of data, ([Illustration 4.31](#)).

ILLUSTRATION 4.31 Data set with Analysis Date and Days Outstanding Columns


	A	B	C	D	E	F	G
	CustID	InvoiceNumber	InvoiceDate	DueDate	InvoiceAmount	AnalysisDate	DaysOutstanding
2	121054	1405	1/8/2025	2/7/2025	\$ 1,169.74	12/31/2025	35
3	121054	1501	2/6/2025	3/8/2025	\$ 286.00	12/31/2025	32
4	121050	1551	2/18/2025	3/20/2025	\$ 760.89	12/31/2025	316
5	121053	1651	3/22/2025	4/21/2025	\$ 298.09	12/31/2025	284
6	121050	1724	4/11/2025	5/11/2025	\$ 572.78	12/31/2025	264
7	121049	1999	6/22/2025	7/22/2025	\$ 795.60	12/31/2025	192
8	121050	2033	7/3/2025	8/2/2025	\$ 819.39	12/31/2025	181

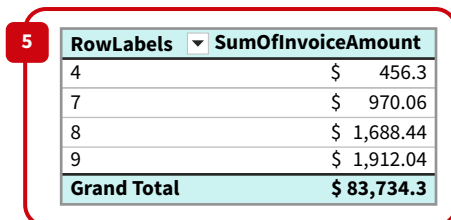
STEP 4: Select the data field titles and data rows and create a PivotTable. This option is located under the **Insert** menu option on Excel (**Illustration 4.32**).

ILLUSTRATION 4.32 Insert PivotTable

Select **New Worksheet**. Your screen will automatically move to the new worksheet. Name this table “BadDebts2025.” The name appears at the top left of your screen.

STEP 5: Select any place in the PivotTable so the **PivotTable Fields** appears on the right of your screen:

- Drag **Invoice Amount** to the **Values** area.
- Drag **Days Outstanding** to **Rows**. The new PivotTable solution should appear (**Illustration 4.33**).

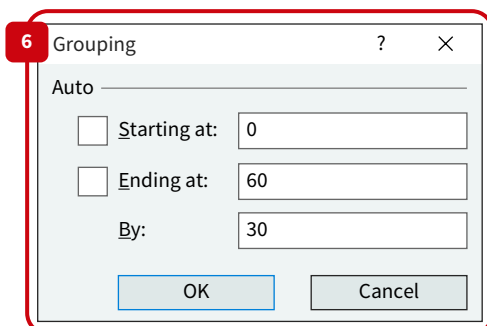


RowLabels	SumOfInvoiceAmount
4	\$ 456.3
7	\$ 970.06
8	\$ 1,688.44
9	\$ 1,912.04
Grand Total	\$ 83,734.3

ILLUSTRATION 4.33 Unpaid Invoices by Days Outstanding

STEP 6: Next, select any row in the **Days Outstanding** column:

- Right click and select **Group**.
- In the dialog box that opens (**Illustration 4.34**), enter a zero (0) in the starting point, 60 in the ending point, and then enter 30 for the **By** option. Select **OK**.

**ILLUSTRATION 4.34** Dialog Box to Group by Days Outstanding

Now the PivotTable has become three rows for the three desired age categories, with the sub-totals already calculated for the **Sum of InvoiceAmount** column. Format the **Sum** column as currency ([Illustration 4.35](#)). Format the amounts as currency.

ILLUSTRATION 4.35 Age Group Totals for Accounts Receivable Invoice Values

RowLabels	SumOfInvoiceAmount
0-29	\$ 30,099.29
30-60	\$ 20,441.46
>60	\$ 33,193.55
Grand Total	\$ 83,734.30

STEP 7: In the cell to the right of the SumofInvoiceAmount title, enter “Estimation %.” In the cell to the right, enter “\$ Uncollectible.” Adjust column widths:

- In the cell to the right of the \$ 20,441.46, enter “0.02,” and in the cell below enter “0.3” for the uncollectible percentages.
- Format the **\$ Uncollectible** column as currency.

STEP 8: In the cells to the right of each of these percentages, enter the following formulas using cell references:

- For the 30-60 day group, the formula is:

$$=(\text{click on cell with } 20441.46) * (\text{click on the cell with the } 0.02)$$

- And for the >60 days group:

$$=(\text{click on the cell with } 33193.55) * (\text{click on cell with the } 0.3)$$

- An example of the formula and totals that result is shown in [Illustration 4.36](#). Add an underline to the cell containing the last formula.

ILLUSTRATION 4.36 Age Group Totals for Bad Debt Estimations

RowLabels	SumOfInvoiceAmount	Estimation %	\$ Uncollectible
0-29	\$ 30,099.29		
30-60	\$ 20,441.46	0.02	\$ 408.83
>60	\$ 33,193.55	0.3	\$ 9,958.07
Grand Total	\$ 83,734.30		

STEP 9: Enter the sum formula in the next cell (D7 in this example solution):

$$=sum(D5:D6)$$

This sum is the desired adjusted year-end credit balance of allowance for uncollectible accounts. This total is \$10,366.89 (also shown in [Illustration 4.37](#)).

STEP 10: Compare the 2025 unadjusted allowance account balance, a \$3,000 credit balance, to the calculated desired ending balance to calculate the amount needed for the adjusting entry to ensure the allowance for uncollectible accounts has that desired credit balance:

- Enter the unadjusted balance in cell D8 and the formula for the calculation in cell D10 (**Illustration 4.37**):

$$=D7-D8$$

- You are now ready to make your adjusting entry:

	Debit	Credit
Bad debt expense	7,366.89	
Allowance for uncollectible accounts		7,366.89

ILLUSTRATION 4.37 Completed Analysis of Bad Debts Expense Calculation

RowLabels	SumOfInvoiceAmount	Estimation %	\$ Uncollectible	
0-29	\$ 30,099.29			
30-60	\$ 20,441.46	0.02	\$ 408.83	
>60	\$ 33,193.55	0.3	\$ 9,958.07	
Grand Total	\$ 83,734.30		\$ 10,366.89	
			(\$ 3,000)	Subtract the unadjusted balance in Allowance
			\$ 7,366.89	Adjusting entry amount

HOW TO 4.2 Create a Frequency Bar Chart in Power BI

Illustration 4.16 can be recreated in Power BI by analyzing the outstanding invoices in accounts receivable by customer and creating a visualization that shows how many outstanding invoices are owed by unique customers at the end of 2024 and 2025 in three groupings: 1-3 invoices, 4-6 invoices, and 7-9 invoices.

What You Will Need: **Data** The How To 4.2 data file.

STEP 1: Upload the data set into Power BI:

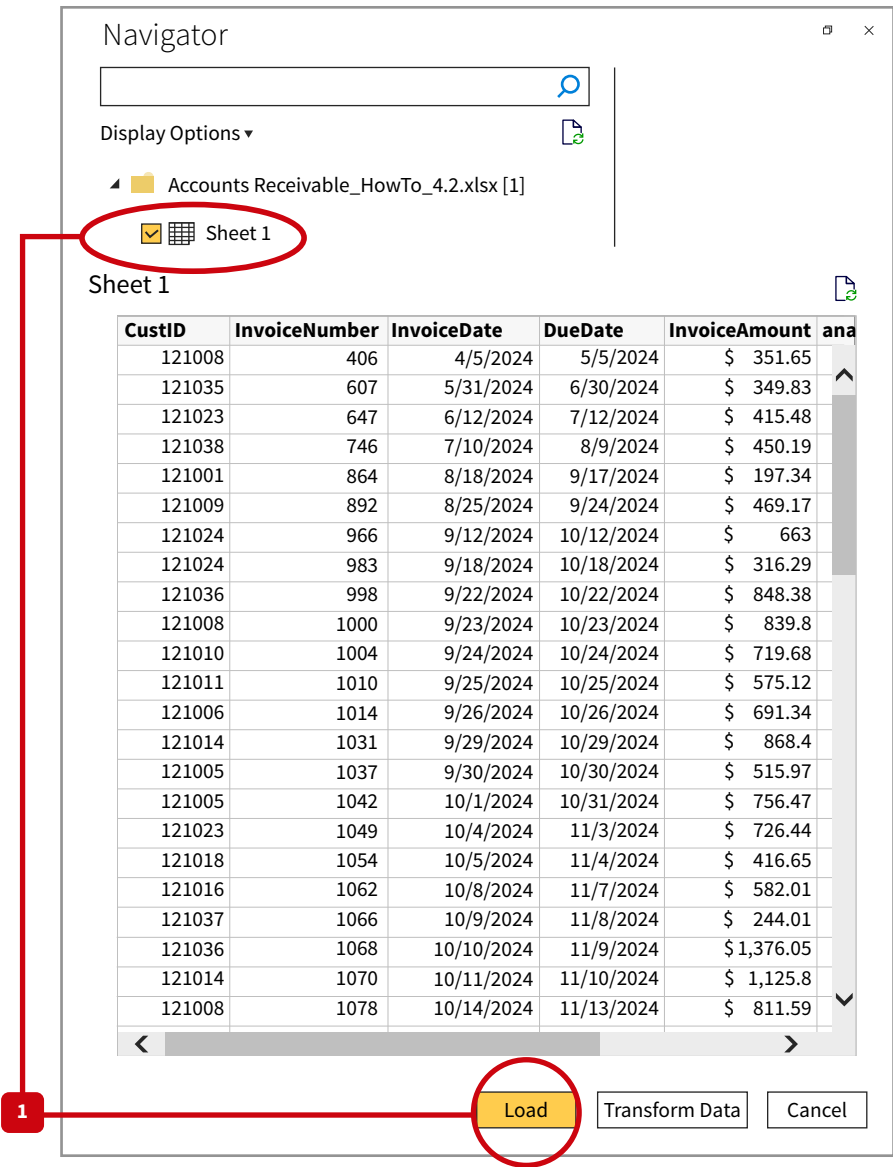
- Open Power BI and Select **Get Data**.
- Select **Import data** from Excel and choose **Excel Worksheet**.
- Navigate to the saved the data set file and open it.

How To



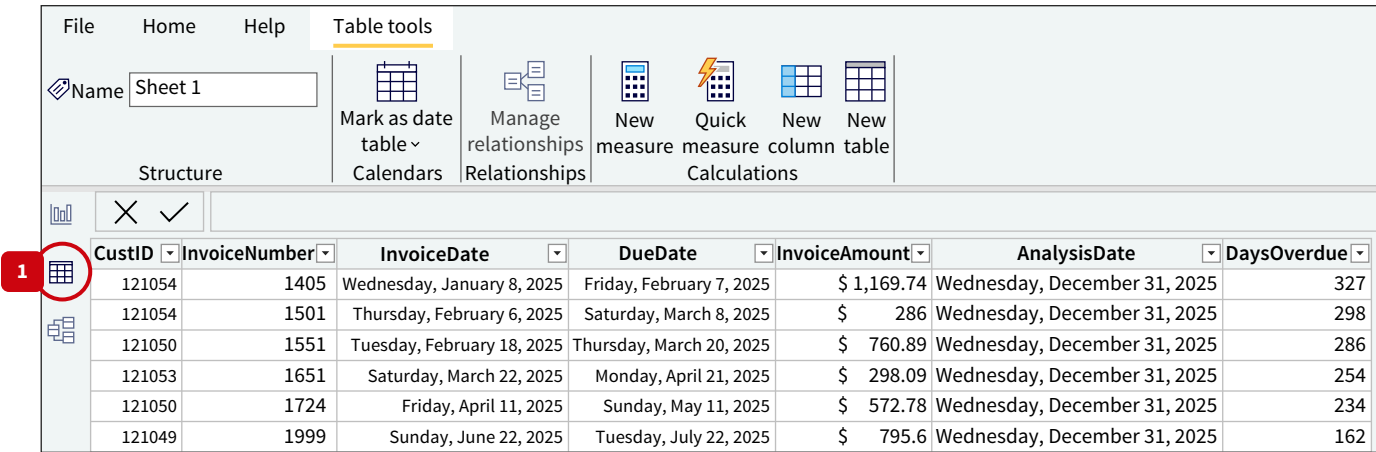
Once the Navigator window opens, select Sheet 1 of the file listed on the left, then select **Load** (Illustration 4.38).

ILLUSTRATION 4.38 Load Data into Power BI



- Verify this import by double-clicking on the data sheet icon on the left side of the screen. Look at the bottom left corner to verify that you have successfully loaded all 201 records of the data set (Illustration 4.39).

ILLUSTRATION 4.39 Verify Excel Import



STEP 2: Select the **AnalysisDate** column.

- In the menu under **Column Tools**, pull down the **Format Field** options and choose the last option “2001 (yyyy)” (**Illustration 4.40**).
- Verify the analysis date field has changed to show 2024 and 2025.

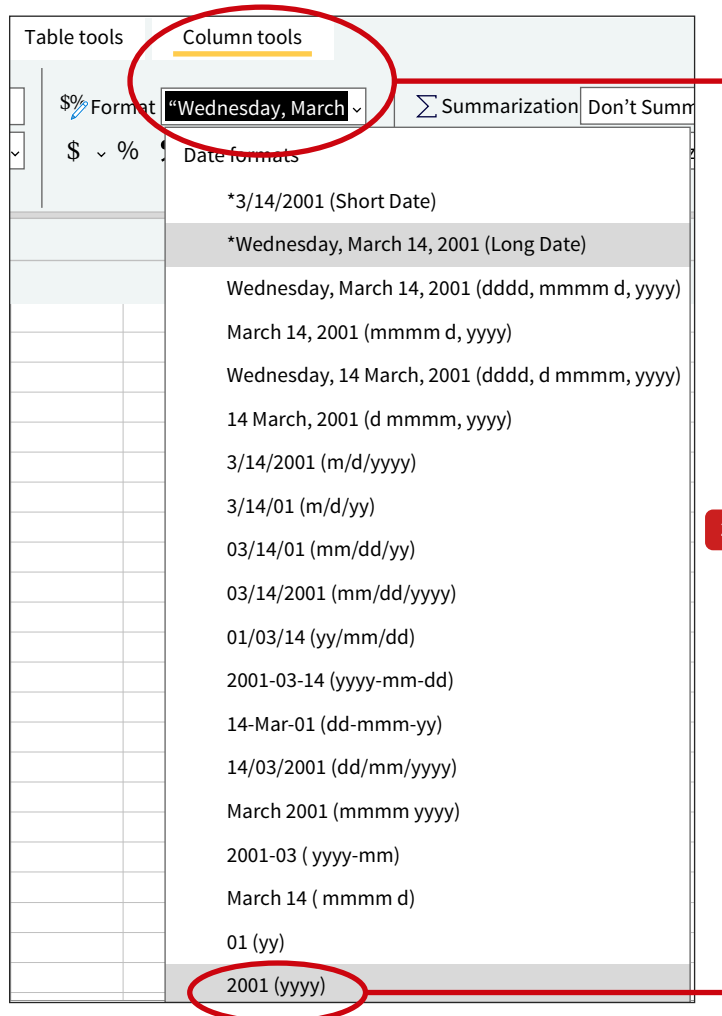


ILLUSTRATION 4.40 Change the AnalysisDate Column Format

STEP 3: Select the **CustID** column by clicking on the column header. Right click and select **Edit Query** (**Illustration 4.41**).

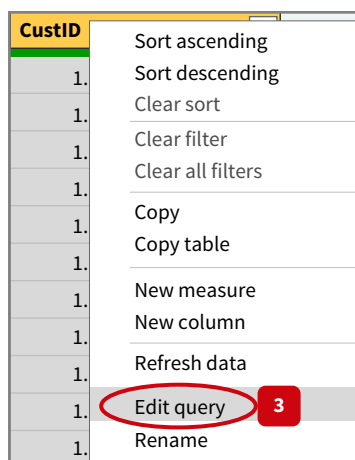


ILLUSTRATION 4.41 Edit Query Selection

- This will open the Power Query Editor window. In this window, the **CustID column** will also be selected.
- Select **Group By** from the top menu, and it will open.
- Select **Advanced**. Right below the CustID field you will see the Add grouping button. Select this to add a second grouping for the field AnalysisDate. Then, below the new grouping box, in the New column name box, enter “CountInvoices” (do not enter the quotes).
- In the pull down menu for **Operation**, select **Count Distinct Rows** and select **OK**. Close and Apply the Power Query Editor ([Illustration 4.42](#)).

ILLUSTRATION 4.42 Group By Customer ID and Count Distinct Invoices

Group By

?

×

Specify the columns to group by and one or more outputs.

☐ Basic

☒ Advanced

CustID

▼

Add grouping

New column name

CountInvoices

Operation

Count Distinct Rows

▼

Column

▼

Add aggregation

OK

Cancel

STEP 4: Select the **Count Invoices** column. Select **Data Groups** from the top column, then select **New Data Group** ([Illustration 4.43](#)).

ILLUSTRATION 4.43 Select Data Groups

File

Home

Help

Table tools

Column tools

Name

CountInvoices

123 Data type

Whole number

▼

\$% Format

Whole number

▼

Σ Summarization

Sum

▼

Sort by column

▼

Data category

Uncategorized

▼

Sort

Groups

Structure

Formatting

Properties

Sort

Groups

✕

✓

CustID	AnalysisDate	CountInvoices
121008	2024	7
121035	2024	4
121023	2024	5

Illustration 4.43 shows the dialog box that opens once you select **Data Groups**:

- In the dialog box select **List** (not bins) for the Group Type.
- Under **Ungrouped values**, hold the shift key down to select values 1 2, and 3.
- Select **Group**. Double click the new group that appears on the right window and change the label to “1-3”.
- Select the **Other** group, and move to the ungrouped left side. Select 4 and 5 together by again holding down the shift key. Select **Group** and move over to the right window. Double click on the label for this field, and type “4-6” (it happens that there are no customers with six outstanding invoices, but the range will be 4-6 regardless).
- Once again, select **Other** and move to the **Ungrouped values** window. Select 7, 8, and 9 by holding down the shift key and selecting **GROUP**.
- Go back to the right and double click on the label for this group. Change it to “7-9.” Select **OK**. (Illustration 4.44).

ILLUSTRATION 4.44 Create Grouping of Number of Invoices

The 'Groups' dialog box is shown with the following configuration:

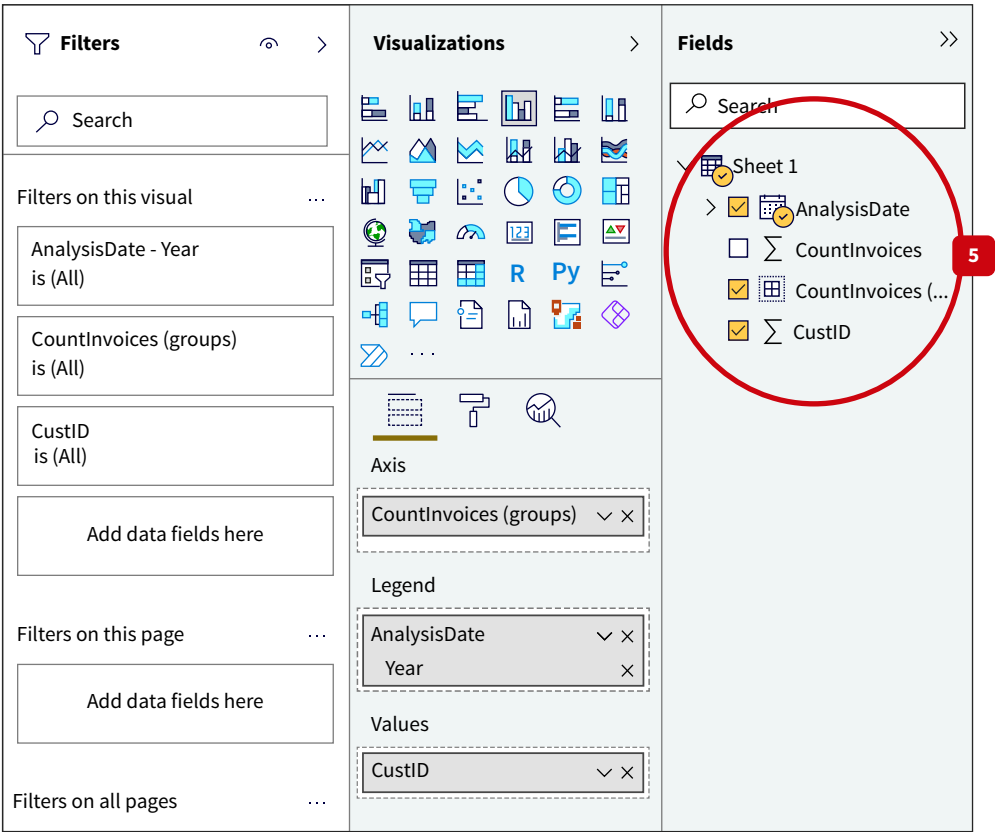
- Name:** CountInvoices (groups)
- Field:** CountInvoices
- Group type:** List
- Ungrouped values:** (Empty list)
- Groups and members:**
 - ▶ 1-3
 - ▶ 4-6
 - ▶ 7-9
 - ▲ Other
 - ☐ Contains all ungrouped values
- Buttons:** Group, Ungroup, Include Other group (checked), OK, Cancel

STEP 5: Now you are ready to create your visualization:

- Select the graphing icon on the left of the screen.
- Select all three variables in your **Fields** list.

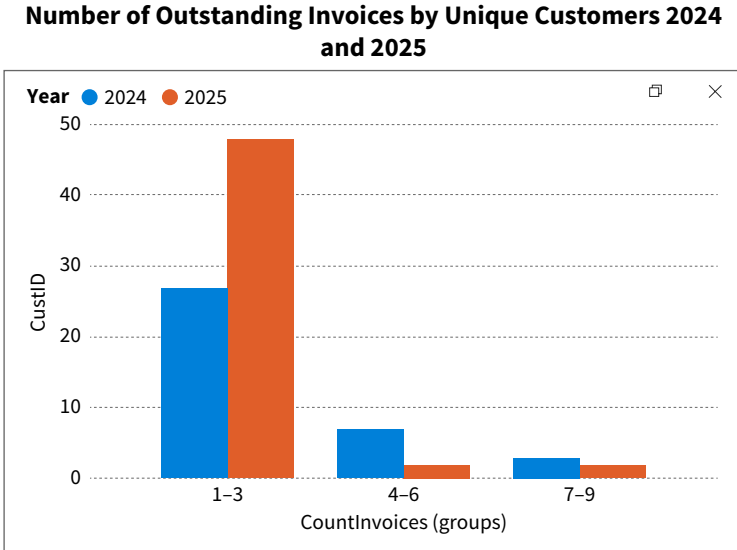
In the **Visualizations** pane, **Count Invoices** (groups) should be in the axis area. You may need to move the **AnalysisDate** field up to the **Legend** field by dragging it. The **Count Invoices** field should be in the **Values** area ([Illustration 4.45](#)).

ILLUSTRATION 4.45 Create the Visualization



Your visualization is complete ([Illustration 4.46](#))!

ILLUSTRATION 4.46 WeMakelt, Inc. Accounts Receivable



Data The Data tag appears when the data required to answer a question or complete an exercise are available on Wiley's online learning platform.

Multiple Choice Questions

1. **(LO 1)** Which of these is *not* an objective of data analysis strategies?
 - a. Collecting data.
 - b. Describing data.
 - c. Diagnosing data.
 - d. Predicting a future value of data.
 - e. Prescribing the impact of a particular strategy on the future value of the data.
2. **(LO 1)** Diagnosing objectives in data analysis strategies involve which of the following?
 - a. Looking for complete independence of variables.
 - b. Looking for causality in regression variables.
 - c. Looking for where to place the blame for bad operational results.
 - d. Looking for associations between data variables.
 - e. Predicting what future outcomes will be in the data.
3. **(LO 1)** The components of an effective data analysis strategy plan consider each of the following *except* for
 - a. analysis controls.
 - b. analysis risks.
 - c. analysis objectives.
 - d. analysis complexity.
4. **(LO 1)** Which of the following specific questions depicts a descriptive analysis objective for accounts receivable balance at year-end?
 - a. How many invoices from the current year were collected during the year?
 - b. What was the total of all sales revenues, cash and credit sales, for the year?
 - c. How many salespersons were employed in New York City this year?
 - d. What is the average days in inventory for each of your product lines?
 - e. How many of the open invoices at the end of the year are past due?
5. **(LO 1)** You are asked to perform an analysis predicting the accounts payable balance at the year-end. Which of the following data sources are best to analyze for this objective?
 - a. A list of authorized new vendors for the year.
 - b. Open vendor invoices that refer to purchases already received.
 - c. A list of checks paid to accounts payable vendors during the year.
 - d. A list of purchase orders issued during the month of December.
 - e. None of these data resources are useful to predict the accounts payable year-end balance.
6. **(LO 1)** Assume you are asked to estimate the amount of uncollectible receivables at year-end. Which of the following is *not* an analysis risk to consider when estimating accounts receivable bad debts at year-end?
 - a. The number of full-time sales employees is reported inaccurately.
 - b. The calculation of days outstanding is inaccurate.
 - c. The company changed collection policies for certain customers.
 - d. There are new customers who are not included in the analysis of aged receivables.
 - e. There are mathematical errors in the summation of receivables.
7. **(LO 1)** Which of the following is an appropriate response to the data risk that the sample selected for analysis is inappropriate?
 - a. Add missing variables to see explanatory power changes.
 - b. Test a sample distribution form for statistical test assumptions.
 - c. Verify the representativeness of your sample.
 - d. Explain the rule used for outlier removal.
 - e. Know your outlier data point.
8. **(LO 1)** Critically thinking through the risks to a data analysis strategy during planning will help avoid
 - a. data selection biases.
 - b. judgment errors from lack of experience.
 - c. interpretation errors from dirty data sets.
 - d. incorrect applications of statistical tests.
 - e. All of the these are possible costs from not considering the risks to data analyses choices.
9. **(LO 1)** A data analyses strategy is more likely to be successful if you critically think through
 - a. the analysis performed by your competitors.
 - b. the results you want to achieve.
 - c. the data that will best inform your questions.
 - d. your objectives for the analysis.
 - e. None of these help you perform better data analyses strategy planning.
10. **(LO 2)** Which of the following refers to the content of a data set's row?
 - a. Data field.
 - b. Attribute.
 - c. Single occurrence of the data record.
 - d. All these define a row of data in a dataset.
11. **(LO 2)** Which of the following could refer to the name of a column in a sales invoices data set?
 - a. Invoice 3570.
 - b. Customers in Orlando, Florida.
 - c. CustomerState.
 - d. BankStatementDate.

12. (LO 2) Which of the following is an example of knowledge you might need to acquire for data analysis strategy planning?

- a. The analysis is appropriate for your data measurement scales.
- b. An understanding of your biases.
- c. Identifying the key stakeholders.
- d. Analysis resources that are available for your compensation.
- e. Compensation correlation with results.

13. (LO 2) Which of the following is *not* a data measure scale that can be used in designing your data analysis strategy?

- a. Optimal data
- b. Ordinal data
- c. Interval data
- d. Ratio data
- e. All of these are data measure scales.

14. (LO 2) Which of the following data measure scales allows for meaningful multiplication of data points?

- a. Interval data
- b. Categorical data
- c. Ordinal data
- d. Ratio data
- e. Group data

15. (LO 3) Descriptive analyses that include median central tendency measures are appropriate for which data measurement scale?

- a. Categorical
- b. Ordinal
- c. Nominal
- d. Interval
- e. All these measurement scales support the median central tendency measure.

16. (LO 3) Diagnostic analyses that include data dispersion measures such as variance and standard deviation are appropriate for which data measurement scale?

- a. Categorical
- b. Ordinal
- c. Ratio
- d. Nominal
- e. All measurement scales are appropriate for variance analysis.

17. (LO 3) Bar chart visualizations are appropriate for data with which measurement scale?

- a. Categorical
- b. Ordinal
- c. Ratio
- d. Nominal
- e. All measurement scales are appropriate for bar charts.

18. (LO 3) Which of the following objective questions reflects a diagnostic purpose for accounts receivable year-end balance?

- a. How many outstanding invoices are there in the year-end accounts receivable balance?
- b. What is the value of the year-end accounts receivable balance?
- c. Are new customers causing the increase in outstanding invoices in the year-end accounts receivable balance compared to last year?

d. What is the median value for outstanding invoices in the year-end accounts receivable balance?

e. What is the invoice that has been outstanding the longest period of time in the year-end accounts receivable balance?

19. (LO 3) Predictive and prescriptive analyses that include creating trendline visualizations are appropriate for which data measurement scale?

- a. Categorical
- b. Ordinal
- c. Nominal
- d. Interval
- e. All of these measurement scales support using trendline visualizations.

20. (LO 2, LO 3) Which of these analysis strategies involve categorical and ratio data measurement scales?

- a. Reconciliations of subsidiary ledger balances to the general ledger balance.
- b. Depreciation schedules for all real estate assets.
- c. Which types of capital sourcing have the highest cost of capital.
- d. Changes to the tax rates applicable to their organization.

21. (LO 4) Managerial accountants would most likely plan data analyses when which of the following occur?

- a. Changes to the tax rates.
- b. Elimination of the financial statement footnote disclosure requirements.
- c. Changes to Securities and Exchange Commission website.
- d. Changes to their organization's performance.

22. (LO 4) AIS accountants would most likely design a data analysis strategy for which of the following opportunities?

- a. New technologies that can provide cost savings for operational processes.
- b. New customers in the marketplace.
- c. Turnover in the regulatory body leadership.
- d. Changes to the footnote disclosures required by generally accepted accounting rules.
- e. Changes in tax rules.

23. (LO 4) Auditors plan data analyses strategies for which of the following situations?

- a. Evaluating the effectiveness of internal controls in their client's accounting system.
- b. Evaluating the representativeness of an account balance on their client's balance sheet or income statement.
- c. Considering the risk involved in providing an audit opinion for a client.
- d. Investigating whether a client's assumptions for recording their lease liabilities is consistent with other companies in their industry.
- e. All of these are examples of situations where auditors plan their data analysis strategies.

24. (LO 4) Financial accountants would most likely plan data analyses strategies for which of the following?

- a. Changes to the management team.
- b. New competitors in the marketplace.

- c. Changes to their assumptions and estimates for end of period adjusting entries.
 - d. Changes to the tax rates applicable to their organization.
 - e. None of these would motivate data analyses by financial accountants.
25. (LO 4) Tax accountants would most likely plan a data analysis strategy that can
- a. describe cash outflows for business expenses during the last year.
 - b. evaluate changes to the financial statement footnote disclosure rules.
 - c. compare advertising expenses to sales revenue increases.
 - d. compare organizational strategy options to predicted operational performance.
 - e. None of these would motivate data analyses by tax accountants.

Review Questions

1. (LO 1) Describe the five components in a data analysis project plan.
2. (LO 1) Discuss why it is important for accounting professionals to focus on the objective and questions to generate data and analysis alternatives.
3. (LO 1) Describe data and analysis risks that should be considered when planning a data analysis project.
4. (LO 2) Define and describe the four measurement scales used as accounting data variables.
5. (LO 2) Explain how selecting certain data fields could be inappropriate for your project plan.
6. (LO 2) Provide examples of two common data risks and a control for each that would reduce the risk.
7. (LO 3) Discuss why it is important to know the measurement scales of variables used in the analysis when determining whether to design a descriptive, diagnostic, predictive or prescriptive data analysis strategy.
8. (LO 3) Discuss why it is important to match the type of analysis performed with the measurement scale of the data.
9. (LO 3) Explain a trend analyses strategy, and provide an example of how an accountant may analyze sales using a trend analysis.
10. (LO 4) Assume you are a systems accountant working in the accounting information systems group at your company. You have been asked to perform data analysis to analyze the number of spam emails that pass through to employees before and after the company's investment in a new firewall. Discuss data risks and identify related controls to minimize those risks in your analysis.
11. (LO 4) Assume you are an auditor working to test internal controls regarding p-card usage at a company. You have the following data fields:
 - EmployeeNumber
 - EmployeeName
 - P-cardTransactionDate
 - P-cardTransaction vendor
 - P-cardTransactionAmount
 Describe a descriptive analysis strategy you can perform using the data strategy of the p-card transaction amount and any one of the other data fields.
12. (LO 4) Assume you are a tax accountant evaluating an individual's financial records for their federal tax compliance. What data risks should be considered?

Brief Exercises

BE 4.1 (LO 1) Put the data analysis project plan components in their proper sequential order, starting with number 1 for the first step.

- ___ a. Embed controls.
- ___ b. Identify project objective and specific questions.
- ___ c. Design data strategy.
- ___ d. Consider risks.
- ___ e. Design analysis strategy.

BE 4.2 (LO 1) Put the steps of creating a data strategy in proper order, starting with number 1 for the first step.

- ___ a. Choose and value the factors upon which to rank the data alternatives.
- ___ b. Evaluate data alternative rankings and select the best data strategy option.
- ___ c. Based on the objective, develop a few data alternatives.
- ___ d. Extract, clean, and transform the data fields to prepare the data for analysis.

BE 4.3 (LO 1) Financial Accounting For each of the following, list which critical thinking element is most clearly involved (S = Stakeholders; P = Purpose; A = Alternatives; R = Risks; K = Knowledge):

- ___ 1. Knowing what valuation procedure to use for U.S. GAAP.
- ___ 2. Considering the impact of the analysis results.
- ___ 3. The basis for the appropriateness of your data selections.
- ___ 4. Not just going with the first idea that you think of for your data and your analysis.
- ___ 5. Performing an analysis that is not appropriate for the measurement scale of your data.

BE 4.4 (LO 2) Match the data measurement scale that could be utilized to each example.

Measurement scales can be used once, more than once, or not at all.

- | | |
|----------------|-------------|
| a. Categorical | c. Interval |
| b. Ordinal | d. Ratio |

Example	Measurement Scale
1. Customer survey ratings.	
2. Account numbers in the chart of accounts.	
3. States or regions of the country.	
4. Book value of depreciable assets.	
5. Transaction dates, such as the dates orders are shipped.	
6. Credits scores of loan applicants.	
7. The VendorID field.	
8. Sales revenues.	
9. The set weight of product boxes shipped.	
10. Supermarket inventory product type, such as produce, pet food, meat, and frozen foods.	

BE 4.5 (LO 2) Auditing Assume you are an audit senior responsible for supervising this year's intern on a data analysis project related to the client's purchases. You provided the intern with a data set and requested that the intern review the data set and be prepared to discuss. An excerpt of the data follows.

PONo	VendID	VendorName	Vendor Quality	Vendor Payterms	PODate	PO ItemNo	ItemCost	ItemQty
2001	783	Pep N Supply	3	1/10 net 30	12/1/2023	23568	41.99	12
2002	783	Pep N Supply	3	1/10 net 30	12/2/2023	23567	31.99	12
2003	784	Playtime Toys	4	1/15 net 30		32425	15.99	11
2004	784	Playtime Toys	4	1/15 net 30	12/4/2023	32426	15.99	10
2005	784	Playtime Toys	4	1/15 net 30	12/5/2023	32427	15.99	25
2006	258	Snappy Supplies	5	net 45	12/6/2023	11246	3.25	26
2007	153		2	2/10 net 60	12/7/2023	11258	333.25	35
	153	Production	2	2/10 net 60	12/8/2023	11259	6.75	45
2009	783	Pep N Supply	3	1/10 net 30	12/9/2023	23566	22.99	100

1. The intern stated, "Reviewing the data set, I classified VendorID as a measured raw data field source." Explain why VendorID should be considered non-measured raw data.
2. The intern stated, "The VendorQuality data field must be considered ratio data because we can analyze this as a variable in our analysis." Explain why the VendorQuality data field should not be considered ratio data.
3. The intern said, "The ItemCost and ItemQty data fields can be combined to create a calculated data field." Explain why this is correct.
4. Based on your review of the excerpt of the data set, what data risks would you expect the intern to identify?
5. Assume you asked the intern to verify the accuracy of the PO number 2006. What would the intern review to determine if the data in the database is accurate?

BE 4.6 (LO 2) For each of the following, identify if the data field is a (MRD) measured raw data field, a (NRD) non-measured raw data field, or a (CAL) calculated field.

- ___ 1. ProductIdentificationCode
- ___ 2. UnitCost for each inventory item
- ___ 3. TotalCost
- ___ 4. GrossProfit
- ___ 5. ProductCategory
- ___ 6. NumberOnHand in the warehouse of each product

BE 4.7 (LO 3) Identify the objective of each of the following analysis strategies as describe, diagnose, predict, or prescribe.

Strategy	Objective
1. Aging of accounts receivable's year-end open invoices on days outstanding to understand what is happening in accounts receivable.	
2. Estimating cost savings and productivity increases from new AIS investments.	
3. Creating dollar value stratified sampling plans for investment asset accounts.	
4. Counting and ranking web server security incidents by week.	
5. Estimating product revenue increases by product from marketing campaigns.	
6. Estimating future cash flows from a continuing current operations strategy.	
7. Calculating before-after cost reductions from system investments.	
8. Testing different tax planning strategies by evaluating future cash savings.	
9. Estimating inventory counts at year-end by sampling a few of the stores and making population inferences.	
10. Frequency counts for unfulfilled or open-partial sales orders so that they can be resolved.	

BE 4.8 (LO 3) For each data structure and analysis listed for descriptive and diagnostic objectives, indicate whether the analysis can be performed with a "Yes" or "No."

- ___ 1. The calculation of the mode on categorical data.
- ___ 2. The variance dispersion of ordinal data.
- ___ 3. The calculation of median on interval data.
- ___ 4. Cumulative frequency distributions on categorical data.
- ___ 5. A visualization of categorical data trends.
- ___ 6. Cross tabulation correlations of ratio data.

BE 4.9 (LO 3) For each data structure and analysis listed for predictive and prescriptive objectives, indicate whether the analysis can be performed with a "Yes" or "No."

- ___ 1. Trendlines on categorical data.
- ___ 2. Regression analysis of ordinal data.
- ___ 3. Optimization modeling of interval data.
- ___ 4. Regression analysis of categorical data.
- ___ 5. What-if analysis of ratio data.
- ___ 6. Trendlines on ratio data.

BE 4.10 (LO 4) Auditing For each of the following data strategies used by auditors, provide at least one risk that should be controlled to rely on the results of the analysis.

- 1. Sampling the clients' transactions by the size of the transactions impacting an account.
- 2. Comparing the estimations and assumptions used by management this year to last year.
- 3. Requesting confirmations of all year-end bank account, accounts receivable, and accounts payable balances.

BE 4.11 (LO 4) Accounting Information Systems For each of the following analyses strategies for AIS projects, identify whether the project's objective is descriptive or diagnostic (D) or predictive or prescriptive (P):

Strategy	Objective
1. Calculating the average amount of time each department in the revenue cycle takes to perform its function per order.	
2. Calculating the cost savings expected from moving to RFID (radio free identification) technologies for inventory identification, pricing, and costing.	
3. Estimating how much server space would be needed for the AIS if sales revenues were to increase by 10% each year for the next five years.	
4. Detecting at which times of day and by which channel the unauthorized access is occurring to the customer relationship management module.	

BE 4.12 (LO 4) Financial Accounting Assume you are a financial accountant tasked with selecting the best financing alternative for your organization. Suggest a data strategy and an analysis strategy for this objective.

Exercises

EX 4.1 (LO 1) Match Project Plan Components Match the data analysis project plan component to each of the following statements:

- | | |
|-------------------------------------|----------------------|
| a. Project objectives and questions | d. Critical risks |
| b. Data strategy | e. Embedded controls |
| c. Analysis strategy | |
- ___ 1. Verifying data values for reasonableness.
 ___ 2. Data formatting that must be transformed before analysis.
 ___ 3. Predicting next year's sales volume.
 ___ 4. Sales volume data.
 ___ 5. Using regression analysis to predict next year's sales volume.
 ___ 6. Diagnosing what is causing the cost overruns in the production process.
 ___ 7. Use of different scales on the y-axis and x-axis across visualizations.

EX 4.2 (LO 1) Managerial Accounting Financial Accounting Design a Data Strategy and an Analysis Strategy Assume your motivation for performing data analysis is to maximize sales revenues. Your specific objective questions are to determine whether sales volume is increasing, stable, or dropping for each of your company's product lines this year compared to last year. Describe one possible data strategy and analysis strategy for this objective.

EX 4.3 (LO 1) Managerial Accounting Accounting Information Systems Match Data Strategy Choices Match each objective to the following data selections.

- | | |
|--|--|
| a. Labor cost data | |
| b. Production run data | |
| c. Past marketing strategy data | |
| d. IT service satisfaction survey data | |
| e. Online market sales data | |
| f. Sales contract data | |
- ___ 1. What is causing our large production materials usage variance?
 ___ 2. Which marketing strategy is most likely to increase our market share?
 ___ 3. Are we in compliance with U.S. GAAP procedures for revenue recognition?
 ___ 4. Which of our product lines could be sold online as well as in the store?
 ___ 5. Is the turnover in our IT staff impacting their service to the other functional areas?
 ___ 6. Are our payroll costs increasing?

EX 4.4 (LO 2) Managerial Accounting Design Data Strategies The following is a list of the available fields in a data set from a pet supply store.

Data Field	Description
1. PONo	Uniquely assigned purchase order number
2. VendorID	Uniquely assigned vendor number
3. VendorName	Vendor name
4. VendorQuality	Quality rating ranging from 1 - 6 where 1 = poor and 6 = exceptional
5. VendorAddress	U.S postal mailing address and street name
6. VendorCity	Mailing address city
7. VendorState	Mailing address state
8. VendorZip	Mailing address zip
9. VendorPayterms	Payment terms negotiated with the vendor
10. PODate	Purchase order date
11. POItemID	Uniquely identifiable item number
12. POItemDescription	Description of item
13. ItemCost	Cost of the inventory item per negotiated terms with vendor
14. ItemQty	Total quantity of items purchased

For each of the following objectives (a, b, c), identify the minimum data fields (1-14) you would select for your data strategy:

- Your objective is to reduce the carbon footprint of your purchases, so you want to identify which vendor transported your purchases across the most miles last year.
- Your objective is to determine which inventory items have the greatest cost increases over the past five years.
- Your objective is to identify the five vendors from whom you made the highest value of purchases last year.

EX 4.5 (LO 2) Prepare a Project Plan You graduate in one month from your undergraduate program. Your objective is to successfully apply to both professional jobs and graduate school. In your application materials, you want to show how your grades improved as you moved from your general education courses to the courses in your major. Describe your data strategy, your analysis strategy, and three possible risks you need to control by providing control suggestions.

EX 4.6 (LO 2) Managerial Accounting Data Strategy Risks and Controls Assume your data analysis objective is to reduce the carbon footprint of your purchases from vendors. Your analysis strategy is to identify which vendor transported your purchases across the most miles last year. You have selected the following data fields for your data strategy.

Data Field	Description
PONo	Uniquely assigned purchase order number
VendorID	Uniquely assigned vendor number
VendorName	Vendor name
VendorAddress	U.S postal mailing address and street name
VendorCity	Mailing address city
VendorState	Mailing address state
VendorZip	Mailing address zip
PODate	Purchase order date

Identify three data risks and data controls you could use for those risks.

EX 4.7 (LO 3) Auditing Managerial Accounting Prepare a Project Plan You are an internal auditor working with your company's outdoor gear sales department to devise a new strategy for sales commissions that incentivizes higher profits than your current flat commission strategy. Describe a data strategy, an analysis strategy and three possible risks you need to control by providing control suggestions.

EX 4.8 (LO 3) Managerial Accounting Prepare a Project Plan You are a managerial accountant for Sihrya's Beauty Supply Store. You have been asked to predict next year's contribution margin. A sample of the available data for your analysis is included here.

Field Label	Field Name in Database
Receipt Number	SaleReceiptNo
Sales Date	Saledate
Inventory Code	InvCode
Number Sold	NoSold
Inventory Description	InvDesc
Inventory Price	InvPrice
Inventory Cost	InvCost

Describe a data strategy, an analysis strategy, and three possible risks you need to control by providing control suggestions.

EX 4.9 (LO 3) Financial Accounting Managerial Accounting Prepare a Project Plan You work as a financial analyst for a large coffee shop chain, such as Starbucks. Your objective is to diagnose which stores have increasing sales. Your data strategy is to use the store sales monthly data from the past two years. Design an analysis strategy, including which risks you should consider and controls that would help mitigate those risks.

EX 4.10 (LO 1-4) Auditing Accounting Information Systems Prepare a Project Plan You are working in the accounting information systems group at your company. You have been asked to evaluate the internal controls associated with authentication of users into the information system. Your manager has asked you to perform a descriptive analysis of failed login attempts. Your manager has provided you with the analysis objective and question. Using the chart, document your data and analysis choices and risks and controls for each.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Evaluate internal controls associated with authentication of users Questions: What are the mean, median, standard deviation, and distribution of failed login attempts during the current year?	1. Data: 2. Analysis:	3. Data: 4. Analysis:	5. Data: 6. Analysis:

EX 4.11 (LO 1-4) Financial Accounting Managerial Accounting Prepare a Project Plan You are a financial analyst at Sihrya's Beauty Salon. The company owner has asked you to perform data analyses to understand the products that contribute to the salon's retail store's profitability. The owner provided you a data dictionary, presented here, describing data that you may consider using in your analysis.

Field Label	Field Name in Database	Field Description
Receipt Number	ReceiptNo	POS assigned receipt number uniquely identifying each sale.
Sales Date	SaleDate	Date of sale per the POS.
Inventory Code	InvCode	Unique inventory identification number for each product in the salon's retail store.
Number Sold	NoSold	Quantity of items sold.
Inventory Description	InvDesc	Description of the inventory item.
Inventory Price	InvPrice	Gross sales price of the inventory item.
Inventory Cost	InvCost	Weighted average cost of the inventory item.

1. State the owner's objective for your data analysis project.
2. Assume your analysis question is to identify the products that have the highest gross profit. Identify the data fields you should include in your analysis. Identify the data analysis choices to answer this analysis question.
3. Identify risks and the related control choices related to the analysis question.

EX 4.12 (LO 1-4) Data Managerial Accounting Select a Data Strategy and Perform an Analysis You are a managerial accountant working at a multi-location pet care retail company. Your supervisor has asked you to compare the dollar amount of purchases made from each vendor in December 2024 compared to December 2025. After discussing with your supervisor, you have identified the following:

Objective: Compare total purchases by vendor in December 2024 and December 2025.

Question: In 2024 and 2025, from which vendors were the most purchases (in dollars) made?

Data: You have access to the following data.

Data Field	Description
PONo	Uniquely assigned purchase order number.
VendorID	Uniquely assigned vendor number.
VendorName	Vendor name.
VendorQuality	Quality rating ranging from 1 - 6 where 1 = poor and 6 = exceptional.
VendorAddress	U.S. postal mailing address and street name.
VendorCity	Mailing address city.
VendorState	Mailing address state.
VendorZip	Mailing address zip.
VendorPayterms	Payment terms negotiated with the vendor.
PODate	Purchase order date.
POItemID	Uniquely identifiable item number.
POItemDescription	Description of item.
ItemCost	Cost of the item per negotiated terms with vendor.
ItemQty	Total quantity of items purchased.

1. Which data items would you use in your data strategy?
2. Design an analysis strategy to identify the vendor or vendors in which the company had a higher dollar amount of purchases in 2025 compared to 2024.
3. Perform the analysis per your analysis strategy. Identify the vendors in which the company had a higher dollar amount of purchases in 2025 compared to 2024.

Problems

PR 4.1 (LO 1-4) Data Auditing Managerial Accounting Complete Project Plan You work in the internal audit group in your organization, and your supervisor has asked you to analyze p-card data. The objective of the analysis is to understand p-card spending in the current year. The questions associated with the objective include:

- With which three vendors does the company spend the most money using p-cards?
- Which employee spends the highest dollar amount of money using the p-card?

Review the data and complete the chart to document your data and analysis strategy choices.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand p-card spending in the current year. Questions: • What are the three vendors where the Company spends the most money using p-cards? • Which employee spends the highest dollar amount of money using the p-card?	Data: Use the data provided in the Excel file. Analysis: Use Excel to create a PivotTable that allows grouping of the p-card spending data by vendor and sort by the highest vendor. Use Excel to create a PivotTable to group p-card spending amount by employee and sort by the highest dollar amount by employee.	1. Data: 2. Analysis:	3. Data: 4. Analysis:

5. Perform the analyses suggested in the chart. Summarize your results.

PR 4.2 (LO 1, 2, 3) Data Financial Accounting Managerial Accounting Complete Steps 2 and 3 of a Project Plan and Perform Analysis. You are a financial analyst at Sihrya's Beauty Salon. The owner has asked you to perform data analyses to understand the products that contribute to the salon's retail store's profitability. The owner gave you a data dictionary, which is presented here, describing data that you may consider using in your analysis.

Field Label	Field Name in Database	Field Description
Receipt Number	ReceiptNo	POS assigned receipt number uniquely identifying each sale.
Sales Date	Saledate	Date of sale per the POS.
Inventory Code	InvCode	Unique inventory identification number for each product in the salon's retail store.
Number Sold	NoSold	Quantity of items sold.
Inventory Description	InvDesc	Description of the inventory item.
Inventory Price	InvPrice	Gross sales price of the inventory item.
Inventory Cost	InvCost	Weighted average cost of the inventory item.

Assume your analysis objective is to identify the products that contribute the most to the salon's retail store profitability. Your specific questions are: Which products have the highest units sold? Which products have a positive gross profit margin? Use the information provided to answer the following questions:

1. Identify the data fields to include in your data strategy.
2. Identify a data strategy that could be used to answer the objective questions.
3. Identify an analysis strategy that could be used to answer the objective questions.
4. Review the data provided to perform the analysis. After performing the analysis, identify the top two products that contribute to the salon's profitability.

PR 4.3 (LO 1-4) Data Managerial Accounting Complete the Project Plan and Perform the Analysis As a financial analyst at a hospitality industry company, you have been asked to design a data analysis strategy to understand the factors that influence a guest's quality rating. Your team administered a survey of guests staying at your properties during the month of June. The database team has linked the survey responses for quality rating of the stay to the guest's location, check-in and check-out date, and spending amounts. They indicated that they only included data related to surveys completed September 1 through September 10.

1. Which data measurement scales are included in the data fields labeled Location, QualityRating, and SpendingAmount?
2. Review the data set, identify data risks present in the data set, and suggest controls for these risks.
3. Assume a member of your team suggested that you use the data to predict the quality rating of future stays. To do this predictive analysis, you would have to use the quality rating data in a predictive analysis strategy. How would you respond to your team member about the appropriateness of using this type of analysis?
4. Use this data set to design and perform a different, original data strategy and an analysis strategy to satisfy the objective question. Document your data strategy, analysis strategy, risks, and controls and present your analysis results.

PR 4.4 (LO 1-4) Data Tax Accounting Complete the Project Plan and Perform the Analysis Beautiful Bites is a chain of bakeries located in Colorado. Its owners are committed to social sustainability values and want to focus their philanthropy in the municipalities where they are collecting the most sales tax. The objective of the data analysis is to determine where most of their customers live. The specific questions is: In which communities do most of our customers reside?

1. Prepare the data analysis project plan.
2. Perform the analysis and summarize the results.

Professional Application Case: Automated Transportation, Inc. (ATI)

Automated Transportation, Inc. is a medium-sized manufacturer of remote-controlled cars, boats, and drones. The company was established five years ago when two brothers decided to build and sell remote control cars to enthusiasts. As the company grew, they expanded their product offerings to include remote control cars, boats, and drones. The brothers serve as the president and CEO of the company, and they now have over 70 employees.

They have two key customer groups, hobbyists and businesses interested in incorporating drones into their business process and supply chain. These two customer bases provide a variety of opportunities for growth. Management values internal control, but since they are busy running the company they have employed accounting personnel to help design the company's processes, policies, and internal controls:

- The company is not required to have a report on internal controls and external auditors are not required to attest to the company's internal controls.
- However, the owners want to be sure the company employees are designing and following policies that will help them maintain efficient and effective operations.
- The owners are also very data-driven and make many of their business decisions only after considering the data collected and analyzed.

The following is an excerpt from the data dictionary designed by the company's information systems personnel and accounting information systems accountants.

Data Field Label	Field Name in Database	Field Description
Invoice number	InvoiceNO	The invoice number, which is hand keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Invoice amount	InvoiceAmt	The amount of the invoice, which is hand-keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Shipment date	ShipDate	The date the product was shipped from the shipping location.

(Continued)



Data Field Label	Field Name in Database	Field Description
Invoice date	InvoiceDate	The date of the invoice, which hand-keyed into the AIS by the AP clerk from a manual invoice mailed to the company by the vendor.
Vendor identification number	VendorID	Unique vendor identification number.
Vendor name	VendorName	Name of the vendor.
Product purchased	ProductID	Product code for the product purchased from the vendor. This product code is consistent with the catalog from the vendor.
Unit cost	UnitCost	The cost per unit of the product purchased from the vendor.
Shipping cost	ShipCost	Total shipping costs.
Flat duty	FlatDuty	The flat duty rate applies to article that are dutiable.
Tariff	TariffAmt	The total dollar amount of a tariff applied to the goods shipped.
Shipping location	ShipLocation	The country in which the goods are shipped from.
Receiving Quality rating	QualityRate	This is a quality rating keyed into the AIS by the receiving team when they receive the goods. The scale is 1 = poor to 5 = excellent quality. The receiving team rates the shipment on packaging, quality of materials, and overall delivery.
Payment terms	PaymentTerms	The agreed-upon payment terms with the vendor. These terms are negotiated by the purchasing manager and keyed into the vendor master file by the purchasing supervisor.
Shipping terms	ShipTerms	Shipping terms—typically FOB destination or FOB shipping.
Payment address	PayAddress	The vendors address where payment is to be mailed.
Purchase order number	PONumber	The unique identifying number assigned to each purchase order issued by the company.
Purchase order date	PODate	The date the purchase order is issued by the company.
Receiving report number	ReceivingNumber	The unique identifying number assigned to each receiving report created by the company's receiving group.
Receiving report date	ReceivingDate	The date the product is received by the company's receiving department.
Quantity received	QtyReceived	The total quantity of items received.
Quantity purchased	QtyPurchased	The total quantity of items on the Purchase Order.
Invoiced quantity	QtyInvoice	The total quantity of items on the invoice, this amount is hand keyed into the AIS by the AP clerk from the manual invoice mailed to the company by the vendor.
Treasurer Approval	Approved	The initials of the Treasurer indicating their approval if the invoice was greater than \$10,000.

PAC 4.1 Accounting Information Systems: Evaluate AIS Internal Controls

Accounting Information Systems As an information systems accountant, the owners have asked to you to design internal controls related to the company's vendor payment system. Remember from your accounting information systems course that there are several important controls in the order-to-pay cycle, including approval of purchases, and approval of vendor invoices. You have implemented the following control:

- The accounting clerk prepares a voucher package containing a disbursement voucher, vendor invoice, purchase order, and receiving document for each payment to be made.
- Invoices where payment exceeds \$10,000 have a second level of approval, which is noted in the information system by the treasurer's initials.

You want to test the operational effectiveness of this control by analyzing data captured by the accounting information system during this process. Assume the objective of your analysis is to understand whether the internal control is operating as designed. Design your data analysis strategy to address this objective. Be sure to document the following components:

1. What is the question your data analysis should address?
2. Using the data dictionary, what measured raw data and non-measured raw data fields should you request?
3. What analysis choices will you make to analyze the data? Specifically, what calculated data fields would you create in your analysis?
4. What are the data risks to your data analysis strategy?
5. What are the analysis risks in your strategy?
6. What controls should you include in your data and analysis strategies to minimize the risks you outlined?

PAC 4.2 Auditing: Select Large and Unusual Purchases

Data Auditing You are a second-year staff working at a public accounting firm assigned to the ATI engagement, which is a new engagement for the firm. Your audit senior has provided you with a transaction file that contains all the company's payments to vendors for the month of January. You have been asked to determine if the company is accurately recording invoice information by designing a data analysis strategy to select purchase transactions for further testing. You must design a data analysis strategy to select purchase transactions for further testing. Your senior asked you to identify transactions that may be outside the normal purchasing behavior. Therefore, you must perform descriptive analyses. Your senior provided you with this partial data analysis plan.

Complete the chart by identifying the risk and related controls for the data and analysis choices determined in the data analysis plan.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Identify purchase transactions for further testing. Questions: Are there purchase transactions that may be considered anomalies?	Data strategy: InvoiceNo, InvoiceDate, QtyInvoice, InvoiceAmt, VendorID, VendorDescription Analysis strategy: Perform descriptive statistics to understand total dollar amount and count of purchases made to each vendor. Perform diagnostic statistics and create a scatterplot whereby the InvoiceDate is on the X-axis and the InvoiceAmount is on the Y-axis to identify any outlier purchases.	1. Data: 2. Analysis:	3. Data: 4. Analysis:

5. Perform descriptive statistical analysis to identify the total dollar amount and count of purchases made to each vendor.
6. Use the provided data set to perform the diagnostic statistical analysis to create a scatterplot where the InvoiceDate is on the X-axis and the InvoiceAmt is on the Y-axis to identify outlier purchases.

PAC 4.3 Financial Accounting: Plan to Identify High Activity Vendors

Financial Accounting The owners of the company want to know which vendors the purchasing group are using and how much they are paying each to negotiate better contract terms with regular vendors. You are performing data analysis to identify the most used vendors. Answer each question with respect to your data analysis.

1. What is the question your data analysis should address?
2. Using the data dictionary, what measured raw data and non-measured raw data fields should you request?
3. What analysis choices will you make to analyze the data? Be specific, what calculated data fields would you create in your analysis?
4. What are the data risks to your data analysis strategy?
5. What are the analysis risks in your strategy?
6. What controls should you include in your analysis to minimize the risks you outlined?

PAC 4.4 Managerial Accounting: Rank Vendor Quality

Data Managerial Accounting Because the company is growing so fast, the owners want to be sure that they are partnering with the right vendors for raw materials. They have asked the receiving team to give each receipt a rating to document the quality of the packaging, the materials in appearance, and overall delivery. For each receipt the receiving group documents a quality rating in the accounting information system. The owners have asked you to determine how the quality rating by vendor has changed since the beginning of the year. Your manager has designed the following partial data analysis strategy. Complete the chart and document the risks and related controls to your analyses.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand the quality ratings of purchases.	Data strategy: VendorID, VendorName, ReceivingNo, ReceivingDate, QualityRate	1. Data:	3. Data:
Questions: What is the average quality rating by vendor?	Analysis strategy: Calculate the average QualityRating by VendorName.	2. Analysis:	4. Analysis:
What is the highest and lowest quality rating by vendor?	Calculate the minimum and maximum QualityRating by VendorName.		
How has the quality rating by vendor changed over time?	Visualize how QualityRating by VendorName has changed over time.		

5. Perform the analyses suggested in the data analyses project plan.

PAC 4.5 Tax Accounting: Plan International Duty Cost Analysis

Tax Accounting The owners want to understand the total dollar amount of customs duty they will owe the U.S. government for the year. They have asked you to perform descriptive analytics to understand the purchases from vendors located in each country. Your company makes purchases from vendors located in several countries, such as China, Mexico, the United States, Japan, South Africa, Israel, Greece, and Egypt.

You and your team have designed a data analysis strategy. You have been asked to consider the risks and related controls in the development of that strategy. Complete the following chart.

Objective and Questions	Data and Analysis Strategies	Risks	Controls
Objective: Understand the purchases made from each country.	Data strategy: ShipLocation, InvoiceAmt, FlatDuty, TariffAmt, PONumber	1. Data:	3. Data:
Questions: What dollar amount of purchases are made from each country?	Analysis strategy: Use a pivot table to drop the ShipLocation into the rows and the sum of InvoiceAmt into values.	2. Analysis:	4. Analysis:
How many purchases were made from vendors in each country?	Use a pivot table to drop the ShipLocation into the rows and count of PONumber into values.		

Le Grind Continuing Case: Design a Data Analysis Plan for Describing and Diagnosing Gross Margin Changes

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